

Lie Algebra and their Representations

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This is the revision note for the course **Lie algebra and their representations**. The introduction section is mostly on motivations to study Lie algebras and one should comfortably skip the part that they find confusing in a first read. The words between quotation marks are the comments by the lecturer.

1 Introduction

Motivations: Recall that connected Lie groups are groups that are also differential manifolds. A class of Lie groups we are interested in is the simple algebraic groups over algebraically closed fields:

$$\begin{aligned} & \text{(the special linear group)} \quad SL(n) = \{A \in Mat_n \mid \det(A) = 1\} \\ & \text{(the special orthogonal group)} \quad SO(n) = \{A \in SL_n \mid AA^T = I\} \\ & \text{(the symplectic group)} \quad SP(2n) = \{A \in GL_{2n} \mid A^T J A = J \text{ where } J \text{ is the canonical symplectic form}\} \\ & \dots \text{ and exactly 5 others as we will see} \end{aligned}$$

Each of these groups has a maximal compact subgroup. For example, $SU(n)$ (the special unitary group) is the maximal compact subgroup of $SL(n)$, while $SO_n(\mathbb{R})$ is the maximal compact group of $SO_n(\mathbb{C})$. The representations of the maximal compact subgroup are the same as the algebraic representations of the simple algebraic group are the same as the finite-dimensional representations of its (the simple algebraic group's) Lie algebra, which are what we study.

Lie groups are in general complicated objects: for example, $SU(2)$ is homotopic to a 3-sphere, and $SU(n)$ is homotopic to a twisted product of a 3-sphere, a 5-sphere and up to a $2n - 1$ -sphere. Thus, studying Lie groups requires some amount of algebraic topology and a lot of algebraic geometry. In this course, we want to replace these “complicated” disciplines with easier discipline of linear algebra.

Therefore, instead of a Lie group G we will be considering $\mathfrak{g} = T_I G$, the tangent space to G at the identity.

Definition 1.1. A **linear algebraic group** is a subgroup of some GL_n defined by polynomial equations in the matrix coefficients.

We see that the examples above are linear algebraic groups.

Proposition 1.1. Linear algebraic groups are precisely affine algebraic groups.

Proof. Note that GL_n can be identified with $\left\{ \begin{pmatrix} A & 0 \\ 0 & \frac{1}{\det(A)} \end{pmatrix} \mid A \in GL_n \right\} \subset Mat_{n+1}$, and so is affine. Since closed subschemes of affine schemes are affine, linear algebraic groups are affine. For the converse direction, we need to show that any affine algebraic group has a faithful finite-dimensional representation, i.e. embeds into some $GL(V)$. A sketch of the proof can be found in [Poo17, theorem 5.3.1]. $\square$

First we explain tangent space in a non-rigorous way like physicists.

Example 1. Let $G = SL_2$ and $|\epsilon| \ll 1$. Then for $g = I + \epsilon \begin{pmatrix} a & b \\ c & d \end{pmatrix} + \text{higher order terms}$ to lie in SL_2 , need $\det(g) = 1$, or

$$1 = 1 + \epsilon(a + d) + \epsilon^2(\dots)$$

So we need to have $a + d = 0$

To do this formally, define the *dual numbers* $E = \mathbb{C}[\epsilon]/(\epsilon^2) = \{a + b\epsilon\}$. For a linear group G , consider $G(E) = \{A \in \text{Mat}_n(E) \mid A \text{ satisfies the equations defining } G \subset GL_n\}$ (recall how we define group schemes as functors). So $SL_2(E) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \mid a, b, c, d \in E, ad - cb = 1 \right\}$. Then the natural map $E \rightarrow \mathbb{C}, \epsilon \mapsto 0$ gives a projection $G(E) \rightarrow G$. Define the *Lie algebra of G* as

$$\mathfrak{g} = \pi^{-1}(I) \cong \{A \in \text{Mat}_n(\mathbb{C}) \mid I + \epsilon A \in G(E)\}$$

(Note that any matrix in $G(E)$ has the form $A + \epsilon B$). In particular $\mathfrak{sl}_2 = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \text{Mat}_2(\mathbb{C}) \mid a + d = 0 \right\}$ is the Lie algebra of SL_2

$G(E)$ is the tangent bundle of G and $\mathfrak{g}$ is the tangent space of G at I .

Example 2. Let $G = GL_n$. We claim that $G(E) = \{A + \epsilon B \mid A, B \in \text{Mat}_n(\mathbb{C}), A^{-1} \text{ exists}\}$. It is clear that $G(E)$ is in the latter set. Conversely, have $(A + \epsilon B)(A^{-1} - A^{-1}BA^{-1}\epsilon) = I$. And the Lie algebra of GL_n is precisely M_n

Example 3. Let $G = SL_n$. Note that $\det(I + A\epsilon) = 1 + \text{trace}(A)\epsilon$. So the Lie algebra of SL_n is precisely $\mathfrak{sl}_n$

Example 4. Let $G = O_n(\mathbb{C})$. Its Lie algebra $\mathfrak{g} = \{A \in \text{Mat}_n(\mathbb{C}) \mid (I + A\epsilon)(I + A\epsilon)^T = I\} = \{A \in \text{Mat}_n(\mathbb{C}) \mid A + A^T = 0\}$ is precisely the group of anti-symmetric matrices. Note $A + A^T = 0 \implies \text{tr}(A) = 0$ since we work over $\mathbb{C}$. So it is also the Lie algebra of SO_n . This is not surprising since O_n has two connected components corresponding to determinant $+1$ and -1 (they are images of each other via a reflection). Since $\mathfrak{g} = T_I G$, it cannot be the component of $\det -1$ as expected.

The above examples motivate us to study the structure of $\mathfrak{g}$ from G being a group. First we note that $\mathfrak{g}$ does not get a multiplication from G since $(I + A\epsilon)(I + B\epsilon) = I + (A + B)\epsilon$. The bilinear map turns out to generalize nicely is the commutator $(P, Q) \mapsto PQP^{-1}Q^{-1}$. If we treat this as a map $G \times G \rightarrow G$ that sends (I, I) to I , it induces a map $T_I G \times T_I G \rightarrow T_I G$ by differentiation.

To make this explicit, let $P = I + A\epsilon, Q = I + B\delta$ with $\epsilon^2 = \delta^2 = 0$ but $\epsilon\delta \neq 0$. Then $PQP^{-1}Q^{-1} = (I + A\epsilon)(I + B\delta)(I - A\epsilon)(I - B\delta) = I + (AB - BA)\epsilon\delta$. Thus, the binary operation on $\mathfrak{g}$ should be $[A, B] = AB - BA$. Also, $(PQP^{-1}Q^{-1}) = QPQ^{-1}P^{-1}$ implies $[A, B] = -[B, A]$ and associativity of multiplication in G implies the Jacobi identity

$$0 = [[X, Y], Z] + [[Y, Z], X] + [[Z, X], Y]$$

This motivates the object of study in this course

Definition 1.2. Let K be a field and $\text{char}(K) \neq 2, 3$. A **Lie algebra** $\mathfrak{g}$ is a vector space over K with a bilinear map (**Lie bracket**) $[\cdot, \cdot]$ with the following properties:

1. $[X, Y] = -[Y, X]$ (*skew-symmetry*)
2. $0 = [[X, Y], Z] + [[Y, Z], X] + [[Z, X], Y]$ (*Jacobi identity*)

Example 5. In example 2 we saw that $\mathfrak{gl}_n = M_n$ with the commutator bracket is the Lie algebra of GL_n .

Example 6. $\mathfrak{sp}_{2n} = \{A \in \mathfrak{gl}_{2n} \mid JA^T + AJ = 0\}$ is the Lie algebra of $SP(2n)$

Example 7. Let $\mathfrak{b}$ be the upper triangular matrices and $\mathfrak{n}$ the strictly upper triangular matrices. They are both Lie algebras under the commutator bracket

Example 8. For any K -vector space V , let $[\cdot, \cdot] : V \times V \rightarrow V$ be the zero map. This is the “abelian Lie algebra”.

Definition 1.3. A **representation** of a Lie algebra $\mathfrak{g}$ on a vector space V is a Lie algebra homomorphism $\phi : \mathfrak{g} \rightarrow GL(V)$. We say that $\mathfrak{g}$ acts on V .

There is always a representation coming from the Lie algebra acting on itself (it is the derivative of the group acting on itself by conjugation)

Definition 1.4. For $x \in \mathfrak{g}$, define $ad(x) : \mathfrak{g} \rightarrow \mathfrak{g}; y \mapsto [x, y]$

Lemma 1.1. $ad : \mathfrak{g} \rightarrow End(\mathfrak{g})$ is a representation

Proof. NTS that $ad([x, y]) = [ad(x), ad(y)] = ad(x)ad(y) - ad(y)ad(x)$. Note that for $z \in \mathfrak{g}$, $ad([x, y])(z) = [[x, y], z]$ and $(ad(x)ad(y) - ad(y)ad(x))(z) = [x, [y, z]] - [y, [x, z]] = -[[y, z], x] - [[z, x], y]$. So this is precisely Jacobi identity. $\square$

Definition 1.5. The **center** of $\mathfrak{g}$ is $\{x \in \mathfrak{g} \mid [x, y] = 0 \ \forall y \in \mathfrak{g}\}$, i.e. the kernel of $ad : \mathfrak{g} \rightarrow End(\mathfrak{g})$

Thus, ad is faithful if and only if $\mathfrak{g}$ has no center.

Is it true that every f.d. Lie algebra injects into some $GL(V)$, i.e. is linear? We see that if $\mathfrak{g}$ has no center then the adjoint representation makes this true. Conversely, inside $\mathfrak{gl}_2$, $\mathfrak{n} = \left\{ \begin{pmatrix} 0 & * \\ 0 & 0 \end{pmatrix} \right\}$ is abelian and so mapped to zero in $\mathfrak{gl}_{\mathfrak{n}}$. So we can't just take ad .

Theorem 1.1. (Ado's theorem) Any finite-dimensional Lie algebra over K is a subalgebra of $\mathfrak{gl}_n$, i.e. admits a faithful finite-dimensional representation.

Remark 1.1. This is the Lie algebra equivalent of the statement we made last time about (affine) algebraic groups embedding into $GL(V)$. That theorem was “easy” because there was a natural representation to look at (functions on the algebraic group). There isn't such a natural object for Lie groups, so Ado's theorem is actually hard.

Example 9. $\mathfrak{g} = \mathfrak{sl}_2$ has basis $e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$. Have $[e, f] = h, [h, e] = 2e$ and $[h, f] = -2f$

A representation of $\mathfrak{sl}_2$ is a triple $E, F, H \in Mat_n$ satisfying the same bracket relations. Where might we find such a thing? In fact, we will get them as derivatives of representations of SL_2 . We will then rederive them from just the linear algebra.

Definition 1.6. If G is an algebraic group, an algebraic representation of G on V is a homomorphism of groups $\rho : G \rightarrow GL(V)$ defined by polynomial equations in the matrix coefficients.

To get a representation of the Lie algebra out of this, we again use the dual numbers $E = K[\epsilon]/(\epsilon^2)$. Have a homomorphism of groups $G(E) \rightarrow GL_V(E)$. Moreover, since $\rho(I) = I$ and it commutes with the projection map, have $\rho(I + A\epsilon) = I + \epsilon \times (\text{some function of } A) = I + \epsilon d\rho A$, say. Then $d\rho$ is the derivative of ρ evaluated at I , and hence $d\rho : \mathfrak{g} \rightarrow \mathfrak{gl}_V$ is a Lie algebra homomorphism, i.e. V is a representation of $\mathfrak{g}$

Example 10. Take $G = SL_2$. Let $L(n)$ be the space of homogeneous polynomials of degree n in two variables x, y , with basis $x^n, x^{n-1}y, \dots, y^n$. Then GL_2 acts on $L(n)$ by change of coordinates: for $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $f \in L(n)$, we have $\rho_n(g)(f) = f(ax + cy, bx + dy)$

In particular, ρ_0 is the trivial representation and ρ_1 is just the usual representation on K^2 . And $\rho_2 \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a^2 & ab & b^2 \\ 2ac & ad+bc & 2bd \\ c^2 & cd & d^2 \end{pmatrix}$. Since GL_2 acts on $L(n)$, we see that SL_2 acts on $L(n)$

Remark 1.2. The proper way to think of this is as follows. GL_2 acts on $\mathbb{P}^1$ and $\mathcal{O}_{\mathbb{P}^1}(n)$ and so on $\Gamma(\mathbb{P}^1, \mathcal{O}(n)) \cong L(n)$, and that's the source of these representations. We can do this sort of thing for all the other algebraic groups we listed previously, using flag varieties instead of $\mathbb{P}^1$ and possibly higher (co)homologies instead of the global sections (this is a theorem of Borel, Wéil, and Bott). That, however, requires algebraic geometry, and gets harder to do in infinitely many dimensions.

We can differentiate the above representations, e.g. for $e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$, we see $\rho(I + \epsilon e)x^i y^j = x^i(\epsilon x + y)^j = x^i y^j + \epsilon j x^{i+1} y^{j-1}$. Hence $d\rho(e)x^i y^j = j x^{i+1} y^{j-1}$. Similarly, we get $d\rho(f)$ and $d\rho(h)$ and deduce that in the action of the Lie algebra,

$$e(x^i y^j) = j x^{i+1} y^{j-1}$$

$$f(x^i y^j) = i x^{i-1} y^{j+1}$$

$$h(x^i y^j) = (i - j)x^i y^j$$

This in fact gives a representation of $\mathfrak{sl}_2$ on $L(n)$. For example, $(\rho(e)\rho(f) - \rho(f)\rho(e))(x^i y^j) = i(j+1)x^i y^j - (i+1)j x^i y^j = (i-j)x^i y^j = \rho(h)(x^i y^j) = \rho([e, f])(x^i y^j)$. On $L(2)$, this is just the adjoint representation.

Based on this, we see that $e = x\partial/\partial y, f = y\partial/\partial x, h = x\partial/\partial x - y\partial/\partial y$ gives a (infinite dimensional) representation of $\mathfrak{sl}_2$ on $k[x, y]$. If $\text{char}(K) = 0$, $L(n)$ is an irreducible representation of $\mathfrak{sl}_2$ since any $x^i y^j$ generates $L(n)$ under the action of $\mathfrak{sl}_2$, and hence of $SL_2(?)$

We have defined a functor from the category of algebraic representations of a linear algebraic group G to the category of Lie-algebra representations of $\text{Lie}(G)$ by $\rho \mapsto d\rho$. However, this functor is not as nice as we want, except for the Lie algebras we care about.

Example 11. Let $G = \mathbb{C}^\times$ and $\mathfrak{g} = \text{Lie}(G) = \mathbb{C}$ with $[x, y] = 0$. A representation of $\mathfrak{g}$ on V is a matrix $A \in \text{End}(V)$ such that $A = \rho(1)$ with $\rho : \mathfrak{g} = \mathbb{C} \rightarrow \text{End}(V)$ the representation. $W \subset V$ is a submodule iff $AW \subset W$ and ρ is isomorphic to $\rho' : \mathfrak{g} \rightarrow \text{End}(V')$ iff A and A' are conjugate as matrices. Therefore, representations of $\mathfrak{g}$ correspond to the Jordan normal form of matrices.

As any linear transformation over $\mathbb{C}$ has an eigenvector, there is always a dim-1 subrep. (the eigenspace) of V . Therefore V is irreducible iff $\dim(V) = 1$. Also, V is completely decomposable iff A is diagonalizable.

As for the representations of $G = \mathbb{C}^\times$, the irreducible representations are $\rho_n : G \rightarrow GL_1 = \text{Aut}(\mathbb{C})$ via $z \mapsto (\text{multiplication by}) z^n$ for $n \in \mathbb{Z}$. Every finite-dimensional representation is a direct sum of these.

Remark 1.3. Proving these statements about representations of G takes some theory of algebraic groups. However, you probably know this for S^1 , which is homotopic to G . Recall that to get complete reducibility in the finite group setting you took an average over the group; you can do the same thing here, because S^1 is compact and so has finite volume. In fact, for the theory that we study, our linear algebraic groups have maximal compact subgroups, which have the same representations, which for this reason are completely reducible. We will not prove that representations of the linear algebraic groups are the same as the representations of their maximal compact subgroups.

Observe that the representations of G and of $\mathfrak{g}$ are not the same. Indeed, $\rho \mapsto d\rho$ will send $\rho_n \mapsto n \in \mathbb{C}$ ($(1 + \epsilon)^n = 1 + n\epsilon$). Therefore, irreducible representations of G embed into irreducible representations of $\mathfrak{g}$, but the map is not remotely surjective.

The above example isn't very surprising, since $\mathfrak{g}$ is also the Lie algebra $Lie(\mathbb{C}, +)$, and we should not expect representations of $\mathfrak{g}$ to coincide with those of $\mathbb{C}^\times$. The surprising result is

Theorem 1.2. *(Lie) $\rho \mapsto d\rho$ is an equivalence of categories $Rep_G \rightarrow Rep_{\mathfrak{g}}$ if G is a simply connected simple algebraic group.*

Remark 1.4. A “simple algebraic group” is not simple in the usual sense: e.g. SL_n has a center, which is obviously a normal subgroup. However, if G is simply connected and simple in the above sense, then the center of G is finite (e.g. $Z(SL_n) = \mu_n$) and the normal subgroups are subgroups of the center.

The map (algebraic groups) $\rightarrow$ (Lie algebras) via $G \mapsto \mathfrak{g} = Lie(G)$ is neither injective nor surjective: If G is an algebraic group, and Z is a finite central subgroup of G , then $Lie(G/Z) = Lie(G)$ (Morally, Z identifies points “far away”, and therefore does not affect the tangent space at I). The image of this map is in fact the “algebraic Lie algebras”. For example, in characteristic p $\mathfrak{sl}_p$ /center cannot be the image of an algebraic group. In general, algebraic groups have a Jordan decomposition – every element can be written as (semisimple) $\times$ (nilpotent), -and therefore the algebraic Lie algebras should have a Jordan decomposition as well. In Bourbaki, you can find an example of a 5-dimensional Lie algebra, for which the semisimple and the nilpotent elements lie only in the ambient $\mathfrak{gl}_V$ and not in the algebra as well.

2 Representations of $\mathfrak{sl}_2$

From now on, all algebras and representations are over $\mathbb{C}$ (or any algebraically closed field with characteristic 0). We will mention which results don't need this assumption.

Recall: $\mathfrak{sl}_2$ has basis $e = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, h = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, f = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$ with $[e, f] = h, [h, e] = 2e$ and $[h, f] = -2f$

The main theorem of this section is the following:

Theorem 2.1. *1. For all $n \geq 0$ there is a unique representation of $\mathfrak{sl}_2$ of dimension $n + 1$ (which is precisely $L(n)$ from last section)*

2. Every f.d. representation of $\mathfrak{g}$ is a direct sum of irreducible representations (i.e. completely decomposable).

To prove this, we introduce **weight space**

Definition 2.1. *Let V be a representation of $\mathfrak{sl}_2$. The λ -weight space of V is $V_\lambda = \{v \in V \mid hv = \lambda v\}$, i.e. the eigenvectors of h with eigenvalue λ*

In $L(n)$, have $L(n)_\lambda = \mathbb{C}x^i y^j$ with $i - j = \lambda$.

Observation: let $v \in V_\lambda$ and consider ev . We have

$$hev = [h, e]v + ehv = 2ev + \lambda ev = (\lambda + 2)ev.$$

That is, if $v \in V_\lambda$ then $ev \in V_{\lambda+2}$ and similarly $fv \in V_{\lambda-2}$. (Clearly iff assuming v is a weight vector)

We will think of this pictorially as a chain

$$\dots \leftrightarrow V_{\lambda-2} \xleftrightarrow{e,f} V_{\lambda} \xleftrightarrow{e,f} V_{\lambda+2} \xleftrightarrow{e,f} \dots$$

For example, in $L(n)$ the chain went from $\lambda = n$ to $\lambda = -n$:

$$V_{-n} = \mathbb{C}y^n \leftrightarrow V_{-n+2} = \mathbb{C}xy^{n-1} \leftrightarrow \dots \leftrightarrow V_n = \mathbb{C}x^n$$

(Note that the string for $L(n)$ consists of $n+1$ elements)

Definition 2.2. If $hv = \lambda v$ and $ev = 0$, i.e. $0 \neq v \in V_{\lambda} \cap \ker(e)$, then we say that v is a **highest-weight vector** of weight λ .

Lemma 2.1. Let V be a representation of $\mathfrak{sl}_2$ and $v \in V$ a highest-weight vector of weight λ . Then $W = \langle v, fv, f^2v, \dots \rangle$ is a subrepresentation.

Proof. NTS $hW \subset W, fW \subset W$ and $eW \subset W$. $fW \subset W$ is by definition. Also, since $v \in V_{\lambda}$, $f^k v \in V_{\lambda-2k}$ and so $hW \subset W$.

Now observe that since v is hw-vector $ev = 0$, $efv = [e, f]v + fev = hv = \lambda v$, $ef^2v = hfv + \lambda fv = (2\lambda - 2)fv \dots$. By induction, one can show that $ef^n v = n(\lambda - n + 1)f^{n-1}v \in W$. So $eW \subset W$ $\square$

Lemma 2.2. Let V be a f.d. representation of $\mathfrak{sl}_2$ and v a hw-vector of weight λ . Then $\lambda \in \mathbb{Z}_{\geq 0}$

Proof. Note all $f^n v$ lie in the eigenspace of h and are linearly independent since they have distinct eigenvalue. Since $\dim V < \infty$, this implies that there is k such that $f^k v \neq 0$ and $f^{k'} v = 0$ for $k' > k$. Then by the proof of lemma 2.1, $0 = ef^{k+1}v = (k+1)(\lambda - k)f^k v \implies \lambda = k$ a nonnegative integer. $\square$

Lemma 2.3. Every f.d. representation V of $\mathfrak{sl}_2$ has a hw-vector.

Proof. We can take an eigenvector v of h since we are over $\mathbb{C}$. Then $v, ev, e^2v \dots$ have distinct eigenvalues and hence linearly independent. Then there is n such that $e^{n-1}v \neq 0, e^n v = 0$ for some n , and therefore $e^{n-1}v$ is a hw-vector. $\square$

Corollary 2.1. If V is a irr. representation of $\mathfrak{sl}_2$ with $\dim V = n+1$, then V has a basis $v_0, v_1, \dots, v_n$ on which $\mathfrak{sl}_2$ acts as $hv_i = (n-2i)v_i; fv_i = v_{i+1}$ with $fv_n = 0$, and $ev_i = i(n-i+1)v_{i-1}$. In particular, there is a unique (up to isom.) $\dim-(n+1)$ irr. representation isomorphic to $L(n)$

Proof. By lemma 2.3, V has a hw-vector v . By lemma 2.1, $W = \langle v, fv, f^2v, \dots \rangle$ is a subrepresentation and so $V = W$. Since $\dim V = n+1$, the string $v, fv, f^2v \dots$ has length $n+1$ and so $f^n v \neq 0, f^{n+1}v = 0$. As shown in the proof of lemma 2.2, have v is of weight n . Now just take $v_0 = v$ and $v_i = f^i v$. It is isomorphic to $L(n)$ via $v_i \mapsto x^{n-i}y^i$ $\square$

We will now show that every finite-dimensional representation of $\mathfrak{sl}_2(\mathbb{C})$ is a direct sum of irreducible representations, or equivalently, the category of finite-dimensional representations of $\mathfrak{sl}_2(\mathbb{C})$ is semisimple, or equivalently, every representation of $\mathfrak{sl}_2(\mathbb{C})$ is completely reducible.

Morally, we've just shown that every such representation consists of "strings" descending from the highest weight, and we will show that these strings don't interact (and so direct sum). We will first show that they don't interact when they have different highest weights. The difficult part is to show this is true when they have the same highest weight.

We will also show that h acts diagonalisably on every finite-dimensional representation, while e and f are nilpotent. In fact, $\text{span}(h) = \left\{ \begin{pmatrix} a & \\ & -a \end{pmatrix} \right\} = \text{Lie} \left\{ \begin{pmatrix} t & \\ & t^{-1} \end{pmatrix} \right\} \subset \text{Lie}(SL_2)$ where $\left\{ \begin{pmatrix} t & \\ & t^{-1} \end{pmatrix} \right\}$

is the maximal torus of SL_2 . Like for $\mathbb{C}^\times$, the representations here should correspond to integers. In some hazy way, the fact that we are picking out the representations of a circle and not any other 1-dimensional Lie algebra is a sign of the Lie algebra remembering something about the group. (Saying what it is remembering and how is a hard question.) ...

Example 12. Recall that $\mathbb{C}[x, y]$ is a representation of $\mathfrak{sl}_2$ via $e = x\partial/\partial y, f = y\partial/\partial x, h = x\partial/\partial x - y\partial/\partial y$ and is a direct sum of the representations $L(n)$

Definition 2.3. Let V be a f.d. representation of $\mathfrak{sl}_2$. Define $\Omega = ef + fe + \frac{1}{2}h^2 \in \text{End}(V)$ to be the *Casimir* of $\mathfrak{sl}_2$

Lemma 2.4. Ω is in the central; that is, $e\Omega = \Omega e, f\Omega = \Omega f$ and $h\Omega = \Omega h$

Proof. In fact, $e\Omega = e^2f + efe + \frac{1}{2}eh^2 = e[e, f] + 2efe + \frac{1}{2}[e, h]h + \frac{1}{2}heh = eh + 2efe - eh + \frac{1}{2}heh = 2efe + \frac{1}{2}heh = [f, e]e + 2efe + \frac{1}{2}h[h, e] + \frac{1}{2}heh = \Omega e$. The rest is similar. $\square$

Also observe $\Omega = [e, f] + 2fe + \frac{1}{2}h^2 = (\frac{1}{2}h^2 + h) + 2fe$. This will be useful later.

Corollary 2.2. If V is an irr. representation of $\mathfrak{sl}_2$, the Ω acts by multiplying a scalar.

Proof. This is Schur's lemma. $\square$

Lemma 2.5. Let $L(n)$ be the irr. representation of $\mathfrak{sl}_2$ with hw n . Then Ω acts on $L(n)$ by $\frac{1}{2}n^2 + n$

Proof. Since Ω commutes with f , it suffices to check this on the hw-vector v . Since $ev = 0$ and $hv = nv$, have $\Omega v = (\frac{1}{2}h^2 + h + 2fe)v = (\frac{1}{2}n^2 + n)v$, as required. $\square$

Remark 2.1. For $L(n)$ and $L(m)$ irr. representations with different hws, Ω acts by different scalar since $\frac{1}{2}n^2 + n$ is increasing in n

Definition 2.4. Let V be a f.d. representation of $\mathfrak{sl}_2$. Define $V^\lambda = \{v \in V | (\Omega - \lambda)^{\dim V} v = 0\}$, the *generalized eigenspace of Ω with eigenvalue λ*

We claim that each V^λ is a subrepresentation of $\mathfrak{sl}_2$. Take $x \in \mathfrak{sl}_2$. Since Ω is central, have $(\Omega - \lambda)^{\dim V} xv = x(\Omega - \lambda)^{\dim V} v = 0$ for $v \in V^\lambda$. So $xv \in V^\lambda$

Now, if $V^\lambda \neq 0$, then V^λ is somehow "glued together" from copies of $L(n)$ for a fixed n and $\lambda = \frac{1}{2}n^2 + n$. This is not hard to be seen from lemma 2.5 and the remark after, but we need to make it precise.

Definition 2.5. Let W be f.d. $\mathfrak{g}$ -module. A **composition series** for W is a sequence of submodules $0 \subset W_0 \subset \dots \subset W_r = W$ such that the quotients W_i/W_{i-1} are irr. $\mathfrak{g}$ -modules.

Lemma 2.6. The composition series always exists

Proof. Take any irr. submodule W' of W . Then by induction on dimensions W/W' has a composition series. Its preimage of the quotient map with W' in the front is what we want. $\square$

So the precise statement is that V^λ **has a composition series with each quotient isomorphic to $L(n)$ for a fixed n** . Indeed, take any irr. subrepresentaion $L(n) \subset V^\lambda$. Then Ω acts on $L(n)$ by $\frac{1}{2}n^2 + n$ and so $\lambda = \frac{1}{2}n^2 + n$ by definition, or in other word n is uniquely determined by λ . Moreover, Ω acts on $V^\lambda/L(n)$ whose generalized eigenvalue is still λ . So we can repeat the argument and get the result.

Proposition 2.1. h acts on V^λ with generalized eigenvalues in the set $\{n, n-2, \dots, -n\}$

Proof. This is a general fact about composite series: let h acts on W and $W' \subset W$ invariant under h , i.e. $hW' \subset W'$. Have

$$\{\text{generalised eigenvalues of } h \text{ acting on } W\} =$$

$$\{\text{generalised eigenvalues of } h \text{ acting on } W'\} \cup \{\text{generalised eigenvalues of } h \text{ acting on } W/W'\}$$

(Compare this with the support of an R -module M $\text{Supp}(M) = \{p \in \text{Spec}(R) | M_p \neq 0\}$). Now since V^λ is composed of $L(n)$, the generalised eigenvalues of h is precisely that set. $\square$

Note also that on the kernel of $e : V^\lambda \rightarrow V^\lambda$ the only generalised eigenvalue of h is n ; that is, $(h - n)^{\dim V^\lambda} x = 0$ for $x \in V^\lambda \cap \ker(e)$. This follows by applying the above observation to the composition series intersected with $\ker(e)z$.

Using induction, we can show the following:

Lemma 2.7. 1. $hf^k = f^kh - 2kf^k$

$$2. e f^{n+1} = f^{n+1}e + (n+1)f^n(h-n)$$

Proposition 2.2. h acts by multiplication by n on $\ker(e : V^\lambda \rightarrow V^\lambda)$. That is, $\ker(e) = (V^\lambda)_n = \{x \in V^\lambda | hx = nx\}$

Proof. For $v \in (V^\lambda)_n$, $ev \in (V^\lambda)_{n+2} = 0$ by proposition 2.1 and so $v \in \ker(e)$.

Conversely, the observation above shows that $\ker(e)$ is in the generalized eigenspace of h of eigenvalue n . So for $v \in \ker(e)$, $(h - n)^{\dim V^\lambda} v = 0$. By lemma 2.7, have

$$(h - n + 2k)^{\dim V^\lambda} f^k v = f^k (h - n)^{\dim V^\lambda} v = 0$$

So $f^k v$ is in the generalized eigenspace of h with eigenvalue $n - 2k$. In particular, have $f^{n+1} v = 0$ by proposition 2.1. By lemma 2.7 (2), we get $(n+1)f^n(h-n)v = 0$ since $v \in \ker(e)$. We claim that for $x \in \ker(e)$, $x \neq 0 \implies f^n x \neq 0$

Take $0 = W_1 \subset \dots \subset W_e = V^\lambda$ a composition series of V^λ with quotients all isomorphic to $L(n)$. Take i such that $x \in W_{i+1}$ but $x \notin W_i$. Then $0 \neq \bar{x} \in W_{i+1}/W_i \cong L(n)$. $e\bar{x} = 0 \implies \bar{x}$ is a hw-vector of $L(n)$, and so $f^n \bar{x} \neq 0 \in L(n)$. Hence $f^n x \neq 0$.

Note $(h - n)v \in \ker(e)$ since $v \in \ker(e)$. By claim, have $(h - n)v = 0$ as required. $\square$

We can now finish the proof of complete reducibility. Choose a basis $w_1, \dots, w_k$ of $\ker(e : V^\lambda \rightarrow V^\lambda)$. Consider the strings $\langle w_i, \dots, f^n w_i \rangle$ generated by w_i . We have seen that they are subrepresentations and the generators are linearly independent. We claim that they have trivial intersections and span V^λ (and so V^λ is a direct sum of them).

This is just linear algebra. Say w_1 is in the string generated by w_2 . Then have $w_1 = \sum_{i=1}^n a_i f^i w_2$. By proposition 2.2, w_j are eigenvectors of h with eigenvalue n and so $f^i w_j$ is eigenvector of h with eigenvalue $n - 2i$. Therefore we must have $a_1 = 1$ and $a_i = 0$ for $i \geq 2$. So $w_1 = w_2$, contradiction. Now take $0 \neq w \in V^\lambda$. Since V^λ is composed of $L(n)$ and so e acts nilpotently, there is $k \geq 0$ such that $e^k w \neq 0$ and $e^{k+1} w = 0$, i.e. $e^k w \in \ker(e)$. (?)

Therefore, h acts semisimply and diagonalisably on V^λ , and therefore on all of V .

Remark 2.2. This is false in characteristic p , i.e. there is a representation of $\mathfrak{sl}_2(\overline{\mathbb{F}_p})$ that does not decompose into a direct sum of irreducibles. For example, consider $L(p)$. It has two irr. subrepresentations $\langle x^p \rangle$ and $\langle y^p \rangle$ of $\mathfrak{sl}_2(\overline{\mathbb{F}_p})$ with trivial actions. But $L(p)$ is not complete reducible because they are the only irr. subrepresentations of $L(p)$

3 Consequences

Now let V, W two representations of a Lie algebra $\mathfrak{g}$. Then $\mathfrak{g} \rightarrow \text{End}(V \otimes W) \cong \text{End}(V) \otimes \text{End}(W)$ via $x \mapsto x \otimes 1 + 1 \otimes x$ is a homomorphism of Lie algebra. (This can be seen by differentiating the diagonal map $G \rightarrow G \times G; g \mapsto (g, g)$). So $V \otimes W$ is also a representation of $\mathfrak{g}$

Remark 3.1. In general, if A is an algebra and V, W are two representations, then $V \otimes W$ is naturally an $A \otimes A$ -representation. To make it an A -representation, need an algebra homomorphism $A \rightarrow A \otimes A$. An object with such a structure (comultiplication, and other properties) is called Hopf algebra. For example, group scheme $G = \mathbb{G}_a$ essentially has comultiplication as above. In some essential way $\mathfrak{g}$ is a Hopf algebra. (Or, rather, a deformation of the universal enveloping algebra of $\mathfrak{g}$ is a Hopf algebra.)

Now take $\mathfrak{g} = \mathfrak{sl}_2$. Since $L(n) \otimes L(m)$ is a $\mathfrak{sl}_2$ -representation, theorem 2.1 gives the decomposition $L(n) \otimes L(m) = \oplus_k a_k L(k)$. We want to determine a_k . One method of doing this would be to compute all the highest-weight vectors.

For example, consider $L(1) \otimes L(n)$ and we compute all its hw-vectors. One is clearly $x \otimes x^n$ of weight $n + 1$, and so it generates a string isomorphic to $L(n + 1)$. The other is $x \otimes x^{n-1}y - y \otimes x^n$ of weight $n - 1$. So $L(1) \otimes L(n) = L(n + 1) \oplus L(n - 1)$ by comparing dimensions. In general we can look for the patterns and write down all the hw-vectors.

For $L(n) \otimes L(m)$, let v_n, v_m denote their hw-vectors respectively. The $v_n \otimes v_m$ is a hw-vector of $L(n) \otimes L(m)$ of weight $n + m$. So a direct summand of $L(n) \otimes L(m)$ is $L(m + n)$. We want to determine the rest.

Definition 3.1. Let V be a f.d. $\mathfrak{sl}_2$ -representation. The **character** of V is $chV = \sum_{n \in \mathbb{Z}} \dim V_n z^n \in \mathbb{N}[z^{\pm 1}]$

Proposition 3.1 (Properties of characters). 1. $chV(1) = \dim V$

$$2. \ chL(n) = z^n + z^{n-2} + \dots + z^{-n} = \frac{z^{n+1} + z^{-n-1}}{z + z^{-1}}$$

$$3. \ chV = chW \text{ iff } V \cong W$$

$$4. \ ch(V \otimes W) = chV \cdot chW$$

5. (The Clebsch-Gordan rule) As a result, we deduce that

$$L(n) \otimes L(m) = \oplus_{k=|n-m|, k \equiv n-m \pmod{2}}^{n+m} L(k)$$

Proof. (1) is clear since h acts diagonally on V .

For (3), note that $chL(n)$ for $n \in \mathbb{N}$ form a $\mathbb{Z}$ -basis of $\mathbb{Z}[z^{\pm 1}]$. Given V, W two $\mathfrak{sl}_2$ -representations, complete reducibility gives

$$V = \oplus_{n \geq 0} a_n L(n)$$

$$W = \oplus_{n \geq 0} b_n L(n)$$

But since $chL(n)$ form a basis, $chV = \sum_{n \geq 0} a_n chL(n)$ determines a_n

For (4), note that $(V \otimes W)_p = \sum_{n+m=p} V_n \otimes W_m$. This is exactly how we multiply polynomials. $\square$

We have a picture for (5) so that we don't have to actually remember it:

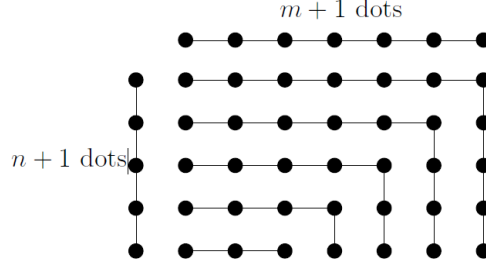


Figure 1: Crystals for $\mathfrak{sl}_2$

The individual strings are $L(k)$ in the decomposition.

One of the points of this course is that this sort of picture-drawing (called *crystals*) lets us decompose tensor products of representations of all other semisimple Lie algebras. (This fact was discovered about 20 years ago by Kashiwara and Lusztig.) (drama drama)

What we will actually show for the other semisimple Lie algebras:

- The category of finite-dimensional representations is completely reducible (i.e. all finite-dimensional representations are semisimple)
- We will parameterise the irreducible representations
- We will compute the character of the irreducible representations
- We will show how to take tensor products of irreducible representations by drawing pictures (crystals)

4 Structure and Classification of Simple Lie Algebras

Definition 4.1. A Lie algebra $\mathfrak{g}$ is simple if it has no proper ideals and $\dim \mathfrak{g} > 1$. $\mathfrak{g}$ is semisimple if it is a direct sum of simple Lie algebras.

Remark 4.1. For abelian Lie algebra with $\dim > 1$, any subspace is an ideal and so it is not simple (I is an ideal of Lie algebra $\mathfrak{g}$ if $[\mathfrak{g}, I] \subset I$). It is sensible to exclude abelian Lie algebras since $\mathbb{C}$ behaves quite differently from $\mathfrak{sl}_2$

Definition 4.2. The derived algebra of $\mathfrak{g}$ is $[\mathfrak{g}, \mathfrak{g}] = \text{Span}[x, y]$ for $x, y \in \mathfrak{g}$

By definition, $[\mathfrak{g}, \mathfrak{g}]$ is an ideal and $\mathfrak{g}/[\mathfrak{g}, \mathfrak{g}]$ is abelian.

Definition 4.3. The central series for $\mathfrak{g}$ is given by $\mathfrak{g}^0 = \mathfrak{g}$, $\mathfrak{g}^n = [\mathfrak{g}^{n-1}, \mathfrak{g}]$, i.e.

$$\mathfrak{g} \supset [\mathfrak{g}, \mathfrak{g}] \supset [[\mathfrak{g}, \mathfrak{g}], \mathfrak{g}] \dots$$

The derived series of $\mathfrak{g}$ is given by $\mathfrak{g}^{(0)} = \mathfrak{g}$, $\mathfrak{g}^{(n)} = [\mathfrak{g}^{(n-1)}, \mathfrak{g}^{(n-1)}]$, i.e.

$$\mathfrak{g} \supset [\mathfrak{g}, \mathfrak{g}] \supset [[\mathfrak{g}, \mathfrak{g}], [\mathfrak{g}, \mathfrak{g}]] \dots$$

Clearly we have $\mathfrak{g}^{(n)} \subset \mathfrak{g}^n$ by induction.

Definition 4.4. $\mathfrak{g}$ is nilpotent if $\mathfrak{g}^n = 0$ for some n , i.e. the central series for $\mathfrak{g}$ terminates. $\mathfrak{g}$ is solvable if $\mathfrak{g}^{(n)} = 0$ for some n , i.e. the derived series for $\mathfrak{g}$ terminates.

Therefore, a nilpotent Lie algebra is always solvable.

Example 13. $\mathfrak{n} = \left\{ \begin{pmatrix} 0 & * & \cdots & * \\ & 0 & \cdots & * \\ & & 0 & \cdots \\ & & & 0 \end{pmatrix} \right\}$ the strictly upper triangular matrices is a nilpotent Lie algebra as easily seen.

Example 14. $\mathfrak{b} = \left\{ \begin{pmatrix} * & * & \cdots & * \\ & * & \cdots & * \\ & & * & \cdots \\ & & & * \end{pmatrix} \right\}$ the upper triangular matrices is a solvable but not nilpotent Lie algebra. In fact, we have $\mathfrak{b}^1 = [\mathfrak{b}, \mathfrak{b}] = \mathfrak{n}$ and $[\mathfrak{n}, \mathfrak{b}] = \mathfrak{n}$

Example 15. Let W be a symplectic vector space with inner product $\langle, \rangle$ (recall that a symplectic vector space is equipped with a nondegenerate bilinear antisymmetric form, and there is essentially one example: given a finite-dimensional vector space L let $W = L + L^*$ with the inner product $\langle L, L \rangle = 0, \langle L^*, L^* \rangle = 0$ and $\langle v, w^* \rangle = -\langle w^*, v \rangle = w^*(v)$)

Define the *Heisenberg Lie algebra* as

$$\mathfrak{H}_W = W \oplus \mathbb{C}c \quad \text{with} \quad [w, w'] = \langle w, w' \rangle c, \quad [c, \mathfrak{H}_W] = 0$$

One easily verifies that this is a nilpotent Lie algebra. In fact, this is the most important nilpotent Lie algebra occurring in nature. For example, let $L = \mathbb{C}$, then $\mathfrak{H}_W = \mathbb{C}p + \mathbb{C}q + \mathbb{C}c$ with $[p, q] = c, [p, c] = [q, c] = 0$ (as we know from classifying the 3-dimensional Lie algebras!). This has a representation on $\mathbb{C}[x]$ via $q \mapsto \text{multiplication by } x, p \mapsto \partial/\partial x, c \mapsto 1$

(For a general vector space L , the basis vectors $v_1, \dots, v_n$ map to multiplication by $x_1, \dots, x_n$ and their duals map to $\partial/\partial x_1, \dots, \partial/\partial x_n$. This is the same as the canonical representation of Weyl algebra.)

Proposition 4.1. 1. *Subalgebras and quotients of solvable Lie algebras are solvable. Subalgebras and quotients of nilpotent Lie algebras are nilpotent.*

2. *Let $\mathfrak{g}$ be a Lie algebra and $\mathfrak{h}$ an ideal. Then $\mathfrak{g}$ is solvable iff $\mathfrak{h}$ and $\mathfrak{g}/\mathfrak{h}$ are solvable. (That is, solvable Lie algebras are built out of abelian Lie algebras, i.e. there is a refinement of the derived series such that the subquotients are 1-dimensional and therefore abelian.)*

3. *$\mathfrak{g}$ is nilpotent iff the center of $\mathfrak{g}$ is nonzero and $\mathfrak{g}/(\text{the center of } \mathfrak{g})$ is nilpotent*

Indeed, if $\mathfrak{g}$ is nilpotent, take the least n to make $\mathfrak{g}^n = 0$. The $0 \neq \mathfrak{g}^{n-1}$ is in the center since $\mathfrak{g}^n = [\mathfrak{g}^{n-1}, \mathfrak{g}] = 0$

4. *$\mathfrak{g}$ is nilpotent iff $\text{ad } \mathfrak{g} \subset \mathfrak{gl}(\mathfrak{g})$ is nilpotent, as we have a natural s.e.s*

$$0 \rightarrow \text{center of } \mathfrak{g} \rightarrow \mathfrak{g} \rightarrow \text{ad } \mathfrak{g} \rightarrow 0$$

The proofs are straightforward and so omitted.

We now state a nontrivial result, which we will not prove.

Theorem 4.1. (Lie's theorem) *Let $\mathfrak{g} \subset \mathfrak{gl}(V)$ be a solvable Lie algebra, and let the base field k be algebraically closed and have characteristic zero. Then there exists a basis $v_1, \dots, v_n$ of V with respect to which $\mathfrak{g} \subset \mathfrak{b}(V)$, i.e. the matrices of all elements of $\mathfrak{g}$ are upper triangular.*

Equivalently, there is a linear function $\lambda : \mathfrak{g} \rightarrow k$ and an element $v \in V$ such that $xv = \lambda(x)v$ for all $v \in \mathfrak{g}$, i.e. v is a common eigenvector for all of $\mathfrak{g}$, i.e. a $\dim - 1$ subrepresentation.

In particular, the only irreducible finite-dimensional representations of $\mathfrak{g}$ are one-dimensional.

Remark 4.2. It is essential to assume k to be algebraically closed with characteristic 0 in Lie's theorem. For example, assume k has characteristic p . Let $\mathfrak{g} = \mathfrak{H}_W$, the dim-3 Heisenberg algebra in example 15. Then $k[x]/x^p$ is a f.d. irreducible representation in the canonical way with dimension markedly bigger than 1

Corollary 4.1. *In characteristic 0, if $\mathfrak{g}$ is a solvable f.d. Lie algebra then $[\mathfrak{g}, \mathfrak{g}]$ is nilpotent.*

Proof. Apply Lie's theorem to the adjoint representation $\mathfrak{g} \rightarrow \text{End} \mathfrak{g}$. With respect to some basis, $\text{ad} \mathfrak{g} \subset \mathfrak{b}(\mathfrak{g})$. Since $[\mathfrak{b}, \mathfrak{b}] \subset \mathfrak{n}$, we see that $[\text{ad} \mathfrak{g}, \text{ad} \mathfrak{g}]$ is nilpotent. Since ad is a representation, have $[\text{ad} \mathfrak{g}, \text{ad} \mathfrak{g}] = \text{ad}[\mathfrak{g}, \mathfrak{g}]$, and so $[\mathfrak{g}, \mathfrak{g}]$ is nilpotent by proposition 4.1 (4). $\square$

Theorem 4.2. (Engel's Theorem). *Let the base field k be arbitrary. $\mathfrak{g}$ is nilpotent iff $\text{ad} \mathfrak{g}$ consists of nilpotent endomorphisms of $\mathfrak{g}$, i.e. ad_x is nilpotent for all $x \in \mathfrak{g}$*

Equivalently, if V is a finite-dimensional representation of $\mathfrak{g}$ such that all elements of $\mathfrak{g}$ act on V by nilpotent endomorphisms, then there exists $v \in V$ such that $xv = 0$ for all $x \in \mathfrak{g}$, i.e. V has trivial dim-1 subrepresentation.

Equivalently, we can write x as a strictly upper triangular matrix.

Definition 4.5. *A symmetric bilinear form $(,) : \mathfrak{g} \times \mathfrak{g} \rightarrow k$ is **invariant** if $([x, y], z) = (x, [y, z])$*

Remark 4.3. If $\mathfrak{a} \subset \mathfrak{g}$ is an ideal and $(,)$ is an invariant bilinear form, then $\mathfrak{a}^\perp$ is an ideal. In fact, for $\in \mathfrak{g}$, has $(\mathfrak{a}, [x, \mathfrak{a}^\perp]) = ([\mathfrak{a}, x], \mathfrak{a}^\perp) = 0$ since $[\mathfrak{a}, x] \subset \mathfrak{a}$, and so $[x, \mathfrak{a}^\perp] \subset \mathfrak{a}^\perp$

Definition 4.6. *If V is a f.d. representation of $\mathfrak{g}$, i.e. if $\mathfrak{g} \rightarrow \mathfrak{gl}(V)$ is a homomorphism, define the **trace form** to be $(x, y)_V = \text{trace}(\rho(x)\rho(y))$*

It is clear that the trace form is symmetric, bilinear and invariant.

Example 16. $(,)_{ad}$ is the "Killing form" (named after a person, not after an action). This is the trace form attached to the adjoint representation. That is, $(x, y)_{ad} = \text{trace}(\text{ad}_x \text{ad}_y)$

Theorem 4.3. (Cartan's criteria) *Let $\mathfrak{g} \subset \mathfrak{gl}(V)$ and let $\text{char} k = 0$. Then $\mathfrak{g}$ is solvable if and only if for every $x \in \mathfrak{g}$ and $y \in [\mathfrak{g}, \mathfrak{g}]$ the trace form $(x, y)_V = 0$. (That is, $[\mathfrak{g}, \mathfrak{g}] \subset \mathfrak{g}^\perp$)*

Proof. If $\mathfrak{g}$ is solvable, by Lie's theorem we can take a basis of V in which x is an upper triangular matrix, y is strictly upper triangular, and then xy and yx both have zeros on the diagonal and thus have trace 0 as required. (That is, if $\mathfrak{g}$ is solvable and nonabelian, all trace forms are degenerate.) The other direction is omitted. $\square$

Corollary 4.2. *$\mathfrak{g}$ is solvable iff $(\mathfrak{g}, [\mathfrak{g}, \mathfrak{g}])_{ad} = 0$*

Proof. If $\mathfrak{g}$ is solvable, apply Lie's theorem to the adjoint representation to get the trace to be 0. Conversely, Cartan's criteria gives $\text{ad} \mathfrak{g} = \mathfrak{g}/(\text{center of } \mathfrak{g})$ is solvable, and so $\mathfrak{g}$ is solvable. $\square$

Example 17. (Not every invariant form is a trace form) Let $\tilde{\mathfrak{H}} = \mathbb{C}\langle c, p, q, d \rangle$ with $[c, \tilde{\mathfrak{H}}] = 0$, $[p, q] = c$, $[d, p] = p$, $[d, q] = -q$. We define a non-degenerate invariant bilinear form on $\tilde{\mathfrak{H}}$. Note $(c, c) = (c, [p, q]) = ([c, p], q) = 0$, $0 = ([p, p], q) = (p, c) = (q, c)$, $(d, p) = (d, [d, p]) = ([d, d], p) = 0 = (d, q)$ and $0 = ([p, p], d) = -(p, p) = (q, q)$ since $(,)$ is invariant. Hence let $(c, d) = ([p, q], d) = (p, q) = 1$. It is clear that $(,)$ is a required bilinear form.

Also we have $\tilde{\mathfrak{H}}^{(1)}[\tilde{\mathfrak{H}}, \tilde{\mathfrak{H}}] = \mathbb{C}\langle c, p, q \rangle$ and $\tilde{\mathfrak{H}}^{(2)} = \mathbb{C}c$ and so $\tilde{\mathfrak{H}}^{(3)} = 0$. Hence $\tilde{\mathfrak{H}}$ is solvable.

Let ρ be the canonical representation of $\mathfrak{H}$ on $k[x]$ in example 15. Let $\rho(d) = x \cdot \partial/\partial x$. This extends ρ to a representation $\tilde{\rho}$ of $\tilde{\mathfrak{H}}$ on $k[x]$

Definition 4.7. The **radical** of $\mathfrak{g}$, $R(\mathfrak{g})$, is the maximal solvable ideal in $\mathfrak{g}$.

Proposition 4.2. 1. $R(\mathfrak{g})$ is the sum of all solvable ideals in $\mathfrak{g}$.

2. $R(\mathfrak{g}/R(\mathfrak{g})) = 0$

Proof. For (1), it suffices to show that the sum of two solvable ideals is solvable. Let I, J be two solvable ideals. We have a canonical isomorphism

$$\frac{I+J}{I} \cong \frac{J}{I \cap J}$$

Now J solvable implies the quotient $\frac{J}{I \cap J}$ solvable. Since I is also solvable, proposition 4.1 gives that $I+J$ is solvable. (2) is trivial. $\square$

Theorem 4.4. In characteristic 0, TFAE:

1. $\mathfrak{g}$ is semisimple
2. $R(\mathfrak{g}) = 0$
3. The Killing form $(\cdot, \cdot)_{ad}$ is non-degenerate. (*the Killing criteria*)

We will prove theorem 4.4 with some efforts.

Definition 4.8. A **derivation** $D : \mathfrak{g} \rightarrow \mathfrak{g}$ is a linear map satisfying $D[x, y] = [Dx, y] + [x, Dy]$

Example 18. ad_x is a derivation for any x by the Jacobian identity. Derivations of the form ad_x for some $x \in \mathfrak{g}$ are called **inner**. Later we will see that all derivations on semisimple Lie algebras are inner.

Remark 4.4. For any Lie algebra $\mathfrak{g}$, have a s.e.s

$$0 \rightarrow R(\mathfrak{g}) \rightarrow \mathfrak{g} \rightarrow \mathfrak{g}/R(\mathfrak{g}) \rightarrow 0$$

where $R(\mathfrak{g})$ is a solvable ideal and $\mathfrak{g}/R(\mathfrak{g})$ is semisimple by theorem 4.4. That is, any Lie algebra has a maximal semisimple quotient, and the kernel is a (maximal) solvable ideal. This shows you how much nicer the theory of Lie algebras is than the corresponding theory for finite groups. In particular, the corresponding statement for finite groups is essentially equivalent to the classification of the finite simple groups.

A theorem that we will neither prove nor use, but which is pretty:

Theorem 4.5. (*Levi's theorem, or Levi decomposition*) In characteristic 0, a stronger result is true. Namely, the exact sequence splits, i.e. there exists a subalgebra $\mathfrak{a} \subset \mathfrak{g}$ such that $\mathfrak{a} \cong \mathfrak{g}/R(\mathfrak{g})$ (This subalgebra is not canonical; in particular, it is not an ideal of $\mathfrak{g}$). That is, $\mathfrak{g}$ can be written as a semidirect product $\mathfrak{g} = R(\mathfrak{g}) \rtimes \mathfrak{a}$

Remark 4.5. This is not true in characteristic p . Let $\mathfrak{g} = \mathfrak{sl}_p(\overline{\mathbb{F}}_p)$. Then $R(\mathfrak{g}) = \overline{\mathbb{F}}_p I$ (note that the identity matrix in dimension p has trace 0), but that there is no complement to $R(\mathfrak{g})$ that is an algebra.

Proof of theorem 4.4. First, note that $R(\mathfrak{g}) = 0$ iff there is no nonzero abelian ideals of $\mathfrak{g}$. The forward direction holds since abelian ideals are solvable. Conversely, for a solvable ideal, the last nonzero term of its derived series is an abelian ideal.

From now on, we use $(\cdot, \cdot)$ to refer to the Killing form $(\cdot, \cdot)_{ad}$

(3) $\implies$ (2): Since $(,)$ is nondegenerate, $\mathfrak{g}^\perp = 0$. Hence by the above observation, it suffices to show that for any abelian ideal $\mathfrak{a}$, $\mathfrak{a} \subset \mathfrak{g}^\perp$.

Write $\mathfrak{g} = \mathfrak{a} + \mathfrak{h}$ where $\mathfrak{h}$ is a vector space complement to $\mathfrak{a}$ (not necessarily an ideal). For $a \in \mathfrak{a}$, ad_a has a block matrix $\begin{pmatrix} 0 & * \\ 0 & 0 \end{pmatrix}$ since $\mathfrak{a}$ is abelian and so ad_a acts by 0 on $\mathfrak{a}$. Similarly, for $x \in \mathfrak{g}$, ad_x has a block matrix $\begin{pmatrix} * & * \\ 0 & * \end{pmatrix}$. Therefore, $(x, a) = \text{trace}(ad_x ad_a) = 0$ for all x and so $a \in \mathfrak{g}^\perp$.

(2) $\implies$ (3): Suppose $\mathfrak{r} \subset \mathfrak{g}^\perp$ (e.g. take $\mathfrak{r} = \mathfrak{g}^\perp$) is a nonzero ideal. Then $ad\mathfrak{r} \subset \mathfrak{gl}(\mathfrak{g})$ and $(x, y)_{ad} = 0$ for $x, y \in \mathfrak{r}$. Hence $ad\mathfrak{r} = \mathfrak{r}/(\text{center of } \mathfrak{r})$ is solvable, and so $\mathfrak{r}$ is a nonzero solvable ideal. Hence $R(\mathfrak{g}) \neq 0$.

For the rest, we need another claim:

Claim: $R(\mathfrak{g}) \supset \mathfrak{g}^\perp \supset [R(\mathfrak{g}), R(\mathfrak{g})]$

Proof of claim. : Note that $\mathfrak{g}^\perp$ is an ideal. Indeed, for $x \in \mathfrak{g}^\perp$ and $y, z \in \mathfrak{g}$, $([x, y], z) = (x, [y, z]) = 0 \implies [x, y] \in \mathfrak{g}^\perp$. To show $\mathfrak{g}^\perp \subset R(\mathfrak{g})$, it is enough to show that $\mathfrak{g}^\perp$ is solvable. But this is clear by Cartan's criteria.

On the other hand, since $R(\mathfrak{g})$ is solvable, the other direction of Cartan's criteria gives that $(x, y)|_{R(\mathfrak{g})} = \text{trace}(ad(x)ad(y)|_{R(\mathfrak{g})}) = 0$ for $x \in [R(\mathfrak{g}), R(\mathfrak{g})]$ and $y \in \mathfrak{g}$. Since $R(\mathfrak{g})$ is an ideal, we see that $ad_x ad_y$ preserves $R(\mathfrak{g})$ and acts trivially on $\mathfrak{g}/R(\mathfrak{g})$ (i.e. $ad_x ad_y : \mathfrak{g} \rightarrow R(\mathfrak{g})$). Hence, by extending a basis of $R(\mathfrak{g})$ to a basis of $\mathfrak{g}$, $(x, y) = \text{tr}(ad_x ad_y) = \text{tr}(ad_x ad_y|_{R(\mathfrak{g})}) = (x, y)|_{R(\mathfrak{g})} = 0$, and so $x \in \mathfrak{g}^\perp$. $\square$

(2)(3) $\implies$ (1): Let $(,)$ be non-degenerate and take a minimal nonzero ideal $\mathfrak{a} \subset \mathfrak{g}$. We claim that $(,)|_{\mathfrak{a}}$ is either 0 or non-degenerate. Indeed, the kernel of $(,)|_{\mathfrak{a}}$ is $\{x \in \mathfrak{a} | (x, \mathfrak{a}) = 0\} = \mathfrak{a} \cap \mathfrak{a}^\perp$ is an ideal, and so is 0 or $\mathfrak{a}$ by minimality of $\mathfrak{a}$. Since $R(\mathfrak{g}) = 0$, $\mathfrak{a}$ is not solvable and $(,)|_{\mathfrak{a}}$ is not identically 0 by Cartan's criteria. So $(,)|_{\mathfrak{a}}$ is non-degenerate.

Therefore, $\mathfrak{g} = \mathfrak{a} \oplus \mathfrak{a}^\perp$ where $\mathfrak{a}$ is a minimal ideal, i.e. simple. Note the ideals of $\mathfrak{a}^\perp$ are ideals of $\mathfrak{g}$, so $R(\mathfrak{g}) = 0 \implies R(\mathfrak{a}^\perp) = 0$. By applying induction on $\mathfrak{a}^\perp$, we get $\mathfrak{g} = \bigoplus_{i=1}^n \mathfrak{a}_i$ where $\mathfrak{a}_i$ are minimal ideals of $\mathfrak{g}$, i.e. simple Lie algebras.

(1) $\implies$ (2): Write $\mathfrak{g} = \bigoplus_i \mathfrak{a}_i$ where $\mathfrak{a}_i$ are simple Lie algebras. If $R(\mathfrak{g}) \neq 0$, take a minimal solvable ideal $\mathfrak{a}$. Note that the direct sum $\mathfrak{g} = \bigoplus_i \mathfrak{a}_i$ is unique up to reordering by considering the projections onto each component, i.e. the minimal ideal $\mathfrak{a} = \mathfrak{a}_i$ for some i . But $\mathfrak{a}$ minimal solvable ideal implies that $[\mathfrak{a}, \mathfrak{a}] = 0$ and so $\mathfrak{a}$ is abelian, contradicting that $\mathfrak{a}$ is a simple Lie algebra. $\square$

Finally, we show that all derivations on semisimple Lie algebras are inner. Let $\mathfrak{g}$ be a semisimple Lie algebra, and let $D : \mathfrak{g} \rightarrow \mathfrak{g}$ be a derivation. Consider the function $l : \mathfrak{g} \rightarrow k$ via $x \mapsto \text{tr}(D \cdot ad_x)$. Since $\mathfrak{g}$ is semisimple, the Killing form is a nondegenerate inner product giving an isomorphism with the dual, so there is $y \in \mathfrak{g}$ such that $l(x) = (y, x)_{ad}$ for all x . Our task is to show that $D = ad_y$, or $E = D - ad_y = 0$.

This is equivalent to showing that $Ex = 0$ for all $x \in \mathfrak{g}$, or equivalently that $(Ex, z)_{ad} = 0$ for all $z \in \mathfrak{g}$. Note that

$$ad_{Ex} = E \cdot ad_x - ad_x \cdot E = [E, ad_x] : \mathfrak{g} \rightarrow \mathfrak{g}$$

since $ad_{Ex}(z) = [Ex, z] = E[x, z] - [x, Ez]$. Therefore

$$(Ex, z)_{ad} = \text{tr}(ad_{Ex} ad_z) = \text{tr}([E, ad_x], ad_z) = \text{tr}(E[ad_x, ad_z])$$

by the property of trace. But by definition of E , $\text{tr}(E \cdot ad_a) = \text{tr}(D \cdot ad_a) - \text{tr}(ad_y \cdot ad_a) = 0$

Remark 4.6. A nilpotent Lie algebra always has non-inner derivations. This was first shown by Jacobson in 1955 in "A note on automorphisms and derivations of Lie algebra"

Example 19. Let $\mathfrak{g} = \langle a, b \rangle$ with $[a, b] = b$. It only has inner derivations. Thus, this condition doesn't characterise semisimplicity.

Remark 4.7. For a simple Lie algebra, any two non-degenerate invariant forms $(,)_1$ and $(,)_2$ are equivalent, and in particular is equivalent to Killing form. To see this, non-degeneracy implies that there is a linear map $T : \mathfrak{g} \rightarrow \mathfrak{g}$ such that $(x, -)_1 = (T(x), -)_2 : \mathfrak{g} \rightarrow k$ for all $x \in \mathfrak{g}$. One can show that T commutes with the adjoint action (the inner derivations). Since $\mathfrak{g}$ is simple, the adjoint representation is irreducible. Then Schur's lemma gives that T is multiplication by a scalar, which implies that $(,)_1, (,)_2$ differ by multiplying that scalar.

Example 20. Later we will see that $\mathfrak{sl}_n(\mathbb{C})$ is simple. We have two nondegenerate bilinear forms on $\mathfrak{sl}_n(\mathbb{C})$: the Killing form $(,)_ad$ and $(A, B) = \text{trace}(AB)$ from the standard representation. Then we can find that $(,)_ad = 2n(,)$ by computing $(H, H)_ad$ where $H = \text{diag}(1, -1, 0, \dots, 0)$

5 Structure Theory

Definition 5.1. A **torus** $\mathfrak{t} \subset \mathfrak{g}$ is an abelian subalgebra s.t. for all $t \in \mathfrak{t}$ the adjoint $\text{ad}_t : \mathfrak{g} \rightarrow \mathfrak{g}$ is a diagonalisable linear map. A **maximal torus** is a torus that is not contained in any bigger torus.

Remark 5.1. Recall a toric variety is a (complex) variety with a dense open torus whose self-action extends to the whole variety. We have seen many examples of toric varieties as geometric quotients where the action of torus is very simple and "diagonal". I guess that is an analogy here.

Example 21. Let $T = (S^1)^r \subset G$ where G is a compact Lie group (or $T = (\mathbb{C}^\times)^r \subset G$ where G is a reductive algebraic group). Then $\mathfrak{t} = \text{Lie}(T) \subset \mathfrak{g} = \text{Lie}(G)$ is a torus, maximal if T is. (This comes from knowing that representations of $\mathfrak{t}$ are diagonalisable by an averaging argument, but we don't really know this.)

Example 22. For $\mathfrak{g} = \mathfrak{sl}_n$ or $\mathfrak{gl}_n$, $\mathfrak{t}$ the diagonal matrices (of trace 0 if inside $\mathfrak{sl}_n$) is a maximal torus.

For a vector space V , let $t_1, \dots, t_r : V \rightarrow V$ be pairwise commuting diagonalisable linear maps; let $\lambda = (\lambda_1, \dots, \lambda_r) \in \mathbb{C}^r$. Set $V_\lambda = \{v \in V | t_i v = \lambda_i v\}$ the simultaneous eigenspace.

Lemma 5.1. $V = \bigoplus_{\lambda \in (\mathbb{C}^\times)^r} V_\lambda$

Proof. Induct on r . For $r = 1$ follows from the requirement that t_1 be diagonalisable.

For $r > 1$, look at $t_1, \dots, t_{r-1}$ and decompose $V = \bigoplus_{\lambda_1, \dots, \lambda_{r-1}} V_{\lambda_1, \dots, \lambda_{r-1}}$.

Since t_r commutes with $t_1, \dots, t_{r-1}$, it preserves this decomposition. Now decompose each $V_{\lambda_1, \dots, \lambda_{r-1}}$ into eigenspaces for t_r □

Let $\mathfrak{t}$ be the r -dimensional abelian Lie algebra with basis $(t_1, \dots, t_r)$. The lemma asserts that V is a semisimple (i.e. completely reducible) representation of $\mathfrak{t}$. $V = \bigoplus V_\lambda$ is the decomposition into isotypic representations. Namely, letting $\mathbb{C}_\lambda$ be the 1-dimensional representation wherein $t_i(w) = \lambda_i w$, we have $\lambda \neq \mu \implies \mathbb{C}_\lambda \not\cong \mathbb{C}_\mu$, and V_λ is $\dim V_\lambda$ copies of $\mathbb{C}_\lambda$

Remark 5.2. Note that every irr. representation of $\mathfrak{t}$ is one-dimensional. Since $\mathfrak{t}$ is abelian, the images $\rho(t_i)$ commute and are diagonalizable, and so can be simultaneously diagonalized. Therefore, after taking a suitable basis,

$$V = \bigoplus_\chi V_\chi \text{ where } \chi : \mathfrak{t} \rightarrow \mathbb{C}^\times \text{ are characters of the torus and } V_\chi = \{v \in V | t(v) = \chi(t)v \ \forall t \in \mathfrak{t}\}$$

But then every $t \in \mathfrak{t}$ acts by multiplication by $\chi(t)$ on V_χ , and so V_χ is a subrep.. Then we have $V = V_\chi$. Also every subspace of V_χ is clearly preserved by $\mathfrak{t}$. This forces V to be one-dimensional.

Another way of saying this is as follows. λ is a linear map $\mathfrak{t} \rightarrow \mathbb{C}$, i.e. an element of the dual space $\lambda \in \mathfrak{t}^*$ (via letting $\lambda(t_i) = \lambda_i$). Therefore, one-dimensional representations of $\mathfrak{t}$ (which are all the irreducible representations of $\mathfrak{t}$) correspond to elements of $\mathfrak{t}^* = \text{Hom}_{\text{vs}}(\mathfrak{t}, \mathbb{C})$. The decomposition

$$V = \bigoplus_{\lambda \in \mathfrak{t}^*} V_\lambda, \text{ where } V_\lambda = \{v \in V \mid tv = \lambda(t)v \ \forall t \in \mathfrak{t}\}$$

is called **the weight space decomposition** of V .

Now, let $\mathfrak{g}$ be a Lie algebra and $\mathfrak{t}$ a maximal torus. The weight space decomposition of $\mathfrak{g}$ is

$$\mathfrak{g} = \mathfrak{g}_0 \oplus \bigoplus_{0 \neq \lambda \in \mathfrak{t}^*} \mathfrak{g}_\lambda$$

where $\mathfrak{g}_0 = \{x \in \mathfrak{g} \mid [t, x] = 0 \ \forall t \in \mathfrak{t}\}$ and $\mathfrak{g}_\lambda = \{x \in \mathfrak{g} \mid \text{ad}_t(x) = [t, x] = \lambda(t)x \ \forall t \in \mathfrak{t}\}$

Definition 5.2. $R = \{\lambda \in \mathfrak{t}^* \mid \mathfrak{g}_\lambda \neq 0, \lambda \neq 0\}$ is the set of **roots** of $\mathfrak{g}$.

We will now compute the root space decomposition for $\mathfrak{sl}_n$.

Example 23. Let $\mathfrak{g} = \mathfrak{sl}_n$. Then the maximal torus $\mathfrak{t}$ = diagonal matrices in $\mathfrak{sl}_n$.

Let $t = \begin{pmatrix} t_1 & & 0 \\ & \ddots & \\ 0 & & t_n \end{pmatrix}$ and E_{ij} the matrix with 1 in the (i, j) th position and 0 elsewhere. Then $[t, E_{ij}] = (t_i - t_j)E_{ij}$. Let $\epsilon_i \in \mathfrak{t}^*$ a dual basis with $\epsilon_i(t) = t_i$. Then $\epsilon_1 + \dots + \epsilon_n = 0$. (Note that $\mathfrak{t} \subset \mathbb{C}^n \implies \mathfrak{t}^*$ is a quotient of $(\mathbb{C}^n)^\times$)

Therefore, we can rewrite above as $[t, E_{ij}] = (\epsilon_i - \epsilon_j)(t)E_{ij}$. Hence the roots of $\mathfrak{sl}_n$ are $\{\epsilon_i - \epsilon_j \mid i \neq j\}$, and $\mathfrak{g}_0 = \mathfrak{t}$ the span of E_{ii} is the maximal torus. And the root space $\mathfrak{g}_{\epsilon_i - \epsilon_j} = \mathbb{C}E_{ij}$ is one-dimensional. So the weight space decomposition is

$$\mathfrak{sl}_n = \mathfrak{t} \oplus \bigoplus_{i \neq j} \mathfrak{g}_{\epsilon_i - \epsilon_j}$$

Example 24. For $\mathfrak{g} = \mathfrak{so}_n$ the antisymmetric matrices we can take the conjugate definition of $\mathfrak{so}_n = \{A \in \mathfrak{gl}_n \mid JA + A^T J = 0\}$ where $J = \begin{pmatrix} 0 & I_n \\ I_n & 0 \end{pmatrix}$. In this version of $\mathfrak{so}_{2n} = \left\{ \begin{pmatrix} A & B \\ C & -A^T \end{pmatrix} \right\}$ where $B = -B^T, C = -C^T$, the maximal torus $\mathfrak{t}$ consists of matrices of form $t = \text{diag}(t_1, \dots, t_n, -t_1, \dots, -t_n)$ for $t_i \in \mathbb{C}$. One can compute that the roots are $\{\pm\epsilon_i \pm \epsilon_j \mid i \neq j\}$ with $\epsilon_i(t) = t_i$ as above. Further we have

$$\mathfrak{g}_{\epsilon_i - \epsilon_j} = \mathbb{C}(E_{ij} - E_{n+j, n+i}) \text{ with } i \neq j$$

$$\mathfrak{g}_{\epsilon_i + \epsilon_j} = \mathbb{C}(E_{i, n+j} - E_{j, n+i}) \text{ with } i < j$$

$$\mathfrak{g}_{-\epsilon_i - \epsilon_j} = \mathbb{C}(E_{n+i, j} - E_{n+j, i}) \text{ with } i < j$$

So the weight space decomposition for $\mathfrak{so}_{2n}$ is

$$\mathfrak{so}_{2n} = \mathfrak{t} \oplus \bigoplus_{i \neq j} \mathfrak{g}_{\epsilon_i - \epsilon_j} \oplus \bigoplus_{i < j} \mathfrak{g}_{\epsilon_i + \epsilon_j} \oplus \bigoplus_{i < j} \mathfrak{g}_{-\epsilon_i - \epsilon_j}$$

For $\mathfrak{so}_{2n+1}$, things are different. $t \in \mathfrak{t}$ is now of form $\text{diag}\{t_1, \dots, t_n, 0, -t_1, \dots, -t_n\}$. The roots are $\{\pm\epsilon_i \pm \epsilon_j \mid i \neq j\} \cup \{\pm\epsilon_i\}$. We call the roots in former set **long roots** because they are longer. The generator of each weight space is the same after reindexed with the extra middle slot. And we have the weight space decomposition

$$\mathfrak{so}_{2n+1} = \mathfrak{t} \oplus \bigoplus_{i \neq j} \mathfrak{g}_{\epsilon_i - \epsilon_j} \oplus \bigoplus_{i < j} \mathfrak{g}_{\epsilon_i + \epsilon_j} \oplus \bigoplus_{i < j} \mathfrak{g}_{-\epsilon_i - \epsilon_j} \oplus \bigoplus_i \mathfrak{g}_{\epsilon_i} \oplus \bigoplus_i \mathfrak{g}_{-\epsilon_i}$$

Example 25. Recall $\mathfrak{g} = \mathfrak{sp}_{2n} = \{A \in \mathfrak{gl}_n \mid JA + A^T J = 0\}$ where $J = \begin{pmatrix} 0 & I_n \\ -I_n & 0 \end{pmatrix}$. Solving this we get $\mathfrak{sp}_{2n} = \left\{ \begin{pmatrix} A & B \\ C & -A^T \end{pmatrix} \right\}$ where $B = B^T, C = C^T$. The maximal torus $\mathfrak{t}$ still consists of matrices of form $t = \text{diag}(t_1, \dots, t_n, -t_1, \dots, -t_n)$ for $t_i \in \mathbb{C}$. The roots are $\{\pm\epsilon_i \pm \epsilon_j \mid i \neq j\} \cup \{\pm 2\epsilon_i\}$. Note that the weight space of $2\epsilon_i$ is generated by $E_{i,n+i}$. The weight space decomposition is

$$\mathfrak{sp}_{2n} = \mathfrak{t} \oplus \bigoplus_{i \neq j} \mathfrak{g}_{\epsilon_i - \epsilon_j} \oplus \bigoplus_{i < j} \mathfrak{g}_{\epsilon_i + \epsilon_j} \oplus \mathfrak{g}_{-\epsilon_i - \epsilon_j} \oplus \bigoplus_i \mathfrak{g}_{2\epsilon_i} \oplus \mathfrak{g}_{-2\epsilon_i}$$

These Lie algebras are called **the classical Lie algebras** (they are automorphisms of vector spaces). The rest of the course will be concerned with explaining how their roots control their representations.

Proposition 5.1. $\mathfrak{sl}_n \mathbb{C}$ is simple.

Proof. The weight space decomposition gives

$$\mathfrak{sl}_N = \mathfrak{g}_0 \oplus \bigoplus_{\lambda \in R} \mathfrak{g}_\lambda$$

where $R = \{\epsilon_i - \epsilon_j \mid i \neq j\}$ and $\mathfrak{g}_{\epsilon_i - \epsilon_j} = \mathbb{C}E_{ij}$

Suppose $\mathfrak{r} \subset \mathfrak{sl}_n$ a nonzero ideal. Choose $0 \neq r \in \mathfrak{r}$ s.t. when we write $r = t + \sum_\lambda e_\lambda$ with $e_\lambda \in \mathfrak{g}_\lambda$, the number of nonzero terms is minimal.

First, suppose $t \neq 0$. Choose $t_0 \in \mathfrak{t}$ s.t. $\alpha(t_0) \neq 0$ for all $\alpha \in R$, i.e. t_0 has distinct eigenvalues. Consider $[t_0, r] = \sum \alpha(t_0)e_\alpha$. Note that this lies in $\mathfrak{r}$, and if it is nonzero, it has fewer nonzero terms than r , a contradiction. Thus, $[t_0, r] = 0 \implies r = t \in \mathfrak{t}$.

Since $t \neq 0$, there is $\alpha \in R$ such that $\alpha(t) \neq 0$ (For $\mathfrak{sl}_n$, R spans $\mathfrak{t}^*$. This is true in general.). Hence it can't have the same eigenvalue since it's trace 0. Then for any $0 \neq e_\alpha \in \mathfrak{g}_\alpha$, $[t, e_\alpha] = \alpha(t)e_\alpha \implies e_\alpha \in \mathfrak{r}$.

Let $\alpha = \epsilon_i - \epsilon_j$. Then $E_{ij} \in \mathfrak{r}$. Now using

$$[E_{ij}, E_{jk}] = E_{ik} \text{ for } i \neq k, [E_{si}, E_{ij}] = E_{sj} \text{ for } s \neq j$$

we get that $E_{ab} \in \mathfrak{r}$ for all $a \neq b$. Finally

$$[E_{i,i+1}, E_{i+1,i}] = E_{ii} - E_{i+1,i+1} \in \mathfrak{r}$$

for all i and so $\mathfrak{r} = \mathfrak{sl}_n$

Remark 5.3. This is actually a combinatorial statement about root spaces; we'll see more about this later.

That leaves us with the case $t = 0$. If $r = cE_{ij}$ has only one nonzero term, we are done exactly as above; therefore, say

$$r = e_\alpha + e_\beta + \sum_{\gamma \in R - \{\alpha, \beta\}} e_\gamma, \alpha \neq \beta$$

Choose $t_0 \in \mathfrak{t}$ s.t. $\alpha(t_0) \neq \beta(t_0)$. Then some linear combination of $[t_0, r]$ and r will have fewer nonzero terms than r , contradiction $\square$

Proposition 5.2. Let $\mathfrak{g}$ be a semisimple Lie algebra. Then maximal tori exist (i.e. are nonzero). Moreover $\mathfrak{g}_0 = \{x \in \mathfrak{g} \mid [t, x] = 0 \ \forall t \in \mathfrak{t}\} = \mathfrak{t}$. (Recall that $\mathfrak{t}$ is not defined to be the maximal abelian subalgebra, but as the maximal abelian subalgebra whose adjoint acts semisimply.)

We won't prove this; the proof involves introducing the Cartan subalgebra and showing that it's the same as a maximal torus. As a result, we will sometimes be calling $\mathfrak{t}$ the **Cartan subalgebra**.

This means that the root space decomposition of a semisimple Lie algebra is

$$\mathfrak{g} = \mathfrak{t} \oplus \bigoplus_{\alpha \in R} \mathfrak{g}_\alpha$$

as we have, of course, already seen for the classical Lie algebras.

Theorem 5.1. (*Structure theorem for semisimple Lie algebras, part I*). Let $\mathfrak{g}$ be a semisimple Lie algebra over $\mathbb{C}$, and let $\mathfrak{t} \subset \mathfrak{g}$ be a maximal torus. Write $\mathfrak{g} = \mathfrak{t} \oplus \bigoplus_{\alpha \in R} \mathfrak{g}_\alpha$. Then

1. $\mathbb{C}R = \mathfrak{t}^*$, i.e. roots span $\mathfrak{t}^*$
2. $\dim \mathfrak{g}_\alpha = 1$
3. If $\alpha, \beta \in R$ and $\alpha + \beta \in R$ then $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] = \mathfrak{g}_{\alpha+\beta}$. If $\alpha + \beta \notin R$ and $\alpha \neq -\beta$ then $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] = 0$
4. $[\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}] \subset \mathfrak{t}$ is one-dimensional, and $\mathfrak{g}_\alpha \oplus [\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}] \oplus \mathfrak{g}_{-\alpha}$ is a Lie subalgebra of $\mathfrak{g}$ isomorphic to $\mathfrak{sl}_2$. (In particular, $\alpha \in R \implies -\alpha \in R$)

These can be easily verified for classical Lie algebras.

Proof. (1) Suppose not, then there exists $t \in \mathfrak{t}$ such that $\alpha(t) = 0$ for all $\alpha \in R$. Then $[t, \mathfrak{g}_\alpha] = 0$ for all α . Also $[t, \mathfrak{t}] = 0$ since $\mathfrak{t}$ is abelian. So t is central in $\mathfrak{g}$. But $\mathfrak{g}$ semisimple $\implies \mathfrak{g}$ has no abelian nonzero ideals by theorem 4.4, in particular no center

For the rest, we will now show a sequence of statements

- (a) $[\mathfrak{g}_\lambda, \mathfrak{g}_\mu] \subset \mathfrak{g}_{\lambda+\mu}$ if $\lambda, \mu \in \mathfrak{t}^*$

Take $x \in \mathfrak{g}_\lambda, y \in \mathfrak{g}_\mu$. Then $[t, [x, y]] = [[t, x], y] + [x, [t, y]] = (\lambda(t) + \mu(t))[x, y] \implies [x, y] \in \mathfrak{g}_{\lambda+\mu}$. This shows that $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] \subset \mathfrak{g}_{\alpha+\beta}$ if $\alpha + \beta \in R$, is 0 if $\alpha + \beta \notin R$, and $[\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}] \subset \mathfrak{t}$

- (b) $(\mathfrak{g}_\lambda, \mathfrak{g}_\mu)_{ad} = 0$ if $\lambda \neq -\mu$; $(\cdot, \cdot)_{ad}|_{\mathfrak{g}_\lambda \oplus \mathfrak{g}_{-\lambda}}$ is non-degenerate.

Still take $x \in \mathfrak{g}_\lambda, y \in \mathfrak{g}_\mu$. Recall $(x, y)_{ad} = \text{tr}(ad_x \cdot ad_y)$. To show it's zero, it is enough to show that $ad_x \cdot ad_y$ is nilpotent. Indeed, for any $\alpha \in R$, $(ad_x \cdot ad_y)^n \mathfrak{g}_\alpha \subset \mathfrak{g}_{\alpha+n(\lambda+\mu)}$ by (a). Since R is a finite set and $\lambda \neq -\mu$, $\mathfrak{g}_{\alpha+n(\lambda+\mu)} = 0$ for n large enough.

On the other hand, $(\cdot, \cdot)_{ad}$ is non-degenerate and $\mathfrak{g} = \bigoplus_{\lambda \in \mathfrak{t}^*} \mathfrak{g}_\lambda \oplus \mathfrak{g}_{-\lambda}$, and so $(\cdot, \cdot)_{ad}|_{\mathfrak{g}_\lambda \oplus \mathfrak{g}_{-\lambda}}$ must be non-degenerate.

- (c) In particular $(\cdot, \cdot)_{ad}|_{\mathfrak{t}}$ is nondegenerate, as $\mathfrak{t} = \mathfrak{g}_0$

Remark 5.4. $(\cdot, \cdot)_{ad}|_{\mathfrak{t}} \neq (\cdot, \cdot)_{ad}$ (which is zero since $\mathfrak{t}$ is abelian).

Therefore, the Killing form defines an isomorphism $v : \mathfrak{t} \rightarrow \mathfrak{t}^*$ via $v(t)(t') = (t, t')_{ad}$. It also defines an induced inner product (well, symmetric bilinear form to be precise – we don't know it's positive) on $\mathfrak{t}^*$, via $(v(t), v(t')) = (t, t')_{ad}$

- (d) $\alpha \in R \implies -\alpha \in R$, since $(\cdot, \cdot)_{ad}$ nondegenerate on $\mathfrak{g}_\alpha \oplus \mathfrak{g}_{-\alpha}$ but $(\mathfrak{g}_\alpha, \mathfrak{g}_\alpha)_{ad} = 0$ if $\alpha \neq 0$. This induces $\mathfrak{g}_{-\alpha} \cong \mathfrak{g}_\alpha^*$ via the Killing form

- (e) $x \in \mathfrak{g}_\alpha, y \in \mathfrak{g}_{-\alpha} \implies [x, y] = (x, y)_{ad} v^{-1}(\alpha)$

By definition, $(t, v^{-1}(\alpha))_{ad} = \alpha(t)$ for all t . Now just note that $(t, [x, y])_{ad} = ([t, x], y)_{ad} = \alpha(t)(x, y)_{ad} = (t, (x, y)_{ad} v^{-1}(\alpha))_{ad}$ for all t and use non-degeneracy. This implies that $[\mathfrak{g}_\alpha, \mathfrak{g}_{-\alpha}] \subset \mathfrak{t}$ is one-dimensional generated by $v^{-1}(\alpha)$

- (f) Let $0 \neq e_\alpha \in \mathfrak{g}_\alpha$ and pick $e_{-\alpha} \in \mathfrak{g}_{-\alpha}$ s.t. $(e_\alpha, e_{-\alpha})_{ad} \neq 0$ by non-degeneracy. Then $[e_\alpha, e_{-\alpha}] = (e_\alpha, e_{-\alpha})_{ad} v^{-1}(\alpha)$. To show that $e_\alpha, e_{-\alpha}$, and their bracket give a copy of $\mathfrak{sl}_2$, we compute that

$$[v^{-1}(\alpha), e_\alpha] = \alpha(v^{-1}(\alpha))e_\alpha = (\alpha, \alpha)e_\alpha, \text{ where } v^{-1}(\alpha) \in \mathfrak{t} \text{ and } (\cdot, \cdot) \text{ is the induced inner product on } \mathfrak{t}^*$$

We will be done if we show that $(\alpha, \alpha) \neq 0$ (then we get a copy of $\mathfrak{sl}_2$ after renormalising).

Proposition 5.3. $(\alpha, \alpha) \neq 0$ for all $0 \neq \alpha \in R$

Proof. Suppose otherwise. Let $m_\alpha = \text{Span}\langle e_\alpha, e_{-\alpha}, v^{-1}(\alpha) \rangle$. If $(\alpha, \alpha) = 0$ then $[v^{-1}(\alpha), e_\alpha] = [v^{-1}(\alpha), e_{-\alpha}]$ and so $[\mathfrak{m}_\alpha, \mathfrak{m}_\alpha] = \mathbb{C}[e_\alpha, e_{-\alpha}] = \mathbb{C}v^{-1}(\alpha)$ and $\mathfrak{m}_\alpha$ is solvable. However, by Lie's theorem this implies that $\text{ad}[\mathfrak{m}_\alpha, \mathfrak{m}_\alpha]$ acts by nilpotents on $\mathfrak{g}$, i.e. $\text{ad}_{v^{-1}(\alpha)}$ is nilpotent. Since $v^{-1}(\alpha) \in \mathfrak{t}^*$, it is also diagonalizable. But the only diagonalisable nilpotent element is 0. Hence $v^{-1}(\alpha) = 0 \implies \alpha = 0$, contradiction. $\square$

Now define

$$h_\alpha = \frac{2}{(\alpha, \alpha)} v^{-1}(\alpha)$$

and rescale $e_{-\alpha}$ such that $(e_\alpha, e_{-\alpha})_{\text{ad}} = 2/(\alpha, \alpha)$ and so $[e_\alpha, e_{-\alpha}] = h_\alpha$. With this one verifies that the map $(e_\alpha, h_\alpha, e_{-\alpha}) \mapsto (e, h, f)$ gives an isomorphism $\mathfrak{m}_\alpha \cong \mathfrak{sl}_2$

Remark 5.5. In $\mathfrak{sl}_n$, the root spaces are spanned by E_{ij} , so we are saying that the subalgebra of matrices of the form $\begin{pmatrix} a & * \\ * & -a \end{pmatrix}$ is a copy of $\mathfrak{sl}_2$, which is obvious. The cool thing is that something like this happens for all the semisimple Lie algebras, and that this lets us know just about everything about them.

We will now use our knowledge of the representation theory of $\mathfrak{sl}_2$ to show $\dim \mathfrak{g}_{-\alpha} = 1$ for $\alpha \in R$. Pick a copy of $\mathfrak{m}_\alpha = \text{Span}\langle e_\alpha, h_\alpha, e_{-\alpha} \rangle \cong \mathfrak{sl}_2$. Suppose $\dim \mathfrak{g}_{-\alpha} > 1$, then the map $\mathfrak{g}_{-\alpha} \rightarrow \mathbb{C}v^{-1}(\alpha)$ via $x \mapsto [e_\alpha, x]$ has a kernel. That is, there is a $y \in \mathfrak{g}_{-\alpha}$ such that $\text{ad}_{e_\alpha}(y) = 0$. Then y is a highest-weight vector w.r.t. the adjoint representation and its weight is determined by

$$[h_\alpha, y] = -\frac{2}{(\alpha, \alpha)} \alpha(v^{-1}(\alpha))y = -2y$$

since $y \in e_{-\alpha}$. So y is of weight -2 . However, we know that in f.d. representations of $\mathfrak{sl}_2$ (which $\mathfrak{g}$ is, since $\mathfrak{g}_\alpha$ acts on it) the highest weights are nonnegative integers, contradiction. $\square$

Before we finish proving the theorem we had (we still need to show $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] = \mathfrak{g}_{\alpha+\beta}$ when $\alpha, \beta, \alpha+\beta \in R$, we show further combinatorial properties of the roots.

Theorem 5.2. (Structure theorem, continued).

1. $\frac{2(\alpha, \beta)}{(\alpha, \alpha)} \in \mathbb{Z}$ for all $\alpha, \beta \in R$
2. If α and $k\alpha$ are in R , then $k = \pm 1$
3. $\bigoplus_{k \in \mathbb{Z}} \mathfrak{g}_{\beta+k\alpha}$ is an irreducible module for $\mathfrak{m}_\alpha$. In particular

$$\{k\alpha + \beta \mid k\alpha + \beta \in R, k \in \mathbb{Z}\} \cup \{0\}$$

is of the form $\beta - p\alpha, \beta - (p-1)\alpha, \dots, \beta + q\alpha$ where $p - q = \frac{2(\alpha, \beta)}{(\alpha, \alpha)}$. We call this set the “ α string through β ”.

Proof. (1) Let q the maximal integer s.t. $\mathfrak{g}_{q\alpha+\beta} \neq 0$ and take $0 \neq x \in \mathfrak{g}_{q\alpha+\beta}$. Then $\text{ad}_{e_\alpha}(x) \in \mathfrak{g}_{(q+1)\alpha+\beta} = 0$ and $[h_\alpha, x] = (q\alpha + \beta)(h_\alpha)x = (2q + \frac{2(\alpha, \beta)}{(\alpha, \alpha)})x$. So x is a hw-vector of weight $2q + \frac{2(\alpha, \beta)}{(\alpha, \alpha)}$. Since the weight in a finite-dimensional representation is an integer, we see that $\frac{2(\alpha, \beta)}{(\alpha, \alpha)} \in \mathbb{Z}$

(3) Structure of $\mathfrak{sl}_2$ -modules tells us that $(ad_{e_{-\alpha}})^r x \neq 0$ for $0 \leq r \leq \frac{2(\alpha, \beta)}{(\alpha, \alpha)} + 2q = N$ and $(ad_{e_{-\alpha}})^{N+1} x = 0$ (string of weight from $-N$ to N). Hence $\{\beta + q\alpha - k\alpha | 0 \leq k \leq N\}$ are all roots (or possibly equal to zero – in any case, they have nonzero root space). This certainly is an irr. $\mathfrak{m}_\alpha$ -module (isom. to some $L(n)$). It is enough to show that there are no other roots in the α string through β . We do this by repeating the same construction from the bottom up:

Let p be the maximal s.t. $\beta - p\alpha \in R$. Take $0 \neq y \in \mathfrak{g}_{\beta-p\alpha}$. Similarly, $ad_{e_{-\alpha}}(y) = 0$ and $[h_\alpha, y] = (\frac{2(\alpha, \beta)}{(\alpha, \alpha)} - 2p)y$, so y is a lowest-weight vector of weight $\frac{2(\alpha, \beta)}{(\alpha, \alpha)} - 2p$. Again, by applying ad_{e_α} repeatedly we get an irr. $\mathfrak{m}_\alpha$ -module with roots from $\beta - p\alpha$ to $\beta + (p - \frac{2(\alpha, \beta)}{(\alpha, \alpha)})\alpha$.

By our choice of p, q , $p - \frac{2(\alpha, \beta)}{(\alpha, \alpha)} \leq q$ and $q + \frac{2(\alpha, \beta)}{(\alpha, \alpha)} \leq p$. Together we get $p - q = \frac{2(\alpha, \beta)}{(\alpha, \alpha)}$ and these two modules coincide.

(2) By (1) we get $\frac{2(\alpha, k\alpha)}{(k\alpha, k\alpha)} = \frac{2}{k} \in \mathbb{Z}$ and $\frac{2(\alpha, k\alpha)}{(\alpha, \alpha)} = 2k \in \mathbb{Z}$. Thus it suffices to show that $\alpha \in R \implies 2\alpha \notin R$. Suppose $2\alpha \in R$. Let $0 \neq x \in \mathfrak{g}_{-2\alpha}$. Then $[e_\alpha, x] \in \mathfrak{g}_{-\alpha}$ has the property that

$$([e_\alpha, x], e_\alpha)_{ad} = (x, [e_\alpha, e_\alpha]) = 0$$

and since $\mathfrak{g}_\alpha$ is one-dimensional and spanned by e_α , we find that $([e_\alpha, x], \mathfrak{g}_\alpha) = 0$. Since $(\cdot, \cdot)_{ad}|_{\mathfrak{g}_\alpha \oplus \mathfrak{g}_{-\alpha}}$ is non-degenerate, $[e_\alpha, x] = 0$, i.e. $x \in \mathfrak{g}_{-\alpha}$ is a hw-vector for $\mathfrak{m}_\alpha$ with highest weight $-2\alpha(h_\alpha) = -4$. Again, this doesn't happen in finite-dimensional representations, so $\mathfrak{g}_{2\alpha} = 0$ and 2α is not a root. (In fact, we could derive this from (3), since we know that there is a 3-dimensional $\mathfrak{sl}_2$ -module running through 0.) $\square$

We finally show that $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] = \mathfrak{g}_{\alpha+\beta}$. Indeed, we have just shown that $\oplus_k \mathfrak{g}_{\beta+k\alpha}$ is an irreducible representation of $\mathfrak{m}_\alpha$, and $ad_{e_\alpha} : \mathfrak{g}_{\beta+k\alpha} \rightarrow \mathfrak{g}_{\beta+(k+1)\alpha}$ is an isomorphism for $k < q$ because of dimension reason (both of dim-1). Since $q \geq 1$, we deduce that $ad_{e_\alpha} \cdot \mathfrak{g}_\beta = \mathfrak{g}_{\beta+\alpha}$

Remark 5.6. The combinatorial statement of (3) is rather clunky. We now “unclunk” it:

For $\alpha \in \mathfrak{t}^*$, define $s_\alpha : \mathfrak{t}^* \rightarrow \mathfrak{t}^*$ by $s_\alpha(v) = v - \frac{2(\alpha, v)}{(\alpha, \alpha)}\alpha$ (a reflection of v about α). We claim that (3) implies (and in fact, is equivalent to in the sense of more general root system) $s_\alpha(\beta) \in R$ whenever $\alpha, \beta \in R$

Indeed, let $\frac{r=2(\alpha, \beta)}{(\alpha, \alpha)}$. NTS $\beta - r\alpha \in R$. But $p - q = r \implies q = p - r \geq r$ and $-p = -r - q \leq -r$. Hence $\beta - r\alpha$ is in the α string through β

Our next goal is to classify the root systems as combinatorial objects. Before we do that, we state some further properties of the roots:

Proposition 5.4. 1. $(\alpha, \beta) \in \mathbb{Q}$ for $\alpha, \beta \in R$

2. If we pick a basis $\beta_1, \dots, \beta_\ell$ of $\mathfrak{t}^*$ with $\beta_i \in R$, then any β expands as $\sum q_i \beta_i$ with $q_i \in \mathbb{Q}$, i.e. $\dim \mathbb{Q}R = \dim_{\mathbb{C}} \mathfrak{t}$

3. $(\cdot, \cdot)$ is positive definite on $\mathbb{Q}R$

Proof. Since $\frac{2(\alpha, \beta)}{(\alpha, \alpha)} \in \mathbb{Z}$, it suffices to show that $(\alpha, \alpha) \in \mathbb{Z}$ for all $\alpha \in R$. Take $h, h' \in \mathfrak{t}$, then $(h, h')_{ad} = \text{tr}(ad_h \cdot ad_{h'}) = \sum_{\alpha \in R} \alpha(h)\alpha(h')$ since $\mathfrak{g} = \mathfrak{t} \oplus \bigoplus_{\alpha \in R} \mathfrak{g}_\alpha$ by the structure theorem and $ad_h, ad_{h'}$ act diagonally w.r.t. this.

Therefore, for $\lambda, \mu \in \mathfrak{t}^*$,

$$(\lambda, \mu) = (v^{-1}(\lambda), v^{-1}(\mu))_{ad} = \sum_{\alpha \in R} \alpha(v^{-1}(\lambda))\alpha(v^{-1}(\mu)) = \sum_{\alpha \in R} (\alpha, \lambda)(\alpha, \mu)$$

In particular, $(\beta, \beta) = \sum_{\alpha \in R} (\alpha, \beta)^2$. Multiplying $\frac{4}{(\beta, \beta)^2}$ to get

$$\frac{4}{(\beta, \beta)} = \sum_{\alpha \in R} \left(\frac{2(\alpha, \beta)}{(\beta, \beta)} \right)^2 \in \mathbb{Z} \implies (\beta, \beta) \in \mathbb{Q}$$

as required

(2) Let B be the Gram matrix of the basis, i.e. $B_{ij} = (\beta_i, \beta_j)$

Fact: for a non-degenerate symmetric bilinear form $(,)$, $\det B \neq 0$

Let $\beta = \sum c_i \beta_i \in R$, then $(\beta, \beta_i) = \sum c_j (\beta_j, \beta_i)$. Therefore, we can recover the vector of c_i as

$$(c_1, \dots, c_\ell) = ((\beta, \beta_1), \dots, (\beta, \beta_\ell))(B^T)^{-1}$$

Since the inner product of roots are all rational by (1), we see that $c_i \in \mathbb{Q}$ as well.

(3) Let $\lambda \in \mathbb{Q}R$. Then $\lambda = \sum c_i \beta_i$ with $c_i \in \mathbb{Q}$, and $(\lambda, \alpha) \in \mathbb{Q}$ for all $\alpha \in \mathbb{Q}$. Moreover, we have seen that $(\lambda, \lambda) = \sum_{\alpha \in R} (\alpha, \lambda)^2 \geq 0$, and if $(\lambda, \lambda) = 0$ then $(\lambda, \alpha) = 0$ for all $\alpha \in R$. Since R span $\mathfrak{t}^*$ and $(,)$ is non-degenerate on $\mathfrak{t}^*$, we deduce that $\lambda = 0$ $\square$

6 Root Systems

Let V be a vector space over $\mathbb{R}$ and $(,)$ an inner-product (i.e. a positive-definite symmetric bilinear form). For $0 \neq \alpha \in V$, write $\alpha^\vee = \frac{2\alpha}{(\alpha, \alpha)}$, so that $(\alpha, \alpha^\vee) = 2$

Define $s_\alpha : V \rightarrow V$ by $s_\alpha(v) = v - (v, \alpha^\vee)\alpha$

Lemma 6.1. s_α is the reflection in the hyperplane orthogonal to α . In particular, all but one of the eigenvalues of s_α are 1, and the remaining one is -1 (with eigenvector α). Thus, $s_\alpha^2 = 1$, or $(s_\alpha + 1)(s_\alpha - 1) = 0$, and $s_\alpha \in O(V, (,))$, the orthogonal group of V defined by the inner product $(,)$.

Proof. Can write $V = \mathbb{R}\alpha \oplus \alpha^\perp$. Then $s_\alpha(\alpha) = -\alpha$ and s_α fixes all $v \in \alpha^\perp$ $\square$

Definition 6.1. A **root system** R in V is a finite set $R \subset V$ such that

1. $0 \notin R$ and $\mathbb{R}R = V$
2. For all $\alpha, \beta \in R$, $(\alpha, \beta^\vee) \in \mathbb{Z}$
3. For all $\alpha \in R$, $s_\alpha(R) \subset R$ (in particular $s_\alpha(\alpha) = -\alpha \in R$)

Moreover, R is called **reduced** if $\alpha, k\alpha \in R \implies k = \pm 1$

Example 26. Let $\mathfrak{g}$ be a semisimple Lie algebra, $\mathfrak{g} = \mathfrak{t} \oplus \bigoplus_{\alpha \in R} \mathfrak{g}_\alpha$. Then we know that $R \subset \mathbb{R}R$ is a reduced root system.

Definition 6.2. Let $W \subset GL(V)$ be the group generated by the reflections s_α for $\alpha \in R$. W is called the **Weyl group** of R .

Observe that $|W| < \infty$: W acts on R by permutations, and since $0 \notin R$, the action is faithful (since $s_\alpha \in W$ for all $\alpha \in R$ and s_α sends α to $-\alpha$. The only possible W -invariant element is 0, but it is not in R). Thus, W injects into $Sym(R)$. So $|R| < \infty \implies |W| < \infty$

Definition 6.3. The **rank** of R is the dimension of V

Definition 6.4. An isomorphism of root systems $\phi : (R, V) \rightarrow (R', V')$ is a linear bijection $\phi : V \rightarrow V'$ s.t. $\phi(R) = R'$. Note that ϕ is not required to be an isometry.

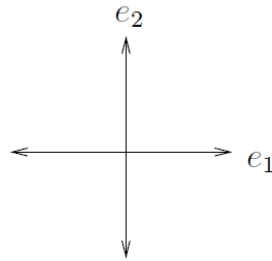
Definition 6.5. If $(R, V), (R', V')$ are root systems, so is their direct sum $(R \sqcup R', V \oplus V')$. A root system which is not isomorphic to a direct sum is called irreducible.

Example 27. Rank 1: $V = \mathbb{R}$, $(x, y) = xy$, $R = \{\alpha, -\alpha\}$ with $\alpha \neq 0$, $W \cong \mathbb{Z}/2$

$$-\alpha \leftarrow \bullet \rightarrow \alpha$$

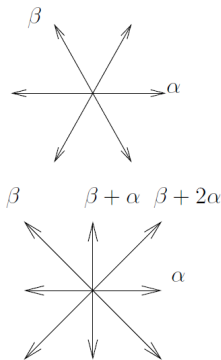
This is the only rank 1 root system

Example 28. Rank 2: $V = \mathbb{R}^2$ with the usual inner product



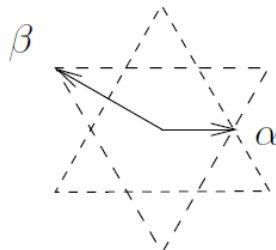
This is $A_1 \times A_1$ (or $A_1 + A_1$), and is not irreducible. The Weyl group is $\mathbb{Z}/2 \times \mathbb{Z}/2$

A_2 : $\alpha = \alpha^\vee, \beta = \beta^\vee, (\alpha, \beta) = -1$. $W = S_3$ is the dihedral group of order 6. This is the root system of $\mathfrak{sl}_3$, as we will see



B_2 : $\alpha = e_1, \beta = e_2 - e_1, (\alpha, \beta) = -1, (\alpha, \alpha) = 1, (\beta, \beta) = 2$. $\alpha, \alpha + \beta$ are short roots, $\beta, 2\alpha + \beta$ are long roots. W is the dihedral group of order 8. This is the root system of $\mathfrak{sp}_4$ and $\mathfrak{so}_5$

G_2 : At this point, we clearly should look at all the dihedral groups, right? Unfortunately, no: the condition $(\alpha, \beta^\vee) \in \mathbb{Z}$ rules out all but the dihedral group of order 12. This is the root system of a Lie group, but not one of the classical ones (symmetries of octonions).



These are the only rank 2 root systems, and A_2, B_2 , and G_2 are irreducible.

Lemma 6.2. *If R is a root system, then $R^\vee = \{\alpha^\vee | \alpha \in R\}$ is also a root system.*

This can be directly verified. For example, we see that $(A_2)^\vee = A_2$ and $(B_2)^\vee \cong B_2$

Definition 6.6. R is **simply laced** if all the roots have the same length.

If R is simply laced, then it is isomorphic (as a root system) to a simply laced root system R' with $(\alpha, \alpha) = 2$ for all $\alpha \in R'$ (i.e. $\alpha = \alpha^\vee$) through rescaling.

We would like to look at roots as coming from a lattice. We will see why length-2 is so useful.

Definition 6.7. A **lattice** L is a finitely-generated free abelian group ($\cong \mathbb{Z}^\ell$ for some ℓ) with bilinear form $L \oplus L \rightarrow \mathbb{Z}$ s.t. $(L \otimes_{\mathbb{Z}} \mathbb{R}, (,))$ is an inner product space.

A **root** of L is $\alpha \in L$ s.t. $(\alpha, \alpha) = 2$. We write $R_L = \{\alpha \in L | (\alpha, \alpha) = 2\}$ for the roots of L .

Note that $\alpha \in R_L \implies s_\alpha(L) \subset L$

Lemma 6.3. R_L is a simply laced root system in $\mathbb{R}_L$

The only unobvious claim is $|R_L| < \infty$. However, R_L is the intersection of a compact set (sphere of radius $\sqrt{2}$) with a discrete set (the lattice), so is finite.

Definition 6.8. We say that L is generated by roots if $\mathbb{Z}R_L = L$. Note that then L is an even lattice, i.e. $(l, l) \in 2\mathbb{Z}$ for $l \in L$

Example 29. Let $L = \mathbb{Z}\alpha$ with $(\alpha, \alpha) = \lambda$. If $\lambda = 2$ then $R_L = \{\pm\alpha\}$; if $\frac{k^2\lambda}{2} \neq 1$ for all k then $R_L = \emptyset$

We now characterise the simply laced root systems.

(1) A_n : Consider $\mathbb{Z}^{n+1} \cong \oplus \mathbb{Z}e_i$ with $(e_i, e_j) = \delta_{ij}$. Define $L = \{l \in \mathbb{Z}^{n+1} | (l, e_1 + \dots + e_{n+1}) = 0\} = \{\sum a_i e_i | \sum a_i = 0\}$. Then $R_L = \{e_i - e_j | i \neq j\}$, $|R_L| = n(n+1)$ and $\mathbb{Z}R_L = L$

Take $\alpha = e_i - e_j$. Then one easily check that $s_{e_i - e_j}$ essentially swaps the i th and j th coordinates of a vector. Therefore, the Weyl group $W = \langle s_{e_i - e_j} \rangle = S_{n+1}$

$R_L, \mathbb{R}L$ is the root system of type A_n . Note that n is the rank of the root system, not the index of the associated Lie algebra. Hence the root system of $\mathfrak{sl}_{n+1}$ has type A_n

(2) D_n : Consider the square lattice $\mathbb{Z}^n$. The roots are $R_L = \{\pm e_i \pm e_j | i \neq j\}$. Set $L = \mathbb{Z}R_L = \{l = \sum a_i e_i | l \in \mathbb{Z}, \sum a_i \text{ is even}\}$.

As before, $s_{e_i - e_j}$ swaps the i th and j th coordinate, while $s_{e_i + e_j}$ flips the sign of the i th and j th coordinates.

$(R_L, \mathbb{R}L)$ is called D_n ; $|R_L| = 2n(n-1)$. The Weyl group is $W = (\mathbb{Z}/2)^{n-1} \rtimes S_n$ where $(\mathbb{Z}/2)^{n-1}$ is the subgroup of even number of sign changes.

We have seen that the root system of $\mathfrak{so}_{2n}$ has type D_n . Note that D_n is irreducible for $n \geq 3$. To see this, notice that by definition roots in different components must be orthogonal (it is a direct sum). Say R is a direct summand and $e_1 - e_2 \in R$. Then for $k \geq 3$, $(e_1 - e_2, \pm e_1 \pm e_k) = \pm 1$ and so $\pm e_1 \pm e_k \in R$. From this we easily deduce that $R = D_n$. On the other hand, we have $D_2 = A_1 \times A_1$. Further we have an isomorphism $D_3 \cong A_3$. We can give an explicit bijection on the roots. However, later this will be done with their Dynkin diagrams

(3) E_8 : Let $\Gamma_n = \{(k_1, \dots, k_n) | \text{either all } k_i \in \mathbb{Z} \text{ or all } k_i \in \mathbb{Z} + \frac{1}{2} \text{ and } \sum k_i \in 2\mathbb{Z}\}$. Suppose $\alpha = (\frac{1}{2}, \dots, \frac{1}{2}) \in \Gamma_n$, and note that $(\alpha, \alpha) = \frac{n}{4}$. Thus, if Γ_n is an even lattice, we must have $8 | n$

For $n > 2$ we will see that the roots of Γ_{8n} is a root system of type D_n . The roots of Γ_8 are $\{\pm e_i \pm e_j, i \neq j\}$ and $\{\frac{1}{2}(\pm e_1 \pm \dots \pm e_8) \text{ with an even number of - signs}\}$

The root system of Γ_8 is called the root system of type E_8 . The number of roots is $R_{E_8} = \binom{8}{2} \cdot 4 + 128 = 240$. The dimension of the associated Lie algebra is $240 + 8 = 248$. (By the structure theorem, $\mathfrak{g} = \mathfrak{t} \oplus \bigoplus_{\alpha \in R} \mathfrak{g}_\alpha$. Since R $\mathbb{C}$ -spans $\mathfrak{t}^*$, have $\dim \mathfrak{t} = \text{rank}(R)$. Now use $\dim \mathfrak{g}_\alpha = 1$)

Remark 6.1. E_8 is weird. Its smallest representation is the adjoint representation of dimension 248. This is quite unlike $\mathfrak{so}_n, \mathfrak{sl}_n$, etc., which had dimension $O(n^2)$ and an n -dimensional representation. There isn't a good understanding of why E_8 is like that.

(4) E_6, E_7 : Note that if R is a root system and $\alpha \in R$ then $\alpha^\perp \cap R$ is also a root system. We will now apply this to $\alpha = (\frac{1}{2}, \dots, \frac{1}{2})$ and $\beta = e_7 + e_8$

Definition 6.9. $\alpha^\perp \cap R_{E_8}$ is a root system of type E_7 .
 $\langle \alpha^\perp, \beta \rangle \cap R_{E_8}$ is a root system of type E_8

Similarly we compute that $|R_{E_7}| = \binom{8}{2} \cdot 2 + \binom{8}{4} = 126$ and $|R_{E_6}| = \binom{6}{2} \cdot 2 + 2 + \binom{6}{3} \cdot 2 = 72$. Their root lattices can be derived from E_8 .

Remark 6.2. These lattices are reasonably natural objects. Indeed, if we took the associated algebraic group and its maximal torus, there is a natural lattice associated with it coming from the fundamental group. Equivalently, we can think of the group homomorphisms to (or from) $\mathbb{C}^\times$. This isn't quite the root lattice, but it's closely related (and we might say more about it later).

Theorem 6.1. (“ADE classification”)

1. The complete list of irreducible simply laced root systems is

$$A_n, n \geq 1, D_n, n \geq 4, E_6, E_7, E_8$$

No two root systems in this list are isomorphic.

2. The remaining irreducible reduced root systems are denoted by

$$B_2 = C_2, B_n, C_n, n \geq 3, F_4, G_2$$

where $R_{B_n} = \{\pm e_i, \pm e_i \pm e_j | i \neq j\}$, $R_{C_n} = \{\pm 2e_i, \pm e_i \pm e_j | i \neq j\}$. It is clear that $R_{B_n} = R_{C_n}^\vee$. The root system of type B_n corresponds to $\mathfrak{so}_{2n+1}$, and the root system of type C_n corresponds to $\mathfrak{sp}_{2n}$. The Weyl groups are

$$W_{B_n} = W_{C_n} = (\mathbb{Z}/2)^n \rtimes S_n$$

The root system F_4 is defined as follows: let $Q = \{(k_1, k_2, k_3, k_4) | \text{all } k_i \in \mathbb{Z} \text{ or all } k_i \in \mathbb{Z} + \frac{1}{2}\}$ and take

$$R_{F_4} = \{\alpha \in Q | (\alpha, \alpha) = 2 \text{ or } (\alpha, \alpha) = 1\} = \{\pm e_i, \pm e_i \pm e_j, i \neq j, \frac{1}{2}(\pm e_1 \pm e_2 \pm e_3 \pm e_4)\}$$

Remark 6.3. The duality between R_{B_n} and R_{C_n} suggests some form of duality between $\mathfrak{so}_{2n+1}$ and $\mathfrak{sp}_{2n}$. We will see it later.

To prove this theorem, we will first choose a good basis for V (the space spanned by the root lattice). Choose $f : V \rightarrow \mathbb{R}$ linear such that $f(\alpha) \neq 0$ for all $\alpha \in \mathbb{R}$.

Definition 6.10. A root $\alpha \in R$ is **positive** if $f(\alpha) > 0$, write $\alpha \in R^+$; **negative** if $f(\alpha) < 0$, write $\alpha \in R^-$

A positive root $\alpha \in R^+$ is **simple** if α is not a sum of two positive roots, write $\alpha \in \Pi$

Example 30. (1) A_n : roots are $\{e_i - e_j, i \neq j\}$. Choose $f(e_1) = n + 1, f(e_2) = n, \dots, f(e_{n+1}) = 1$. Then $R^+ = \{e_i - e_j, i < j\}$

Since $f(R) \in \mathbb{Z}, f(\alpha) = 1 \implies \alpha$ simple. Hence $\{e_i - e_{i+1} | i = 1, \dots, n\} \subset \Pi$. On the other hand, it is clear that the positive integer span of these is R^+ , so in fact we have equality.

(2) B_n : $R = \{\pm e_i, \pm e_i \pm e_j, i \neq j\}$. Still take $f(e_1) = n, \dots, f(e_n) = 1$. Then $R^+ = \{e_i, e_i \pm e_j, i < j\}$ and $\Pi = \{e_1 - e_2, \dots, e_{n-1} - e_n, e_n\}$

(3) C_n : take the same f as in B_n . $R^+ = \{2e_i, e_i \pm e_j, i < j\}$ and $\Pi = \{e_1 - e_2, \dots, e_{n-1} - e_n, 2e_n\}$

(4) D_n : $R = \{\pm e_i \pm e_j, i \neq j\}$. Take the same f . $R^+ = \{e_i \pm e_j, i < j\}$ and $\Pi = \{e_1 - e_2, \dots, e_{n-1} - e_n, e_{n-1} + e_n\}$ (Easy to check that $e_{n-1} + e_n$ is simple by $f(e_{n-1} + e_n) = 3$)

(5) E_8 : set $f(e_1) = 28 = 1 + 2 + \dots + 7$ and $f(e_i) = 9 - i$ for $i = 2, \dots, 8$, then

$$R^+ = \{e_i \pm e_j, i < j, \frac{1}{2}(e_1 \pm e_2 \dots \pm e_8) \text{ with an even number of - signs}\}$$

$$\Pi = \{e_2 - e_3, \dots, e_7 - e_8, \frac{1}{2}(e_1 + e_8 - e_2 - \dots - e_7), e_7 + e_8\}$$

Remark 6.4. So far we have made two choices. We have chosen a maximal torus, and we have chosen a function f . **Neither of these matters.** We may or may not prove this about the maximal torus later; as for f , note that the space of functionals f is partitioned into cells within which we get the same notions of positive roots. The Weyl group acts transitively on these cells, which should show that the choice of f does not affect the structure of the root system we are deriving.

Proposition 6.1. 1. $\alpha, \beta \in \Pi \implies \alpha - \beta \notin R$

2. $\alpha, \beta \in \Pi, \alpha \neq \beta \implies (\alpha, \beta^\vee) \leq 0$ (i.e. the angle between simple roots is obtuse).

3. Every $\alpha \in R^+$ can be written as $\alpha = \sum k_i \alpha_i$ where $k_i \in \mathbb{Z}_{\geq 0}$ and α_i simple roots

4. Simple roots are linearly independent.

5. $\alpha \in R^+, \alpha \notin \Pi \implies$ there is $\alpha' \in \Pi$ such that $\alpha - \alpha' \in R^+$

6. R is irreducible iff Π is indecomposable (i.e. $\Pi \neq \Pi_1 \sqcup \Pi_2$ with $(\Pi_1, \Pi_2) = 0$)

Proof. (1) Suppose $\alpha - \beta \in R$. WTS assume that $\alpha - \beta$ is positive. Then $\alpha = \beta + (\alpha - \beta)$. (2) If $(\alpha, \beta^\vee) > 0$, then $s_\beta(\alpha) = \alpha - (\alpha, \beta^\vee)\beta \in R \implies \alpha$ or β can be written as a sum of two positive roots. (3) is by induction on $f(\alpha)$. (4) is trivial since simple roots are positive. (5) is by (2). (6) is clear. $\square$

We will now find another way to represent the root system. Let $\Pi = \{\alpha_1, \dots, \alpha_l\}$ and $a_{ij} = (\alpha_i^\vee, \alpha_j)$

Definition 6.11. $A = (a_{ij})$ is the **Cartan matrix**.

Proposition 6.2. 1. $a_{ij} \in \mathbb{Z}, a_{ii} = 2$ and $a_{ij} \leq 0$ for $i \neq j$

2. $a_{ij} = 0$ iff $a_{ji} = 0$

3. $\det A > 0$

4. All principal subminors of A have positive determinant.

Proof. The first two should be obvious from the above. To prove the determinant, note that

$$A = \begin{pmatrix} \frac{2}{(\alpha_1, \alpha_1)} & & 0 \\ & \ddots & \\ 0 & & \frac{2}{(\alpha_l, \alpha_l)} \end{pmatrix} \cdot ((\alpha_i, \alpha_j))$$

The first matrix is diagonal with positive entries, the second one is the Gram matrix of a positive definite bilinear form and so has positive determinant.

The last property follows because any principal subminor of A (i.e. obtained by removing the same set of rows and columns) has the same form as A . $\square$

We will represent A by a **Dynkin diagram**. The vertices correspond to simple roots. Between vertices α_i and α_j there are $a_{ij}a_{ji}$ lines (note that this value can be 0 or 1 in the simply laced diagrams, may be 2 in B_n or C_n , and may be 3 in G_2). If there are 2 or 3 lines, we put an arrow towards the short root.

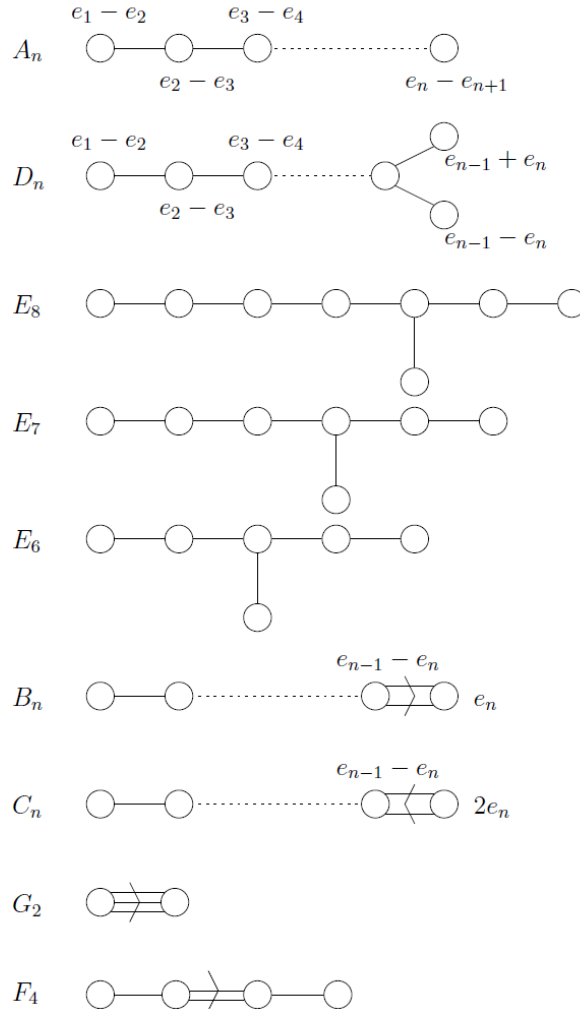


Figure 2: Dynkin diagrams (simply laced first).

If (R, V) is an irreducible root system with positive roots R^+ , then there is a unique root $\theta \in R^+$ s.t. $\forall \alpha \in \prod, \theta + \alpha \notin R^+$. This is called the **highest root**.

At the moment, the easiest way to prove this is to examine all the root systems we have. Later we'll give a uniform proof (and show the relationship between θ and the adjoint representation).

Example 31. In A_n , the highest root is $e_1 - e_{n+1}$

Define the **extended Cartan matrix** $\tilde{A}$ by setting $\alpha_0 = -\theta$ (and then $\tilde{A} = (a_{ij})_{0 \leq i, j \leq l}$ where $a_{ij} = (\alpha_i^\vee, \alpha_j) = \frac{2(\alpha_i, \alpha_j)}{(\alpha_i, \alpha_i)}$)

The associated diagram is called the extended or **affine Dynkin diagram**.

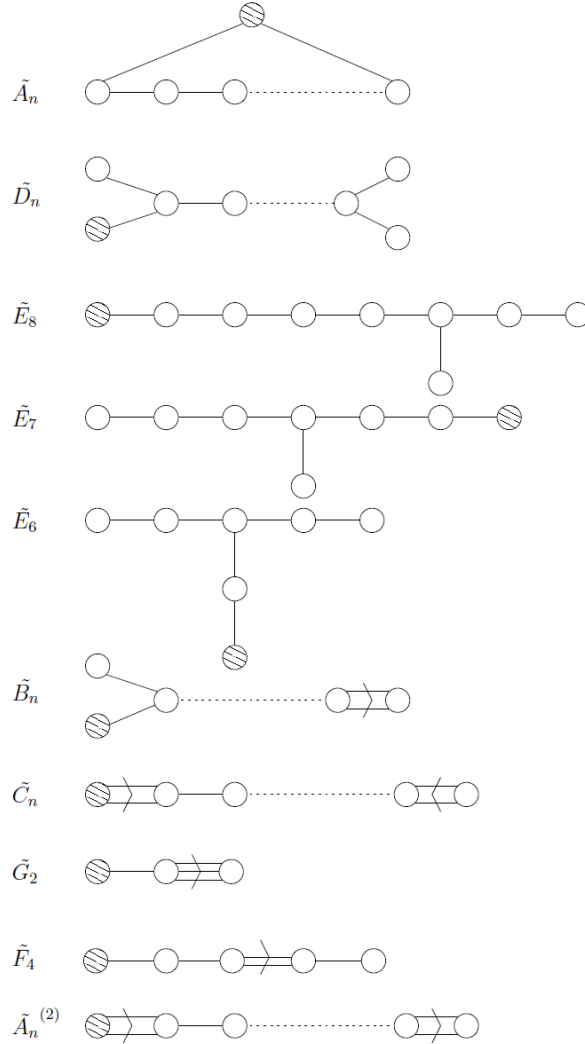


Figure 3: Affine Dynkin diagrams.

Example 32. A_1 has Dynkin diagram $A = (2)$ and affine Dynkin diagram $\tilde{A} = \begin{pmatrix} 2 & -2 \\ -2 & 2 \end{pmatrix}$

$$A_n \text{ has } A = \begin{pmatrix} 2 & -1 & & \\ -1 & 2 & \ddots & \\ & \ddots & \ddots & -1 \\ & & -1 & 2 \end{pmatrix} \text{ and } \tilde{A} = \begin{pmatrix} 2 & -1 & 0 & & -1 \\ -1 & 2 & -1 & & \\ 0 & -1 & 2 & \ddots & 0 \\ & & \ddots & \ddots & -1 \\ -1 & & 0 & -1 & 2 \end{pmatrix} \text{ Notice that } \tilde{A} \text{ satisfies}$$

almost the same conditions as the Cartan matrix, i.e. (1) $a_{ij} \in \mathbb{Z}$, $a_{ij} \leq 0$ for $i \neq j$, $a_{ii} = 2$ (2) $a_{ij} = 0$ iff $a_{ji} = 0$ (3) $\det \tilde{A} = 0$, and every principal subminor of $\tilde{A}$ has positive determinant.

The third property follows since $\tilde{A}$ is related to the Gram matrix for $\alpha_0, \dots, \alpha_l$, which are not linearly independent.

Example 33. For B_n , the highest root is $e_1 + e_2$ and so $\alpha_0 = -e_1 - e_2$. We compute that $A =$

$$\begin{pmatrix} 2 & -1 & & \\ -1 & 2 & \ddots & \\ & \ddots & \ddots & -1 \\ & & -2 & 2 \end{pmatrix} \text{ and } \tilde{A} = \begin{pmatrix} 2 & 0 & -1 & \\ 0 & 2 & -1 & \\ -1 & -1 & 2 & \ddots \\ & \ddots & \ddots & -1 \\ & & -2 & 2 \end{pmatrix}$$

For C_n , the highest root is $2e_1$ and so $\alpha_0 = -2e_1$. The Dynkin diagram is $A =$

$$\begin{pmatrix} 2 & -1 & & \\ -1 & 2 & \ddots & \\ & \ddots & \ddots & -2 \\ & & -1 & 2 \end{pmatrix}$$

and the affine Dynkin daigram $\tilde{A} =$

$$\begin{pmatrix} 2 & -1 & & \\ -2 & 2 & -1 & \\ & -1 & 2 & \ddots \\ & \ddots & \ddots & -2 \\ & & -1 & 2 \end{pmatrix}$$

For D_n , the highest root is again $e_1 + e_2$ and so $\alpha_0 = -e_1 - e_2$. The Dynkin diagram is $A =$

$$\begin{pmatrix} 2 & -1 & & \\ -1 & 2 & \ddots & -1 \\ & \ddots & \ddots & 0 \\ & -1 & 0 & 2 \end{pmatrix} \text{ and the affine Dynkin diagram is } \tilde{A} = \begin{pmatrix} 2 & 0 & -1 & \\ 0 & 2 & -1 & \\ -1 & -1 & 2 & \ddots & -1 \\ & \ddots & \ddots & 0 \\ & -1 & 0 & 2 \end{pmatrix}$$

Note that the Dynkin diagram of A^T is the Dynkin diagram of A with the arrows reversed.

These diagrams (plus their transposes) are (almost) all the non-garbage Lie algebras out there. (We'll be slightly more precise about the meaning of "non-garbage" later.)

Theorem 6.2. *An irreducible (i.e., connected) Dynkin diagram is one of $A_n, B_n, C_n, D_n, E_6, E_7, E_8, F_2$, and G_2*

Proof. We will only give a sketch of proof step by step.

(1) First, we classify the rank-2 Dynkin diagrams, i.e. their Cartan matrices: $A = \begin{pmatrix} 2 & -a \\ -b & 2 \end{pmatrix}$ with $\det A = 4 - ab > 0$. This leaves the options $(a, b) = (0, 0)(A_1 \times A_1), (1, 1)(A_2), (2, 1)$ or $(1, 2)(B_2)$, and $(3, 1)$ or $(1, 3)(G_2)$

(2) Observe any subdiagram of a Dynkin diagram is a Dynkin diagram. In particular, since the affine Dynkin diagrams have determinant 0, they are not subdiagrams of a Dynkin diagram

(3) A Dynkin diagram contains no cycles. Indeed, let α_i be simple roots and take $\alpha = \sum_i \alpha_i / (\alpha_i, \alpha_i)$. Then

$$0 < (\alpha, \alpha) = n + \sum_{i < j} \frac{2(\alpha_i, \alpha_j)}{\sqrt{(\alpha_i, \alpha_i)(\alpha_j, \alpha_j)}} = n - \sum_{i < j} \sqrt{a_{ij}a_{ji}}$$

(note that $(\alpha_i, \alpha_j) < 0$). On the other hand, if we had a cycle on n simple roots, it would have at least n edges, giving $\sum_{i < j} \sqrt{a_{ij}a_{ji}} \geq n$, contradiction.

(4) If a diagram is simply laced, it is A, D , or E :

Suppose it's not of type A . Since $\tilde{D}_4$ is not a subdiagram, any branchings are at most 3-way; and since $\tilde{D}_n$ is not a subdiagram for $n > 4$, there is exactly one branch point. Therefore, the diagram is T-shaped, with p, q , and r vertices in each leg. We call this shape $T_{p,q,r}$; e.g. $E_8 = T_{5,3,2}$.

Now using the fact that $\det T_{p,q,r} = pq + qr + rp - pqr > 0$, we conclude that D_n, E_6, E_7, E_8 are the only possibilities. (e.g. $\det E_8 = 15 + 10 + 6 - 30 = 1$)

(5) We now deal with the non-simply-laced diagrams.

If there is a triple bond in the diagram, then the Dynkin diagram is G_2 .

Finally, if there is a double bond in the diagram, then by $\tilde{C}_n$ and $\tilde{A}_n^{(2)}$ there's only one of them, and the diagram does not branch by $\tilde{B}_n$. If the double bond is in the middle, we have F_4 (since we don't have $\tilde{F}_4$); otherwise, we have B_n or C_n . $\square$

Remark 6.5. We compute the determinant of the Cartan matrix of A_n : $A_1 : \det(2) = 2$, $A_2 : \det \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} = 3$, $A_3 : \det = 4, \dots$

This comes out of the following: $SLn + 1 = \{A : \det A = 1\}$ has center consisting of $(n + 1)$ th roots of unity. Note that the order of the center is equal to the determinant of the Cartan matrix. This is true in general: **the order of the center of a simply connected algebraic group with Lie algebra $\mathfrak{g}$ and associated Cartan matrix A is equal to $\det A$.**

In particular, this means that the algebraic group attached to E_8 has no center, since $E_8 : \det = 1$.

7 Existence and Uniqueness

A. Independence of choices

Theorem 7.1. *All maximal tori are conjugate. (Recall that we are in the setting of $\text{char } k = 0$, $\bar{k} = k$.) That is, if $\mathfrak{t}$ and $\mathfrak{t}_0$ are two maximal tori of $\mathfrak{g}$, there is $g \in \text{Aut}(\mathfrak{g})^0$ s.t. $g\mathfrak{t} = \mathfrak{t}_0$. (Note that $\text{Aut}(\mathfrak{g})$ is itself an algebraic group and $\text{Aut}(\mathfrak{g})^0$ is the connected component of 1. Also $\text{Lie}(\text{Aut}(\mathfrak{g})) = \mathfrak{g}$)*

The proof is not very hard and involves a little bit of algebraic geometry; we won't give it.

Theorem 7.2. *All positive root system R^+ are conjugate. That is, let (V, R) be a root system. Let $f_1, f_2 : V \rightarrow \mathbb{R}$ satisfy $f_i(\alpha) \neq 0$ for $\alpha \in R$ and define R_1^+, R_2^+ . Then there is a unique $w \in W$ (the Weyl group) such that $wR_1^+ = R_2^+$. Consequently, $w \prod_1 = \prod_2$ and we get the same Cartan matrix out of the two choices.)*

The proof is quite easy, but we still won't give it.

Corollary 7.1. $\mathfrak{g}$ determines the Cartan matrix, i.e. the Dynkin diagram, regardless of the choices (of $f : V \rightarrow \mathbb{R}$ and the maximal torus $\mathfrak{t}$) we made.

B: Uniqueness

Theorem 7.3. *For $i = 1, 2$, let $\mathfrak{g}_i$ be semisimple Lie algebras with Cartan subalgebras $\mathfrak{t}_i$, root systems R_i , positive root systems R_i^+ , simple roots $\prod_i$ and Cartan matrices A_i . Suppose that, after reordering indices, $A_1 = A_2$. Then there exists an isomorphism of Lie algebras $\phi : \mathfrak{g}_1 \rightarrow \mathfrak{g}_2$ such that $\phi(\mathfrak{t}_1) = \mathfrak{t}_2$, $\phi(R_1) = R_2$, etc.*

We will not prove this theorem, but we will say more about it.

Example 34. Note that for D_2 and $A_1 \times A_1$, their Dynkin diagrams both consist of two isolated nodes, and so $D_2 \cong A_1 \times A_1$. Similarly, D_3 and A_3 's Dynkin diagrams are both a chain with three nodes, and so $D_3 \cong A_3$.

C: Existence

Theorem 7.4. *Let A be a Cartan matrix. There exists a semisimple Lie algebra whose Cartan matrix is A .*

Remark 7.1. We already know this for the infinite families, but a nice proof would not be case-by-case.

We will now talk about existence and uniqueness. Let $\mathfrak{g}$ be a semisimple Lie algebra with $\mathfrak{g} = \mathfrak{t} \oplus \bigoplus_{\alpha \in R} \mathfrak{g}_\alpha$ and simple roots $\Pi = \{\alpha_1, \dots, \alpha_l\}$. Choose

$$0 \neq E_i \in \mathfrak{g}_{\alpha_i}, F_i \in \mathfrak{g}_{-\alpha_i} \text{ s.t. } (E_i, F_i)_{ad} = \frac{2}{(\alpha_i, \alpha_i)}, H_i = \frac{2v^{-1}(\alpha_i)}{(\alpha_i, \alpha_i)} = v^{-1}(\alpha_i^\vee)$$

(can take this since $(\cdot, \cdot)_{ad}|_{\mathfrak{g}_\lambda \oplus \mathfrak{g}_{-\lambda}}$ is non-degenerate by theorem 5.1). Define $a_{ij} = (\alpha_i^\vee, \alpha_j) = \frac{2(\alpha_i, \alpha_j)}{(\alpha_i, \alpha_i)}$

We have the following relations:

- $[H_i, H_j] = 0$ (the torus commute)
- $[H_i, E_j] = \alpha_j(H_i)E_j = \frac{2\alpha_j(v^{-1}(\alpha_i))}{(\alpha_i, \alpha_i)}E_j = a_{ij}E_j$
- similarly, $[H_i, F_j] = -a_{ij}F_j$
- $[E_i, F_j] = 0$ for $i \neq j$ since $\alpha_i - \alpha_j \notin R$ by the property of simple roots
- $[E_i, F_i] = (E_i, F_i)v^{-1}(\alpha_i) = H_i$ by theorem 5.1 (e)

“Depending on your mood, these are either some of the relations or all of the relations”. We will see what this means.

Let $\mathfrak{n}^+ = \bigoplus_{\alpha \in R^+} \mathfrak{g}_\alpha$, $\mathfrak{n}^- = \bigoplus_{\alpha \in R^+} \mathfrak{g}_{-\alpha}$. Then $\mathfrak{g} = \mathfrak{t} + \mathfrak{n}^+ + \mathfrak{n}^-$ (note that $\mathfrak{n}^+$ and $\mathfrak{n}^-$ are not ideals)

Lemma 7.1. E_i generate $\mathfrak{n}^+$, F_i generate $\mathfrak{n}^-$. Hence E_i, F_j generate $\mathfrak{g}$ as Lie algebras.

Proof. For a root $\alpha \in R^+$, write $\alpha = \sum k_i \alpha_i$ where $k_i \in \mathbb{Z}_{\geq 0}$ and define the height of α to be $h(\alpha) = \sum k_i$. Then induct on the height of α . First, since $\mathfrak{g}_{\alpha_i}$ is one-dimensional, it is generated by E_i . If $\alpha \in R^+$ is not simple, by definition there is α_i such that $\alpha - \alpha_i = \beta \in R^+$. By inductive hypothesis, $\mathfrak{g}_\beta$ is generated by E_j . But then $\mathfrak{g}_\alpha = [\mathfrak{g}_\beta, \mathfrak{g}_{\alpha_i}]$ is also generated by E_j

The corresponding claim about F_i follows immediately by taking $-R^+$ as the positive roots.

Finally, $[E_i, F_i] = H_i$, and $H_i = cv^{-1}(\alpha_i)$ generate $\mathfrak{t}$ since the simple roots α_i generate $\mathfrak{t}^*$ and v is an isomorphism. $\square$

Let A be a “generalized Cartan matrix”, i.e, $a_{ii} = 2$, $a_{ij} = 0$ iff $a_{ji} = 0$ and $a_{ij} \in \mathbb{Z}_{\leq 0}$ for $i \neq j$ (but we do not include requirements of positive definiteness). Let

$$\tilde{\mathfrak{g}} = \langle E_i, F_i, H_i \mid [H_i, H_j] = 0, [H_i, E_j] = a_{ij}E_j, [H_i, F_j] = -a_{ij}F_j, [E_i, F_j] = 0 \text{ for } i \neq j \text{ and } [E_i, F_i] = H_i \rangle$$

Clearly, if A is the Cartan matrix of $\mathfrak{g}$, then $\tilde{\mathfrak{g}} \twoheadrightarrow \mathfrak{g}$. Are these all the relations? Not quite.

Let $\tilde{\mathfrak{g}} = \tilde{\mathfrak{g}} / \langle (ad_{E_i})^{1-a_{ij}} E_j, (ad_{F_i})^{1-a_{ij}} F_j \rangle$ (e.g. if $a_{ij} = 0$ then E_i, E_j commute, if $a_{ij} = -1$ then $[E_i, E_j] = 0$). These are called the “Serre relations”

Remark 7.2. They are called the Serre relations because they were discovered by Harish-Chandra and Chevalley.

Example 35. For $\mathfrak{sl}_n$, $\mathfrak{g}_{\epsilon_i - \epsilon_j} = \mathbb{C}E_{ij}$. Note that $[E_{i,i+1}[E_{i,i+1}, E_{i+1,i+2}]] = [E_{i,i+1}, E_{i,i+2}] = 0$ and $a_{i,i+1} = -1$

Theorem 7.5. 1. If A is indecomposable, $\tilde{\mathfrak{g}}$ has a unique maximal ideal, and $\bar{\mathfrak{g}}$ is the quotient of $\tilde{\mathfrak{g}}$ by its maximal ideal, so is simple. (But not necessarily finite dimensional!)

2. Hence, if $\mathfrak{g}$ is a finite-dimensional semisimple Lie algebra with Cartan matrix A , then the map $\tilde{\mathfrak{g}} \rightarrow \mathfrak{g}$ via $E_i \mapsto E_i, F_i \mapsto F_i, H_i \mapsto H_i$ (where on the left-hand side we have the canonical generators of $\tilde{\mathfrak{g}}$ and on the right hand side we have the elements of the root spaces we found above) factors through $\bar{\mathfrak{g}}$ and is surjective, so gives an isomorphism $\bar{\mathfrak{g}} \cong \mathfrak{g}$.

The second part of the theorem clearly implies the uniqueness. We also see that existence is equivalent to the statement

$$\bar{\mathfrak{g}} \text{ finite dimensional} \iff A \text{ is a Cartan matrix}$$

Definition 7.1. $\bar{\mathfrak{g}}$ is called a **Kac-Moody algebra** (this time because they were independently discovered by Kac and Moody).

Most of the representation theory we will be talking about holds verbatim for the Kac-Moody algebras. However, note that all we have given for them is a finite presentation, which is not a very nice object. In particular, the root spaces of these algebras are not 1-dimensional. The dimension of the root space for any given root is computable but horrible, unless the matrix A is an affine Dynkin diagram. In that case, the Lie algebra is approximately $\mathbb{C}c + \mathfrak{g} \otimes \mathbb{C}[t, t^{-1}]$

The Weyl group has a corresponding presentation:

Theorem 7.6. (Presentation of Weyl group) Write $s_i = s_{\alpha_i}$, then

$$W = \langle s_i | (s_i)^2 = 1, (s_i s_j)^{m_{ij}} = 1 \rangle$$

where m_{ij} are defined in terms of the Cartan matrix as follows:

$$a_{ij}a_{ji} \quad 0 \ 1 \ 2 \ 3$$

$$m_{ij} \quad 2 \ 3 \ 4 \ 6$$

E.g., if i and j are not connected in the Dynkin diagram then $s_i s_j = s_j s_i$, and if they are connected by a single edge, then $s_i s_j s_i = s_j s_i s_j$. These are known as the **braid relations**. In A_{n-1} they look like this:

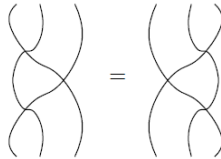


Figure 4: Braid relations in A_{n-1} . We are asserting that the permutations are equal.

We will not prove the Weyl group of each root system is isomorphic to W in theorem 7.6 but let's look at some examples

Example 36. In A_2 with simple roots α, β , $s_\alpha s_\beta$ is rotation anticlockwise by $2/3\pi$, and so $m = 2$, meanwhile $a_{12}a_{21} = 1$

In B_2 , $s_\alpha s_\beta$ is rotation anticlockwise by $\pi/2$ and so $m = 4$, meanwhile $a_{12}a_{21} = 2$

In G_2 , $s_\alpha s_\beta$ is rotation anticlockwise by $\pi/3$ and so $m = 6$, meanwhile $a_{12}a_{21} = 3$

Remark 7.3. To show existence, we know that $\mathfrak{g} = \mathfrak{t} \oplus \bigoplus_{\alpha \in R} \mathfrak{g}_\alpha$, so in principle existence should only require picking a basis E_α and computing $[E_\alpha, E_\beta] = c_{\alpha\beta} E_{\alpha+\beta}$. How hard is it to write down $c_{\alpha\beta}$?

For a long time it was thought to be quite hard; however, with the discovery of the affine Lie algebras, it turned out to be quite easy. We might even get to doing it for the ADE algebras.

Remark 7.4. So far the status of existence of a finite-dimensional simple Lie algebra corresponding to an abstract root system for us is as follows: we have shown that there exists a simple Lie algebra corresponding to each generalised Cartan matrix, but not that it is finite-dimensional. Suppose that for the $\mathfrak{g}$ that we have defined we write $\bar{\mathfrak{g}} = \mathfrak{t} \oplus \bigoplus_{\alpha \in R} \bar{\mathfrak{g}}_\alpha$ (where the torus comes from the root system). We can ask two questions: (1) what is the relationship between R and the root system we started with?; (2) What is the dimension of $\mathfrak{g}_\alpha$? While it is true that $W \cdot \prod \subset R$, in general there is not an equality between them. We call the roots in $W \cdot \prod$ “real roots” R_{re} ; for the real roots, $\dim \mathfrak{g}_\alpha = 1$. For finite-dimensional Lie algebras, all roots are real. If A is an affine Cartan matrix, then the semidefinite nature of it means that there exists a unique (up to scaling) root of length 0, called the imaginary root. The imaginary roots are pleasant and describable objects. For a general, indefinite, Cartan matrix, there exists an algorithm for computing $\dim \mathfrak{g}_\alpha$, but the numbers are not very meaningful.

8 Representations

Let $\mathfrak{g}$ be a semisimple Lie algebra. $\mathfrak{g} = \mathfrak{t} \oplus \bigoplus_{\alpha \in R} \mathfrak{g}_\alpha = \mathfrak{g} \oplus \mathfrak{n}^+ \oplus \mathfrak{n}^-$. Let V be a finite-dimensional representation of $\mathfrak{g}$. (We don’t quite need finite-dimensional; we’ll see exactly what we need a bit later.)

Proposition 8.1. (1) $V = \bigoplus_{\lambda \in \mathfrak{t}^*} V_\lambda$ where $V_\lambda = \{v \in V | t(v) = \lambda(t)v \ \forall t \in \mathfrak{t}\}$ (this is the weight space decomposition with respect to $\mathfrak{t}$)

(2) If $V_\lambda \neq 0$, then $\lambda(h_\alpha) = (\lambda, \alpha^\vee) \in \mathbb{Z}$ for all $\alpha \in R$ (Recall that $h_\alpha = v^{-1}(\alpha^\vee)$ is an element of $(\mathfrak{sl}_2)_\alpha$)

Proof. Since V is a finite-dimensional representation of $\mathfrak{g}$, it is a finite-dimensional representation of $(\mathfrak{sl}_2)_\alpha$. Therefore, h_α acts diagonalisably on V with $\lambda(h_\alpha) \in \mathbb{Z}$. The first assertion follows because h_α span $\mathfrak{t}$. $\square$

Definition 8.1. $Q = \mathbb{Z}R = \bigoplus_{\alpha_i \in \prod} \mathbb{Z}\alpha_i$ is the **lattice of roots**, $P = \{\gamma \in \mathbb{Q}R | (\gamma, \alpha^\vee) \in \mathbb{Z} \ \forall \alpha \in R\}$ (the dual lattice) is the **lattice of weights**. Note that the condition can be checked by computing $(\gamma, \alpha_i^\vee)$ for $\alpha_i \in \prod$

Since, for $\alpha, \beta \in R$ $(\alpha, \beta^\vee) \in \mathbb{Z}$, we have $Q \in P$. Also, if V is a finite-dimensional representation of $\mathfrak{g}$ and $V_\lambda \neq 0$, then $\lambda \in P$ since $(\lambda, \alpha^\vee) \in \mathbb{Z}$

Remark 8.1. (1) $|P/Q| < \infty$. In fact, $|P/Q| = \det A$ where A is the Cartan matrix.

(2) $W(P) \subset P$, where P is the Weyl group

Example 37. In $\mathfrak{sl}_2$, $R = \pm\alpha$, $Q = \mathbb{Z}\alpha$ with $(\alpha, \alpha) = 2$. So $P = \mathbb{Z}\frac{\alpha}{2}$ and $|P/Q| = 2 = \det A$

Definition 8.2. For V a finite-dimensional representation of $\mathfrak{g}$, define the **Weyl character**

$$chV = \sum_{\lambda \in R} \dim V_\lambda \cdot e^\lambda \in \mathbb{Z}[P]$$

Here e^λ is a formal symbol ($e^\lambda e^\mu = e^{\lambda+\mu}$) and form a basis for $\mathbb{Z}[P]$

Example 38. In $\mathfrak{sl}_2$, $P = \mathbb{Z}\frac{\alpha}{2}$. Write $e^{\frac{\alpha}{2}} = z$. Then $chL(n) = z^n + \dots + z^{-n} = \frac{z^{n+1} - z^{-n-1}}{z - z^{-1}}$. (Note that $\frac{\alpha}{2} = h^*$).

Example 39. Let $\mathfrak{g} = \mathfrak{sl}_3$ and V the adjoint representation. Then

$$chV = 2 + z + w + z^{-1} + w^{-1} + zw + (zw)^{-1}$$

where $z = e^{\alpha_1}, w = e^{\alpha_2}$

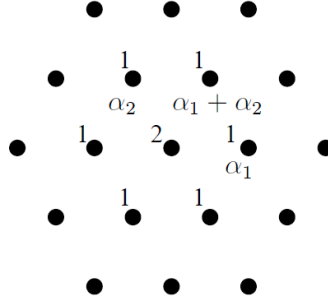


Figure 5: Root lattice for $\mathfrak{sl}_3$, with dimensions of weight spaces. Note that the outer numbers are 1, and the picture is W -invariant ($W = S_3$ here).

Proposition 8.2. *If V is a finite-dimensional representation of $\mathfrak{g}$, then $\dim V_\lambda = \dim V_{w\lambda}$ for all $w \in W$, i.e. chV is W -invariant, $chV \in \mathbb{Z}[P]^W$. (In particular, for $\mathfrak{sl}_n$ this means that chV are symmetric functions.)*

We will sketch three proofs, of which the third one is the easiest, but the first two reveal what is actually going on.

Sketch 1. If G is an algebraic group with Lie algebra $\mathfrak{g}$ then (by a theorem) $W = N(T)/T$ (where T is the maximal torus in G whose Lie algebra is $\mathfrak{t}$, and $N(T)$ is its normaliser).

Example 40. If $G = SL_n$ with T the diagonal matrices, then $N(T)$ are matrices with only one nonzero entry in each row and column, and $N(T)/T = S_n$ the permutation matrices.

That is, there is $\bar{w} \in N(T)$ s.t. $\bar{w}T = w$

If G acts on V (as it always does when G is simply connected), then $\bar{w}(V_\lambda) = V_{w\lambda}$, since

$$t\bar{w}v = \bar{w}(\bar{w}^{-1}t\bar{w})v = \bar{w}(\lambda(\bar{w}^{-1}t\bar{w})v) = w\lambda(t)\bar{w}v$$

That is, W is almost a subgroup of G . However, it is definitely not actually a subgroup. E.g. in $\mathfrak{sl}_2$, take $\bar{s} = \begin{pmatrix} & -1 \\ 1 & \end{pmatrix}$. Then $\bar{s}^2 = \begin{pmatrix} -1 & \\ & -1 \end{pmatrix} \in T$ is not equal to 1. That is, W does not embed into G and does not act on V . (There is a small 2-group, $\approx (\mathbb{Z}/2)^l$, that intervenes. This 2-group is also involved in determining the coefficients $c_{\alpha\beta}$ $\square$)

Sketch 2. We can mimic the above argument in $\mathfrak{g}$. Notice that

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 1 & -1 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix} = \exp(f)\exp(-e)\exp(f)$$

where the matrix exponential is defined in the usual power series way. Therefore, for each root α we define $\dot{s}_\alpha = \exp(f_\alpha)\exp(-e_\alpha)\exp(f_\alpha)$

Remark 8.2. (1) If V is a finite-dimensional representation of $\mathfrak{g}$, then e_α, f_α act nilpotently on V , so $\dot{s}_\alpha : V \rightarrow V$ is a well-defined-finite sum.

(2) $\dot{s}_\alpha^2 = \epsilon_\alpha : V_\lambda \rightarrow V_\lambda$, where $\epsilon_\alpha^2 = 1$ and ϵ_α acts by multiplication by $(-1)^m$ on V_m .

(3) $\dot{s}_\alpha V_\lambda = V_{s_\alpha \lambda}$

We did not really need V to be finite-dimensional; what we need instead is for each e_α and f_α to act **locally nilpotently**. We say $x : V \rightarrow V$ acts locally nilpotently if for all $v \in V$ we have $x^n v = 0$ for some n . This is equivalent to saying that as an $\mathfrak{sl}_2$ -module, V splits into a (possibly infinite) direct sum of finite-dimensional $\mathfrak{sl}_2$ -modules. Such a V is called **integrable** (because you can “integrate” the Lie algebra action to a group action).

In the rest of the course, you can replace all instances of “finite-dimensional” by “integrable”, and then all theorems and proofs will work for the Kac-Moody algebras as well. $\square$

Sketch 3. If we’re just working with $\mathfrak{sl}_2$ anyway, we shouldn’t need all the other stuff.

Consider V as a representation of $(\mathfrak{sl}_2)_\alpha \times \mathfrak{t}$. Then V splits up into a direct sum of strings, each of which is of the form

$$\lambda \quad \lambda - \alpha \quad \dots \quad \lambda - m\alpha$$

where $m = \lambda(h_\alpha) = (\lambda, \alpha^\vee)$ and so $s_\alpha(\lambda) = \lambda - m\alpha$. So s_α will just flip the string from left to right (as the reflection did in $\mathfrak{sl}_2$) and it is s_α -invariant $\square$

Definition 8.3. For $\lambda, \mu \in \mathfrak{t}^*$, write $\mu \leq \lambda$ if $\lambda - \mu = \sum k_i \alpha_i$ with $k_i \in \mathbb{Z}_{\geq 0}$. The points $\{\mu \in P \mid \mu \leq \lambda\}$ are the lattice points in an obtuse cone.

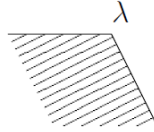


Figure 6: Cone $\leq \lambda$ for $\mathfrak{sl}_3$

Definition 8.4. If V is a representation of $\mathfrak{g}$, we say that $\lambda \in P$ is a **highest root** if $V_\lambda \neq 0$, and $V_\mu \neq 0 \implies \mu \leq \lambda$

We say $v \in V_\lambda$ is a **singular vector** if $v \neq 0$ and $e_\alpha v = 0$ for all $\alpha \in R^+$. (Note that $e_\alpha v \in V_{\alpha+\lambda}$, so if λ is a highest root, then all nonzero vectors of V_λ are singular)

We say that λ is an **extremal weight** if $w\lambda$ is a highest weight for some $w \in W$.

Example 41. In the adjoint reps. of $\mathfrak{sl}_3$, the highest weight is $\alpha_1 + \alpha_2$ (see figure 5), and the entire outer hexagon are extremal weights.

Theorem 8.1. $\mathfrak{g}$ a semisimple Lie algebra

1. “Complete reducibility”: if V is a finite-dimensional representation of $\mathfrak{g}$, then V is a direct sum of irreducible representations of $\mathfrak{g}$.

Set $P^+ = \{\lambda \in P \mid (\lambda, \alpha^\vee) > 0 \ \forall \alpha \in R^+\}$ (of course, it’s enough to check this on $\alpha \in \Pi$). This is the cone of dominant (psitive) weights. The assertion of the next parts of the theorem is that P^+ parametrises the finite-dimensional irreducible representations of $\mathfrak{g}$, with $\lambda \mapsto L(\lambda)$. More precisely,

2. Let V be a finite-dimensional irreducible representation of $\mathfrak{g}$, and $v \in V_\lambda$ a singular vector. Then

(a) $\dim V_\lambda = 1$

(b) $V_\mu \neq 0 \implies \mu \leq \lambda$, i.e. λ is a highest weight (and v a hw-vector)

(c) $(\lambda, \alpha_i^\vee) \in \mathbb{N}$ for all i , i.e. $\lambda \in P^+$

Moreover, if W is another finite-dimensional irreducible representation of $\mathfrak{g}$ with highest weight λ , and $w \in W_\lambda$, $w \neq 0$, then there exists a unique isomorphism $V \cong W$ mapping $v \mapsto w$. (Of course, given they are irreducible representations, if there is an isomorphism, there is a one-dimensional family of them, so we would pin down the exact one by sending $v \mapsto w$. That is, uniqueness is not the nontrivial statement here.)

3. Given $\lambda \in P^+$, there exists a finite-dimensional irreducible representation with highest weight λ , called $L(\lambda)$.

4. A formula for $\text{ch}L(\lambda)$ (the Weyl character formula), to be stated later.

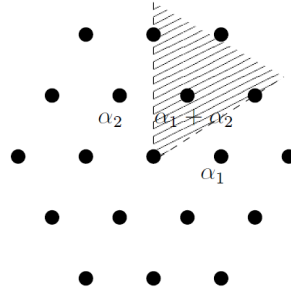


Figure 7: Cone of positive weights for $\mathfrak{sl}_3$. The adjoint representation's highest weight is $\alpha_1 + \alpha_2$.

Remark 8.3. For Kac-Moody algebras, we need a slight modification, since it is no longer obvious that we will have highest-weight vectors. We need to work in the category of integrable highest-weight representations instead.

We will prove this theorem later, but let's see some applications first.

Corollary 8.1. $\text{ch}L(\lambda) = e^\lambda + \sum_{\mu < \lambda} a_\mu e^\mu$ (a_μ are to be determined) hence $\{\text{ch}L(\lambda) | \lambda \in P^+\}$ are linearly independent (they have different leading terms). On the other hand,

$$\text{ch}L(\lambda) = m_\lambda + \sum_{\mu < \lambda} a_{\mu\lambda} m_\mu \in \mathbb{Z}[P]^W$$

where $m_\mu = \sum_{\gamma \in W\mu} e^\gamma$. Since m_λ form a basis of $\mathbb{Z}[P]^W$, $\text{ch}L(\lambda)$ are a basis of $\mathbb{Z}[P]^W$ as well.

Corollary 8.2. If V, W are f.d. representations of $\mathfrak{g}$, then $V \cong W \iff \text{ch}V = \text{ch}W$

Proof. Complete reducibility + characters of irreducibles are a basis (the same as for $\mathfrak{sl}_2$) □

Definition 8.5. Define $w_i \in P$ the dual basis to the simple roots w.r.t. the inner product, i.e. $(w_i, \alpha_j^\vee) = \delta_{ij}$. These are called the **fundamental weights**.

Then $P^+ = \oplus_i \mathbb{Z}_{\geq 0} w_i = \{\sum n_i w_i | n_i \in \mathbb{Z}_{\geq 0}\}$. This is a **fundamental domain** for the action of W . Each irreducible representation will have a unique highest weight lying in this cone, which is what parametrises the representation.

Example 42. For $\mathfrak{sl}_3$, we compute that $w_1 = \frac{1}{3}(2e_1 - e_2 - e_3) = \frac{1}{3}(2\alpha_1 + \alpha_2)$ and $w_2 = \frac{1}{3}(e_1 + e_2 - 2e_3) = \frac{1}{3}(\alpha_1 + 2\alpha_2)$; For $\mathfrak{sl}_n$, $w_k = e_1 + \dots + e_k - \frac{k}{n}(e_1 + \dots + e_n)$ for $1 \leq k \leq n-1$

Remark 8.4. For $\lambda \in P$, note that $\lambda = \sum \lambda(h_i)w_i$ where $h_i = \mu^{-1}(\alpha_i^\vee)$ since $\lambda(h_i) = (\lambda, \alpha_i^\vee)$

Remark 8.5. For any $\mathfrak{g}$, $L(0)$ is the trivial representation.

For any simple $\mathfrak{g}$, the adjoint representation $V = \mathfrak{g}$ is irreducible. The decomposition $\mathfrak{g} = \mathfrak{t} \oplus \bigoplus_{\alpha \in R} \mathfrak{g}_\alpha$ means that a highest weight λ is a root such that $\lambda + \alpha_i \notin R$ for all i , i.e. $\lambda = \theta$, the highest root in R^+ . (Therefore, the theorem of the highest weight implies that θ is unique, as promised.)

Example 43. Recall in A_{n-1} the highest root is $\theta = e_1 - e_n$, the simple roots are $\alpha_i = e_i - e_{i+1}$ and $h_i = v^{-1}(\alpha_i^\vee)$ is a diagonal matrix with $+1$ in the i th position, -1 in the $(i+1)$ th position, and 0 elsewhere). Have $\theta(h_1) = 1, \theta(h_{n-1}) = 1$, and $\theta(h_i) = 0$ else. We can label the Dynkin diagram by $\theta(h_i)$:

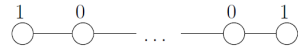


Figure 8: Adjoint representation of $\mathfrak{sl}_n$; labels are $\theta(h_i)$, giving a decomposition of θ into fundamental weights.

For B_n , the highest root is $e_1 + e_2$, and h_i is a diagonal matrix with $+1$ in the i th position, -1 in the $(i+1)$ th position, and 0 elsewhere for $1 \leq i \leq n-1$ and $h_n = 2E_{nn}$. So $\theta(h_2) = 1, \theta(h_i) = 0$ else. This can be read from the affine Dynkin diagram as well

Example 44. Other representations of $\mathfrak{sl}_n$: we have the standard (irr.) representation, $\mathbb{C}^n$, with weight basis $v_1, \dots, v_n$ and weights $e_1, \dots, e_n$ (note that $\sum e_i = 0$ on $\mathfrak{t}$). And e_1 is the highest root since $e_1 > e_2 > \dots > e_n$ as $e_i - e_{i+1}$ are simple roots. Therefore $\mathbb{C}^n = L(e_1)$ and so

$$ch\mathbb{C}^n = e^{e_1} + \dots + e^{e_n} = z_1 + \dots + z_n$$

where $z_i = e^{e_i} \in \mathbb{Z}[P] = \mathbb{Z}[z_1^\pm, \dots, z_n^\pm]/(z_1 \cdots z_n = 1)$

Now, if V and W are representations of $\mathfrak{g}$, then so is $V \otimes W$ (via $x \in \mathfrak{g}$ acts by $x \otimes 1 + 1 \otimes x$). In particular, $V \otimes V$ is a representation. Note that $\sigma : V \otimes V \rightarrow V \otimes V$, $a \otimes b \mapsto b \otimes a$, commutes with the $\mathfrak{g}$ -action, so eigenspaces of σ are $\mathfrak{g}$ -modules. That is, S^2V and $\wedge^2V$ are $\mathfrak{g}$ -modules. In general, these may not be irreducible, but they are for $\mathfrak{sl}_n$ (provided V is irreducible, of course!), as we check below:

Consider $\wedge^s \mathbb{C}^n$ for $s \leq n-1$. This has weight basis $v_{i_1} \wedge \dots \wedge v_{i_s}$ for $i_1 < \dots < i_s$, and the weight of this is $e_{i_1} + \dots + e_{i_s}$ as is clear from the action. Therefore,

$$E_i(v_{i_1} \wedge \dots \wedge v_{i_s}) = 0 \text{ for all } i \iff v_{i_1} \wedge \dots \wedge v_{i_s} = v_1 \wedge \dots \wedge v_s$$

where $E_i = E_{i, i+1}$ is the simple root vector of $\alpha_i = e_i - e_{i+1}$ and hence $E_i(v_j) = \begin{cases} v_i & \text{if } j = i+1 \\ 0 & \text{else} \end{cases}$
 $(E_1(v_{i_1} \wedge \dots \wedge v_{i_s}) = 0 \text{ for all } i \text{ forces } i_j - 1 \in \{i_1, \dots, i_s\} \text{ for all } j)$. So $v_1 \wedge \dots \wedge v_s$ is the only singular vector. Note its weight is $w_s = e_1 + \dots + e_s$ and is the s -th fundamental weight as $(w_s, e_i - e_{i+1}) = \delta_{si}$. Therefore, $\wedge^s \mathbb{C}^n = L(w_s)$ is a fundamental representation. In particular $\wedge^{n-1} \mathbb{C}^n \cong (\mathbb{C}^n)^* = L(w_{n-1})$

Consider the symmetric algebra $S^m \mathbb{C}^n$. It has weight basis $v_{i_1} \dots v_{i_m}$ with $i_1 \leq \dots \leq i_s$ with weight $w = e_{i_1} + \dots + e_{i_m}$. Similarly, we notice that

$$E_i(v_{i_1} \dots v_{i_m}) = 0 \text{ for all } i \iff v_{i_1} \dots v_{i_m} = v_1^m$$

i.e. $S^m \mathbb{C}^n$ is an irr.representation isomorphic to $L(me_1)$

For $\wedge^s \mathbb{C}^n$ we have seen the weights are $e_{i_1} + \dots + e_{i_s}$ where $1 \leq i_1 < \dots < i_s \leq n$. Hence $ch(\wedge^s \mathbb{C}^n) = \sum_{1 \leq i_1 < \dots < i_s \leq n} e^{e_{i_1} + \dots + e_{i_s}}$. Then generating function is $\sum_{s \geq 0} ch(\wedge^s \mathbb{C}^n) q^s = \prod_{i=1}^n (1 + z_i q)$ where $z_i = e^{e^i}$

Similarly, the weights for $S^m \mathbb{C}^n$ are $e_{i_1} + \dots + e_{i_m}$ where $1 \leq i_1 \leq \dots \leq i_m \leq n$ and so $ch(S^m \mathbb{C}^n) = \sum_{1 \leq i_1 \leq \dots \leq i_m \leq n} e^{e_{i_1} + \dots + e_{i_m}}$. Its generating function is $\sum_{m \geq 0} (ch S^m \mathbb{C}^n) q^m = \prod_{i=1}^n \frac{1}{1 - z_i q}$ where $z_i = e^{e^i}$

Remark 8.6. If V, W are $\mathfrak{g}$ -modules, then $V^* \otimes W \cong Hom(V, W)$ as $\mathfrak{g}$ -modules via $v^* \otimes w \mapsto (v \mapsto v^*(v)w)$. Note that if V is a $\mathfrak{g}$ -module, then V^* is a $\mathfrak{g}$ -module with the action $x\lambda(v) = -\lambda(xv)$. The $-$ sign comes from differentiating the group action $(x \cdot g)(v) = g^{-1}(xv)$

Now take $V = \mathbb{C}^n$ and $\mathfrak{g} = \mathfrak{sl}_n$. Then have $V^* \otimes V \cong Hom(V, V) = Mat_n(\mathbb{C}) \cong \mathfrak{sl}_n \oplus \mathbb{C}$, adjoint + trivial representations (where the trivial representation comes from $Id \in Hom_{\mathfrak{g}}(V, V)$). In contrast $V \otimes V = S^2 V \oplus \wedge^2 V$ always

Example 45. For $\mathfrak{so}_n$ with $n = 2l$ or $2l + 1$, the Cartan subalgebra consists of matrix of the form $diag\{x_1, -x_1, \dots\}$ with 0 at the end if n is odd. For the standard representation $V = \mathbb{C}^n$, it follows that the weights are $\pm e_1, \dots, \pm e_l$ if n is even and $\pm e_1, \dots, \pm e_l, 0$, and the highest weights are again e_1 . So $\mathbb{C}^n = L(e_1)$ and we have $ch(\mathbb{C}^n) = \sum (e^{e_i} + e^{-e_i})$ if n is even and $ch(\mathbb{C}^n) = 1 + \sum (e^{e_i} + e^{-e_i})$ if n is odd.

$V \cong V^*$ via the form defining $\mathfrak{g}$ ($x \mapsto (x, -)$). We now decompose the $\mathfrak{so}_n$ -module $V \otimes V$. First we have $V \otimes V = S^2 V \oplus \wedge^2 V$. Note that there is an isomorphism $\wedge^2 V \cong \mathfrak{so}_n$ via $u \wedge v \mapsto (x \mapsto (u, x)v - (v, x)u)$. So the $\mathfrak{so}_n$ -module $\wedge^2 V$ is isomorphic to the adjoint representation, and so the highest weight is precisely the highest root $e_1 + e_2 = w_2$ and so $\wedge^2 V = L(w_2)$. On the other hand, we have $S^2 V = S_0^2 V \oplus \mathbb{C}$ where $S_0^2 V$ is traceless symmetric tensors and $\mathbb{C}$ is the trivial representation coming from the identity. Using example 44, we get $S^2 V \cong L(2w_1) \oplus \mathbb{C}$, and so $V \otimes V \cong L(w_2) \oplus L(2w_1) \oplus \mathbb{C}$

We will now devote ourselves to proving Theorem 8.3 classifying all the representations of simple Lie algebras

Let $\mathfrak{g}$ be any Lie algebra with a non-degenerate invariant bilinear form $(,)$, e.g. $\mathfrak{g}$ semisimple and $(,) = (,)_{ad}$. Let $x_1, \dots, x_N$ be a basis of $\mathfrak{g}$ and $x^1, \dots, x^N$ a dual basis (i.e. $(x_i, x^j) = \delta_{ij}$)

Definition 8.6. $\Omega = \sum x_i x^i$ is the *Casimir* of $\mathfrak{g}$.

Remark 8.7. For the moment, Ω is an operator on any representation of $\mathfrak{g}$. However, we will in a moment define the “universal enveloping algebra of $\mathfrak{g}$ ”, and that is where Ω naturally lives.

Lemma 8.1. For $x \in \mathfrak{g}$, have $[x, \Omega] = 0$

Proof 1, coordinate-based. $[\Omega, x] = [\sum x_i x^i, x]$. Since ad_x is a derivation, we can rewrite it as $\sum [x_i, x] x^i + \sum x_i [x^i, x]$. Write $[x^i, x] = \sum a_{ij} x^j$ and $[x_i, x] = \sum b_{ij} x_j$. Then

$$a_{ij} = ([x^i, x], x_j) = (x^i, [x, x_j]) = -([x_j, x], x^i) = -b_{ji}$$

$$\text{Hence } [\Omega, x] = \sum [x_i, x] x^i + \sum x_i [x^i, x] = \sum b_{ij} x_j x^i + \sum x_i a_{ij} x^j = \sum (b_{ji} + a_{ij}) x_i x^j = 0 \quad \square$$

Proof 2, without coordinates. We have maps of $\mathfrak{g}$ -modules $\mathbb{C} \hookrightarrow End(\mathfrak{g}) \cong \mathfrak{g}^* \otimes \mathfrak{g} \cong \mathfrak{g} \otimes \mathfrak{g}$ (the first map is via $\lambda \mapsto \lambda Id$, i.e. $1 \mapsto \sum x_i \otimes x^i$). $\mathfrak{g}$ acts on V gives $\mathfrak{g} \rightarrow End(V)$. Finally, we have a map of $\mathfrak{g}$ -modules $End(V) \otimes End(V) \rightarrow End(V)$ by multiplication. Together they give

$$\mathbb{C} \hookrightarrow End(\mathfrak{g}) \cong \mathfrak{g}^* \otimes \mathfrak{g} \cong \mathfrak{g} \otimes \mathfrak{g} \rightarrow End(V) \otimes End(V) \rightarrow End(V)$$

sending $1 \mapsto \Omega$. Therefore, ω generates a trivial $\mathfrak{g}$ -submodule of $End(V)$, which is equivalent to saying $[\Omega, x] = 0$ for all $x \in \mathfrak{g}$. $\square$

Now let $\mathfrak{g}$ be semisimple and $\mathfrak{g} = \mathfrak{t} \oplus \bigoplus_{\alpha \in R} \mathfrak{g}_\alpha$ and $(\cdot, \cdot)_{ad}$ the Killing form. Choose a basis $u_1, \dots, u_l$ for $\mathfrak{t}$ and x_α of $\mathfrak{g}_\alpha$. Choose a dual basis $u^1, \dots, u^l$ for $\mathfrak{t}$ and $x^{-\alpha} \in \mathfrak{g}_{-\alpha}$ so that $(x_\alpha, x^{-\alpha}) = 1$ ($x^{-\alpha} \in \mathfrak{g}_{-\alpha}$ is by theorem 5.1 (b)). Normalize x_α so that $(x_\alpha, x_{-\alpha}) = 1$, i.e. $x^{-\alpha} = x_{-\alpha}$. Then $[x_\alpha, x_{-\alpha}] = (x_\alpha, x_{-\alpha})_{ad} v^{-1}(\alpha) = v^{-1}(\alpha)$ by theorem 5.1 (e). Now

$$\Omega = \sum u_i u^i + \sum_{\alpha \in R^+} (x_\alpha x_{-\alpha} + x_{-\alpha} x_\alpha) = \sum u_i u^i + 2 \sum_{\alpha \in R^+} x_{-\alpha} x_\alpha + \sum_{\alpha \in R^+} v^{-1}(\alpha)$$

Definition 8.7. $\rho = \frac{1}{2} \sum_{\alpha \in R^+} \alpha$

Remark 8.8. Later we will see what ρ actually is. If you like algebraic geometry, this is the square root of the canonical bundle on the flag variety.

$$\text{Then } \Omega = \sum u_i u^i + 2 \sum_{\alpha \in R^+} x_{-\alpha} x_\alpha + 2v^{-1}(\rho)$$

Lemma 8.2. *Let V be a $\mathfrak{g}$ -module, $v \in V$ a singular vector with weight λ (i.e. $\mathfrak{n}^+ v = 0$ and $tv = \lambda(t)v$). Then $\Omega v = (|\lambda + \rho|^2 - |\rho|^2)v$*

Proof. Apply the most recent version of ω :

$$\Omega v = \left(\sum \lambda(u_i) \lambda(u^i) + \lambda(2v^{-1}(\rho)) \right) v + 2 \sum_{\alpha \in R^+} x_{-\alpha} x_\alpha v = (\lambda, \lambda) v + 2(\lambda, \rho) v$$

since v is a singular vector. (note that $v^{-1}(\lambda) = \sum \lambda(u^i) u_i = \sum \lambda(u_i) u^i$ and so $(\lambda, \lambda) = (v^{-1}(\lambda), v^{-1}(\lambda))_{ad} = \sum \lambda(u_i) \lambda(u^i)$) $\square$

In particular, if V is irreducible, Ω acts on V by $(\lambda, \lambda) + 2(\lambda, \rho)$ by Schur's lemma.

9 PBW Theorem

Let $\mathfrak{g}$ be any Lie algebra over k

Definition 9.1. *The **universal enveloping algebra** of $\mathfrak{g}$ $U\mathfrak{g}$ is the associative algebra over k generated by $\mathfrak{g}$ with relations $[x, y] = xy - yx$ for all $x, y \in \mathfrak{g}$*

More formally, if V is a vector space over k , then

$$TV = k \oplus V \oplus V \otimes V \oplus \dots = \bigoplus_{n \geq 0} V^{\otimes n}$$

is the **tensor algebra** of V , and is the free associative algebra generated by V . Let J be the two-sided ideal in $T\mathfrak{g}$ generated by $x \otimes y - y \otimes x = [x, y]$ for all $x, y \in \mathfrak{g}$; then $U\mathfrak{g} = T\mathfrak{g}/J$

The Casimir operator naturally lives in $U\mathfrak{g}$; in fact, as we've shown, in the center $Z(U\mathfrak{g})$.

Observe that $T\mathfrak{g}$ is a graded algebra, but that the relations we are imposing on it are not homogeneous. Consequently, $U\mathfrak{g}$ is not graded, but only filtered. Define $\mathcal{U}_n$ to be the span of products of $\leq n$ elements of $\mathfrak{g}$ (so $\mathcal{U}_0 = k, \mathcal{U}_1 = k + \mathfrak{g}$ and so on), then $\mathcal{U}_n \subset \mathcal{U}_m$ for $n \leq m$ and $\mathcal{U}_n \cdot \mathcal{U}_m \subset \mathcal{U}_{n+m}$. By abuse of notation, for $x, y \in U\mathfrak{g}$ we write $[x, y]$ to mean $xy - yx$. We can proceed by induction to show that for $x \in \mathcal{U}_n, y \in \mathcal{U}_m, [x, y] \in \mathcal{U}_{n+m-1}$ and so $[\mathcal{U}_n, \mathcal{U}_m] \subset \mathcal{U}_{n+m-1}$

Definition 9.2. *For a filtration $F_0 \subset F_1 \subset \dots$, we write $gr(F) = \bigoplus F_i / F_{i-1}$. This is the **associated graded algebra**.*

Theorem 9.1. (*Poincaré-Birkhoff-Witt*)

$$gr(U\mathfrak{g}) = \oplus \mathcal{U}_i / \mathcal{U}_{i-1} \cong S\mathfrak{g}, \text{ the symmetric algebra of } \mathfrak{g}$$

Equivalently, if $x_1, \dots, x_n$ is a basis for $\mathfrak{g}$, then $\{x_1^{a_1} \dots x_n^{a_n}\}$ is a basis for $U\mathfrak{g}$. In particular, $\mathfrak{g} \hookrightarrow U\mathfrak{g}$

Note that there is a map $\phi : \mathfrak{g} \rightarrow gr(U\mathfrak{g})$ via $x \mapsto x + \mathcal{U}_0$. Then $\phi(x)\phi(y) - \phi(y)\phi(x) = [x, y] + \mathcal{U}_1 = 0$. By universal property of symmetric algebra, ϕ extends to $S\mathfrak{g} \rightarrow gr(U\mathfrak{g})$, which is clearly surjective.

Therefore, the hard part of PBW theorem is injectivity. We will not prove the theorem.

Remark 9.1. The theorem should imply that $U\mathfrak{g}$ is a deformation of $S\mathfrak{g}$ (i.e. if we consider $xy - yx = t[x, y]$ then for $t = 0$ we get $S\mathfrak{g}$ whereas for $t = 1$ - and, in fact, for generic t by renormalising - we get $U\mathfrak{g}$). The shortest proof of the theorem is indeed via deformation theory.

10 Back to Representations

First note that If V is a representation of $\mathfrak{g}$ and $v \in V$, then the $\mathfrak{g}$ -submodule of V generated by v is $U\mathfrak{g} \cdot v$ (more like notations)

Definition 10.1. V is a **highest weight module** for $\mathfrak{g}$ if there is $v \in V$ a singular vector ($\mathfrak{n}^+ v = 0$ and $tv = \lambda(t)v$ for $t \in \mathfrak{t}$ and some $\lambda \in \mathfrak{t}^*$) such that $V = U\mathfrak{g} \cdot v$. In fact, we have $U\mathfrak{n}^- \cdot v = V$: PBW theorem tells us that if x_i is a basis of $\mathfrak{g}$, then $x_1^{a_1} \dots x_n^{a_n}$ is a basis for $U\mathfrak{g}$. After taking a suitable basis, we have $U\mathfrak{g} = U\mathfrak{n}^- \otimes U\mathfrak{t} \otimes U\mathfrak{n}^+$. Now just notice that by definition $\mathfrak{n}^+ \cdot v = 0$ and so $U\mathfrak{n}^+ v = U\mathfrak{t}v = \mathbb{C}v \subset U\mathfrak{n}^- \cdot v$

Remark 10.1. If V is irreducible and finite-dimensional, it is a highest weight module (as it has a singular vector v , and $U\mathfrak{g} \cdot v$ is a submodule).

Proposition 10.1. Let V be a highest weight module for $\mathfrak{g}$ (not necessarily finite-dimensional), and let v_Λ be a highest weight vector with highest weight $\Lambda \in \mathfrak{t}^*$.

1. $\mathfrak{t}$ acts diagonally on V , and so $V = \oplus_{\lambda \in D(\Lambda)} V_\lambda$, where

$$D(\Lambda) = \{\Lambda - \sum k_i \alpha_i \mid k_i \in \mathbb{Z}_{\geq 0}\} = \{\mu \in \mathfrak{t}^* \mid \mu \leq \Lambda\}$$

(the “descent” of Λ)

2. $V_\Lambda = \mathbb{C}v_\Lambda$, and all other weight spaces are finite-dimensional.

3. V is irreducible if and only if all singular vectors are in V_Λ .

4. Ω acts on V by $|\Lambda + \rho|^2 - |\rho|^2$

5. If v_λ is a singular vector, then $|\Lambda + \rho|^2 = |\lambda + \rho|^2$

6. If $\Lambda \in \mathbb{R}R$, then there exist only finitely many λ such that V_λ contains a singular vector.

7. V contains a unique maximal proper submodule I . I is graded by $\mathfrak{t}$, $I = \oplus (I \cap V_\lambda)$, and I is the sum of all proper submodules of V .

Remark 10.2. The assumption in (6) is in fact unnecessary, but there isn't quite a one-line proof in that case. Note that if V is finite-dimensional, then we in fact know that all weights lie in $\mathbb{Q}R$, but we are not assuming finite dimensionality here.

Proof. As $V = U\mathfrak{n}^- \cdot v_\Lambda$, expressions $e_{-\beta_1} \dots e_{-\beta_l} v_\Lambda$ spans V where β_i is a positive root and $e_{-\beta_i} \in \mathfrak{g}_{-\beta_i}$ is a generator. The weight of such an expression is $\Lambda - \beta_1 - \dots - \beta_l$ (note that $te_\alpha v = e_\alpha tv + [t, e_\alpha]v = (\lambda(t) + \alpha(t))e_\alpha v$)

This implies (1) immediately, and (2) upon noticing that there are finitely many ways of writing $\lambda - \beta_1 - \dots - \beta_l$ for $\beta_i \in R^+$

(3) If $v_\lambda \in V_\lambda$ is a singular vector, then $N = U\mathfrak{g} \cdot v_\lambda = U\mathfrak{n}^- v_\lambda$ is a submodule of V with weights in $D(\Lambda)$. Since $\lambda \neq \Lambda$, have $D(\lambda) \subsetneq D(\Lambda)$ whence N is a proper submodule. Thus, if there is a singular vector outside V_Λ , then V is not irreducible.

Conversely, if N is a proper submodule, then $\mathfrak{t}N \subset N$ and so N is graded by $\mathfrak{t}$, and its weights lie in $D(\Lambda)$. Let $\lambda = \Lambda - \sum k_i \alpha_i$ with $\sum k_i$ minimal. Since N is a proper submodule, $\lambda \neq \Lambda$ and so $\sum k_i > 0$, and in that case any nonzero vector in N_λ is singular (since $\lambda + \alpha_i$ is not a weight of N for any simple α_i).

(7): Any proper submodule of V is $\mathfrak{t}$ -graded and does not contain v_Λ . The sum of all proper submodules still does not contain v_Λ , so it is the maximal proper submodule.

(4): for the singular vector v_Λ we have $\Omega v_\Lambda = (|\Lambda + \rho|^2 - |\rho|^2)v_\Lambda$. Since $V = U\mathfrak{g}v_\Lambda$ and Ω commutes with all elements in $U\mathfrak{g}$ by lemma 8.1, Ω acts on V by the same scalar

(5) is now immediate (apply the same observation to any other singular vector).

(6) If V_λ has a singular vector, by (5) have $|\Lambda + \rho|^2 = |\lambda + \rho|^2$. This means that λ lies in the intersection of a sphere in $\mathbb{R}R$ (compact) and $D(\Lambda)$ (discrete), so there are only finitely many such λ □

Definition 10.2. Let $\Lambda \in \mathfrak{t}^*$. A **Verma module** $M(\Lambda)$ with highest weight Λ and highest weight vector v_Λ is a universal module with highest weight Λ . That is, given any other highest weight module W with highest weight Λ and highest weight vector w , there exists a unique map $M(\Lambda) \rightarrow W$ sending $v_\Lambda \mapsto w$.

Remark 10.3. Verma modules were also discovered by Harish-Chandra.

Proposition 10.2. Let $\Lambda \in \mathfrak{t}^*$. Then

1. There exists a unique Verma module $M(\Lambda)$.
2. There exists a unique irreducible highest-weight module of weight Λ , called $L(\Lambda)$.

Proof. Uniqueness follows from universal property. For existence, take

$$M(\Lambda) = U\mathfrak{g} \otimes_{U\mathfrak{b}} \mathbb{C}_\Lambda$$

where $\mathfrak{b} = \mathfrak{n}^+ + \mathfrak{t}$ and $\mathbb{C}_\Lambda$ is the $\mathfrak{b}$ -module on which $\mathfrak{n}^+ \cdot v = 0$ and $t \cdot v = \Lambda(t)v$.

Remark 10.4. You should think of this as the induced representation.

More concretely,

$$M(\Lambda) = U\mathfrak{g} / (\text{left ideals generated by } u - \Lambda(u) \text{ for all } u \in U\mathfrak{b})$$

(where Λ is extended to act on $\mathfrak{b}$ by 0 on $\mathfrak{n}^+$).

That is, $M(\Lambda)$ is the module generated by $\mathfrak{g}$ acting on 1 with relations $\{\mathfrak{n}^+ 1 = 0, t \cdot 1 = \Lambda(t)1 \text{ for all } t \in \mathfrak{t}\} = J_\Lambda$, and only the relations these imply.

If W is any other highest weight module of highest weight Λ , then $W = U\mathfrak{g}/J$ for some $J \subset J_\Lambda$, i.e. $M(\Lambda) \twoheadrightarrow W$.

In particular, an irreducible highest-weight module must be of the form $M(\Lambda)/I(\Lambda)$ where $I(\Lambda)$ is a maximal proper submodule of $M(\Lambda)$. Since we showed that there exists a unique maximal proper submodule, $L(\Lambda)$ is unique. □

Proposition 10.3. Let $R^+ = \{\beta_1, \dots, \beta_r\}$. Then $e_{-\beta_1}^{k_1} \dots e_{-\beta_r}^{k_r} v_\Lambda$ is a basis for $M(\Lambda)$

Proof. Immediate from the difficult direction of PBW theorem. $\square$

Corollary 10.1. Any irreducible finite-dimensional $\mathfrak{g}$ -module is of the form $L(\Lambda)$ for some $\Lambda \in P_+$

Proof. We know that the highest weight must lie in P^+ by $\mathfrak{sl}_2$ theory. $\square$

Example 46. Let $\mathfrak{g} = \mathfrak{sl}_2$. The Verma module $M(\Lambda)$ looks like this:

$$\begin{array}{ccccccc} \bullet & - & \bullet & - & \bullet & - & \dots \\ v_\Lambda & & Fv_\Lambda & & F^2v_\Lambda & & \end{array}$$

i.e. an infinite string.

Recall that in lemma 2.1 we have shown that $ef^n v = n(\lambda - n + 1)f^{n-1}v$, where v is a hw-vector of weight λ . Hence for $\lambda \notin \mathbb{Z}_{\geq 0}$, $M(\lambda) = L(\lambda)$ is the irr. hw-module of weight λ . If $\lambda \in \mathbb{Z}_{\geq 0}$, $M(\lambda)$ contains a unique proper submodule, spanned by $f^{\lambda+1}v, f^{\lambda+2}v, \dots$ since $ef^{\lambda+1}v = 0$

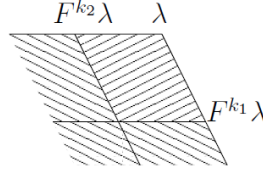


Figure 9: Some of the singular vectors in a representation of $\mathfrak{sl}_3$ – note that the quotient isn't W -invariant, so there must be others.

Now if $\mathfrak{g}$ is any simple Lie algebra with settings after remark 7.1, similarly we have $E_i F_i^n v_\Lambda = n(\Lambda(H_i) + 1 - n)v_\Lambda$ and $E_j F_i^n v_\Lambda = F_i^n E_j v_\Lambda = 0$ for $i \neq j$. So if $\Lambda(H_i) \in \mathbb{Z}_{\geq 0}$, $F_i^{\Lambda(H_i)+1} v_\Lambda$ is a singular vector of $M(\Lambda)$

Further we will see that $chM(\lambda) = e^\lambda \prod_{\alpha \in R^+} (1 + e^{-\alpha} + \dots) = e^\lambda \prod_{\alpha \in R^+} \frac{1}{1 - e^{-\alpha}}$

We will now try to show that $L(\Lambda)$ is finite-dimensional if $\Lambda \in P^+$

Proposition 10.4. If $\Lambda \in P^+$, then $L(\Lambda)$ is integrable, i.e. E_i, F_i act locally nilpotently

Proof. If V is any highest-weight module with highest weight Λ , E_i acts locally nilpotently as $E_i V_\Lambda \subset V_{\Lambda+\alpha_i}$ and weights of V lie in $D(\Lambda)$. We must show that F_i acts locally nilpotently.

If $F_i^{\Lambda(H_i)+1} v_\Lambda \neq 0$, we have seen that $F_i^{\Lambda(H_i)+1} v_\Lambda$ is also a singular vector of weight $< \Lambda$. This contradicts to irreducibility of $L(\Lambda)$ by proposition 10.1 (3). Hence $F_i^{\Lambda(H_i)+1} v_\Lambda = 0$ for all i $\square$

In the NNR notes we have seen a trick $a^k b = \sum_{i=0}^k ad_a^i b a^{k-i}$. Now the Serre relations give $(ad_{e_{\alpha_i}})^{1-a_{ij}} e_{\alpha_j} = 0$, where $A = (a_{ij})_{i,j}$ is the Cartan matrix. Using this trick repeatedly and that F_i acts locally nilpotently, we see that $F_i^N e_{-\beta_1} \dots e_{-\beta_r} v_\Lambda = 0$ for N sufficiently large.

Corollary 10.2. $\dim L(\Lambda)_\mu = \dim L(\Lambda)_{w\mu}$ for $w \in W$

Proof. When we were proving Proposition 8.2, we only needed local nilpotence of E_i and F_i to conclude invariance under $w = s_{\alpha_i} = s_i$ (see the discussion after remark 8.2). Since W is generated by s_i , we conclude invariance under all of W . $\square$

Theorem 10.1. (*Cartan's theorem*). If $\mathfrak{g}$ is finite-dimensional and $\Lambda \in P^+$, then $L(\Lambda)$ is finite-dimensional.

Remark 10.5. This is clear pictorially - $L(\Lambda)$ is W -invariant and sits in the cone $D(\Lambda)$. However, it's faster to work through it formally.

Proof. First we show that if $\alpha \in R^+$ then $e_{-\alpha}$ acts nilpotently on $L(\Lambda)$, i.e. all (not just the simple) local $\mathfrak{sl}_2$'s act integrably. In fact, $e_{-\alpha}^n v_\Lambda = 0$ for $n = (\Lambda, \alpha^\vee) + 1$, since otherwise we have $L(\Lambda)_{\Lambda - n\alpha} \neq 0$ and so $s_\alpha(\Lambda - n\alpha)$ is a weight in $L(\Lambda)$. However,

$$s_\alpha(\Lambda - n\alpha) = s_\alpha(\Lambda) + n\alpha = \Lambda - (\Lambda, \alpha^\vee)\alpha + n\alpha = \Lambda + \alpha > \Lambda$$

and so cannot be a weight in $L(\Lambda)$ as Λ is the highest weight. By the above observation we see that $e_{-\alpha}$ acts locally nilpotently on all of $L(\Lambda) = U\mathfrak{n}^- v_\Lambda$. We deduce that $L(\Lambda)$ has only finitely many weights. By proposition 10.1 (2), all weight spaces are f.d. (more directly, $e_{-\beta_1}^{k_1} \dots e_{-\beta_r}^{k_r} v_\Lambda$ is a basis for $L(\Lambda)$ where $\beta_i \in R^+$. Of course, there is only finite ways to write each weight into the form $\Lambda - k_1\beta_1 - \dots - k_r\beta_r$, and so $L(\Lambda)$ is f.d. $\square$

Lemma 10.1. $s_i(R^+ \setminus \{\alpha_i\}) = R^+ \setminus \{\alpha_i\}$

Proof. Take $\alpha \in R^+ \setminus \{\alpha_i\}$ and we can write $\alpha = \sum_{j \neq i} k_j \alpha_j$ for $k_i \geq 0$ where $i \neq j$ and $k_j > 0$ for some j . Note $s_i(\alpha)$ has the same coefficients of α_j for $j \neq i$. Therefore, the coefficient of $s_i(\alpha)$ on α_j is positive and so $s_i(\alpha) \in R^+$ $\square$

Lemma 10.2. Recall $\rho = \frac{1}{2} \sum_{\alpha \in R^+} \alpha$. Then $\rho(H_i) = (\rho, \alpha_i^\vee) = 1$ for all i , i.e. $\rho = w_1 + \dots + w_l$

Proof. Observe that by lemma 10.1

$$s_i \rho = s_i \left(\frac{1}{2} \alpha_i + \sum_{\alpha \in R^+ \setminus \{\alpha_i\}} \frac{1}{2} \alpha \right) = \rho - \alpha_i$$

On the other hand, have $s_i \rho = \rho - (\rho, \alpha_i^\vee) \alpha_i$. So $(\rho, \alpha_i^\vee) = 1$ as required. $\square$

Lemma 10.3. (*Key lemma*) Let $\Lambda \in P^+$, $\mu + \rho \in P^+$, $\mu \leq \Lambda$. If $|\Lambda + \rho| = |\mu + \rho|$, then $\Lambda = \mu$

Remark 10.6. This, again, is geometrically obvious from the fact that both $D(\Lambda)$ and a sphere are convex.

Proof. Note that

$$0 = (\Lambda + \rho, \Lambda + \rho) - (\mu + \rho, \mu + \rho) = (\Lambda - \mu, \Lambda + \mu + 2\rho)$$

Write $\Lambda - \mu = \sum k_i \alpha_i$. Since $\Lambda + (\mu + \rho) \in P^+$ and $(\rho, \alpha_i) > 0$ by lemma 10.2, must have $k_i = 0$ for all i $\square$

Theorem 10.2. (*Weyl complete reducibility*). Let $\mathfrak{g}$ be a semisimple Lie algebra, $\text{char } k = 0$, $\bar{k} = k$. Then every finite-dimensional $\mathfrak{g}$ -module V is a direct sum of irreducibles.

Proof. Recall that V is completely reducible as an $(\mathfrak{sl}_2)_\alpha$ -module and $V = \oplus V_\lambda$

Consider $V^{\mathfrak{n}^+} = \{x \in V | \mathfrak{n}^+ x = 0\}$. By Engel's theorem, $V^{\mathfrak{n}^+} \neq 0$.

Since $[\mathfrak{t}, \mathfrak{n}^+] \subset \mathfrak{n}^+$, $\mathfrak{t}$ acts on $V^{\mathfrak{n}^+}$, so $V^{\mathfrak{n}^+} = \oplus_{\mu \in P} V_\mu^{\mathfrak{n}^+}$

Claim: For every $v_\mu \in V_\mu^{\mathfrak{n}^+}$, the module $L = U\mathfrak{g}v_\mu$ is irr. (and so $L = L(\mu)$)

Proof of claim. L is a highest weight module with highest weight μ since $\mathfrak{n}^+v_\mu = 0$ and so $U\mathfrak{g}v_\mu = U\mathfrak{n}^-v_\mu$, so we must only show that it has no other singular vectors of different weight. If λ is the weight of a singular vector in L , then $\lambda \leq \mu$. Then by proposition 10.1 (5), have $|\mu + \rho| = |\lambda + \rho|$. Since V , and therefore L , is finite-dimensional, we must have $\lambda, \mu \in P^+$. By the key lemma, $\lambda = \mu$ $\square$

Consequently, $V' = U\mathfrak{g}V^{\mathfrak{n}^+}$ is completely reducible: if $\{v_1, \dots, v_r\}$ is a weight basis (i.e. generators of each weight space) for $V^{\mathfrak{n}^+}$ with weight $\lambda_1, \dots, \lambda_r$, then $V' = \oplus L(\lambda_i)$ by the claim. It remains to show that $N = V/V' = 0$

Suppose not. N is f.d., so $N^{\mathfrak{n}^+} \neq 0$. Let $v_\lambda \in N_\lambda^{\mathfrak{n}^+}$ be a singular vector with $\lambda \in P^+$. Lift v_λ to $\bar{v}_\lambda \in V_\lambda$. Since $v_\lambda \neq 0$ in N , we must have $E_i \bar{v}_\lambda \neq 0 \in V_{\lambda+\alpha_i}$ for some i (otherwise, $\bar{v}_\lambda$ is a singular vector and so $\lambda = \lambda_i$ for some i)

Note that $E_i v_\lambda = 0$, so $E_i \bar{v}_\lambda \in V'$. Consequently, the Casimir Ω acts on $E_i \bar{v}_\lambda$ by $|\bar{\lambda} + \rho|^2 - |\rho|^2$ for some $\bar{\lambda} \in \{\lambda_1, \dots, \lambda_r\}$. On the other hand, since v_λ is singular in N , Ω acts on v_λ by $|\lambda + \rho|^2 - |\rho|^2$. Since Ω commutes with E_i , we get $|\bar{\lambda} + \rho| = |\lambda + \rho|$. Moreover, $\lambda + \alpha_i$ is a weight in $L(\bar{\lambda})$, and so $\lambda < \lambda + \alpha_i \leq \bar{\lambda}$. This, however, contradicts the key lemma. $\square$

Remark 10.7. The usual proof (in particular, Weyl's own proof) of this theorem involves lifting the integrable action to the group, and using the maximal compact subgroup (for which we have complete reducibility because it has positive, finite measure). Somehow, the key lemma is the equivalent of a positive-definite metric.

Lemma 10.4. *Let $\Lambda \in \mathfrak{t}^*$ and $M(\Lambda)$ the Verma module. Then*

$$chM(\Lambda) = e^\Lambda \prod_{\alpha \in R^+} \frac{1}{1 - e^{-\alpha}}$$

Proof. Again, we have seen that $e_{-\beta_1}^{k_1} \dots e_{-\beta_r}^{k_r} v_\Lambda$ is a basis for $M(\Lambda)$ where $R^+ = \{\beta_1, \dots, \beta_r\}$. Therefore the dimension of each weight space of $M(\Lambda)$ is the number of the ways to write the weight in the form $\Lambda - \sum k_i \beta_i$. And this is precisely the corresponding coefficients in $e^\Lambda \prod_{\alpha \in R^+} \frac{1}{1 - e^{-\alpha}}$ $\square$

Write $\Delta = \prod_{\alpha \in R^+} (1 - e^{-\alpha})$. We have shown that $chM(\Lambda) = e^\Lambda / \Delta$

Lemma 10.5. *For $w \in W$, $w(e^\rho \Delta) = \det w \cdot e^\rho \Delta$, where $\det : W \rightarrow \mathbb{Z}/2$ gives the determinant of w acting on $\mathfrak{t}^*$ (± 1)*

Proof. Since W is generated by simple reflections, it suffices to check that $s_i(e^\rho \Delta) = -e^\rho \Delta$. Indeed

$$s_i(e^\rho \Delta) = e^{\rho - \alpha_i} s_i((1 - e^{-\alpha_i}) \prod_{\alpha \in R^+ \setminus \{\alpha_i\}} (1 - e^{-\alpha})) = e^{\rho - \alpha_i} (1 - e^{\alpha_i}) \prod_{\alpha \in R^+ \setminus \{\alpha_i\}} (1 - e^{-\alpha}) = -e^\rho \Delta$$

by lemma 10.1 $\square$

Lemma 10.6. *1. For any highest weight module $V(\Lambda)$ with highest weight Λ there exist coefficients $a_\lambda \geq 0$ with $\lambda \leq \Lambda$ s.t.*

$$chV(\Lambda) = \sum_{\lambda \leq \Lambda, |\lambda + \rho| = |\Lambda + \rho|} a_\lambda chL(\lambda)$$

2. There exists coefficients $b_\lambda \in \mathbb{Z}$ with $b_\Lambda = 1$ s.t.

$$chL(\Lambda) = \sum_{\lambda \leq \Lambda, |\lambda + \rho| = |\Lambda + \rho|} b_\lambda chM(\lambda)$$

We write $B(\Lambda) = \{\lambda \leq \Lambda \mid |\lambda + \rho| = |\Lambda + \rho|\}$. Recall that $B(\Lambda)$ is finite

Remark 10.8. We have shown that $B(\Lambda)$ is finite when $\Lambda \in \mathbb{R}R$, but this is true in general.

Proof. First, note that (1) implies (2). Indeed, totally order $B(\Lambda)$ so that $\lambda_i \leq \lambda_j \implies i \leq j$. Applying (1) to the Verma modules gives a system of equations relating $chM(\lambda)$ and $chV(\lambda)$ which is upper-triangular with 1's on the diagonal (recall that $L(\Lambda)$ is a maximal quotient of $M(\Lambda)$). Inverting this system gives (2). (Note in particular that the coefficients in (2) may be negative.)

We now prove (1). Recall that the weight spaces of a highest weight module are finite-dimensional. We induct on $\sum_{\mu \in B(\Lambda)} V(\Lambda)_\mu$

If $V(\Lambda)$ is irreducible, we clearly have (1). Otherwise, there exists $\Lambda \neq \mu \in B(\Lambda)$ with a singular vector $v_\mu \in V(\Lambda)_\mu$ by proposition 10.1. Pick μ with the largest height of $\Lambda - \mu$ (recall the height of $\sum k_i \alpha_i$ is $\sum k_i$). Then $U\mathfrak{g}v_\mu \subset V(\Lambda)$ has no singular vectors, and is therefore irreducible, i.e. $L(\mu)$. Set $\overline{V(\Lambda)} = V(\Lambda)/L(\mu)$, and we see that $\overline{V(\Lambda)}$ is a highest-weight module with a smaller value of $\sum_{\gamma \in B(\Lambda)} \dim V(\Lambda)_\gamma$, and $chV(\Lambda) = chL(\mu) + ch\overline{V(\Lambda)}$ by the property of characters. $\square$

We now compute $chL(\Lambda)$ for $\Lambda \in P^+$. We know

$$chL(\Lambda) = \sum_{\lambda \leq \Lambda, |\lambda + \rho| = |\Lambda + \rho|} b_\lambda chM(\lambda) = \sum b_\lambda e^{\lambda/\Delta}$$

By corollary 10.2 have $w(chL(\Lambda)) = chL(\Lambda)$ for all $w \in W$ and $w(e^\rho \Delta) = \det w \cdot e^\rho \Delta$. Therefore

$$e^\rho \Delta chL(\Lambda) = \sum b_\lambda e^{\lambda + \rho}$$

is W -antiinvariant. We can therefore rewrite

$$\sum_{\lambda \in B(\Lambda)} b_\lambda e^{\lambda + \rho} = \sum_{(\text{orbits of } W \text{ on } B(\Lambda)) + \rho \text{ and } \lambda \text{ a representative}} b_\lambda \sum_{w \in W} \det w \cdot e^{w(\lambda + \rho)}$$

Now if $\lambda \in \mathbb{R}R$ (which is true since $\Lambda \in P^+$), then $W(\lambda + \rho)$ intersects $\{x \in \mathbb{R}R | (x, \alpha_i^\vee) \geq 0\}$ in exactly one point (this set is a fundamental domain for the W -action, and we won't have boundary problems because $(\rho, \alpha_i^\vee) = 1 \geq 0$). Therefore, we can take a dominant weight as the orbit representative. Note, however, that there is only one dominant weight in $B(\Lambda)$, namely, Λ ! (This was the Key Lemma 10.3.) Therefore, there is only one orbit (i.e. $B(\Lambda)$ is W -transitive) and so $b_\lambda = b_\Lambda = 1$ and we derive that

Theorem 10.3. (Weyl Character Formula).

$$chL(\Lambda) = \frac{\sum_{w \in W} \det w \cdot e^{w(\Lambda + \rho)}}{e^\rho \prod_{\alpha \in R^+} (1 - e^{-\alpha})} = \sum_{w \in W} \det w \cdot chM(w(\Lambda + \rho) - \rho)$$

Example 47. Let $\mathfrak{g} = \mathfrak{sl}_2$ and write $z = e^{\alpha/2}$. Then $\mathbb{C}[P] = \mathbb{C}[z, z^{-1}]$ and $\rho = z$. Then by the formula

$$chL(m\alpha/2) = chM(m\alpha/2) - chM(-(m+2)\alpha/2) = z^m / (1 - z^{-2}) - z^{-m-2} / (1 - z^{-2}) = (z^{m+1} - z^{-m-1}) / (z - z^{-1})$$

as before

Remark 10.9. The second expression for the character of $L(\Lambda)$ looks like an Euler characteristic. In fact, it is: there is a complex of Verma modules whose cohomology are the irreducible modules. This is known as the ‘‘Bernstein-Gelfand-Gelfand’’ (BGG) resolution. The first expression comes from something like a cohomology of coherent line bundles, and is variously known as Euler-Grothendieck, Atiyah-Bott, and Atiyah-Singer.

Corollary 10.3. Since $L(0) = \mathbb{C}$, $chL(0) = 1$ and so

$$e^\rho \prod_{\alpha \in R^+} (1 - e^{-\alpha}) = \sum_{w \in W} detw \cdot e^{w\rho}$$

This is known as the **Weyl denominator identity**.

Example 48. Let $\mathfrak{g} = \mathfrak{sl}_n$, then the Weyl denominator identity is equivalent to the Vandermonde determinant $\prod_{i < j} (z_j - z_i)$ where $z_i = e^{-e_i}$, since $\alpha \in R^+$ is of form $e_i - e_j$ with $i < j$ and so $(1 - e^{-\alpha}) = e^{e_j} (z_j - z_i)$

Corollary 10.4. (Weyl dimension formula)

$$dimL(\Lambda) = \prod_{\alpha \in R^+} (\alpha, \Lambda + \rho) / (\alpha, \rho)$$

Example 49. Let $\mathfrak{g} = \mathfrak{sl}_3$ of type A_2 . Then $R^+ = \{\alpha, \beta, \alpha + \beta\}$ and $\rho = \alpha + \beta = w_1 + w_2$ (w_i the dual basis). Let $\Lambda = m_1 w_1 + m_2 w_2$. We compute $dimL(\Lambda) = \frac{1}{2}(m_1 + 1)(m_2 + 1)(m_1 + m_2 + 2)$

“Just to check that you too certainly know the dimensions of the irreducible representations, let me state the obvious exercise.” Compute the dimensions of all the finite-dimensional irreducible representations of B_2 and G_2 . (This is a wonderful material for exam, as always)

Remark 10.10. Let $w \in W$ be written as $w = s_{i_1} s_{i_2} \dots s_{i_r}$ where s_{i_k} are simple reflections. Then $detw = (-1)^r$. The minimal r such that w can be written in this form is called **the length** for w , denoted $l(w)$. The Monoid Lemma asserts that you can get from one minimal-length expression for w to another by repeatedly applying the braid relations. Note that $l(w) = \#\{R^- \cap w^{-1}R^+\} = l(w^{-1})$

We now go back to our proof of Weyl dimension formula

Proof. We know $chL(\Lambda)(1) = dimL(\Lambda)$. So we would like to set $e^\lambda \mapsto 1$ but that leads to a 0/0 expression in the Weyl character formula, which is hard to evaluate. Therefore, we will start by defining the homomorphism $F_\mu : \mathbb{C}[P] \rightarrow \mathbb{C}(q); e^\lambda \mapsto q^{-(\lambda, \mu)}$. We would like to evaluate $F_0(chL(\Lambda))$, since $F_0(e^\lambda) = 0$ for all λ

We apply μ to the Weyl denominator identity and get

$$q^{-(\rho, \mu)} \prod_{\alpha \in R^+} (1 - q^{(\alpha, \mu)}) = \sum_{w \in W} detw \cdot q^{-(w\rho, \mu)} = \sum_{w \in W} detw \cdot q^{-(\rho, w^{-1}\mu)}$$

since $detw = detw^{-1}$ and $(wx, y) = (x, w^{-1}y)$ (i.e. the Weyl group is a subgroup of the orthogonal group of the inner product).

Now apply F_μ to the Weyl character formula to get

$$F_\mu(chL(\Lambda)) = \frac{\sum_{w \in W} detw \cdot q^{-(w(\Lambda + \rho), \mu)}}{q^{-(\rho, \mu)} \prod_{\alpha \in R^+} (1 - q^{(\alpha, \mu)})}$$

provided that $(\alpha, \mu) \neq 0$ for all $\alpha \in R^+$

Now take $\mu = \rho$ (recall that $(\rho, \alpha_i) > 0$ for simple roots α_i). Then

$$F_\rho(ch(L(\Lambda))) = \sum dimL(\Lambda)_\lambda q^{-(\lambda, \rho)} = \frac{q^{-(\rho, \Lambda + \rho)} \prod_{\alpha \in R^+} (1 - q^{(\alpha, \Lambda + \rho)})}{q^{-(\rho, \rho)} \prod_{\alpha \in R^+} (1 - q^{(\alpha, \rho)})}$$

where we used our expression for the Weyl denominator identity above taking $\mu = \Lambda + \rho$ and applied it to the numerator.

Setting $q = 1$ and applying L'Hôpital's rule gives,

$$\dim L(\Lambda) = \prod_{\alpha \in R^+} \frac{(\alpha, \Lambda + \rho)}{(\alpha, \rho)}$$

□

Remark 10.11. “You are now in a position to answer any question you choose to answer, if necessary by brute force.” For example, let us decompose $L(\lambda) \otimes L(\mu) = \sum m_{\lambda\mu}^v L(v)$; we want to compute the **Littlewood-Richardson** coefficients $m_{\lambda\mu}^v$ (recall that we had the Clebsch-Gordan rule for them in $\mathfrak{sl}_2$ in proposition 3.1 (5))

Define $\bar{\cdot} : \mathbb{Z}[P] \rightarrow \mathbb{Z}[P]$, $e^\lambda \mapsto e^{-\lambda}$ and $CT : \mathbb{Z}[P] \rightarrow \mathbb{Z}$, $e^\lambda \mapsto 0$ if $\lambda \neq 0$ and $e^0 \mapsto 1$. Define $(,) : \mathbb{Z}[P] \times \mathbb{Z}[P] \rightarrow \mathbb{Z}$ by $(f, g) = \frac{1}{|W|} CT(f \bar{g} \Delta \bar{\Delta})$ where $\Delta = \prod_{\alpha \in R^+} (1 - e^{-\alpha})$

Claim: Let $\chi_\lambda = chL(\lambda)$, $\chi_\mu = chL(\mu)$, then $(\chi_\lambda, \chi_\mu) = \delta_{\lambda\mu}$ (If our Lie algebra has type A , then χ_λ are the Schur functions, and this is their orthogonality.)

Given this claim we have $m_{\lambda\mu}^v = (\chi_\lambda \chi_\mu, \chi_v)$ (recall that $chL(\lambda) \otimes L(\mu) = chL(\lambda) \cdot chL(\mu)$) is algorithmically computable. It was a great surprise to everyone when it was discovered that actually there is a simpler and more beautiful way of doing this.

Proof of claim. By definition and Weyl character formula

$$(\chi_\lambda, \chi_\mu) = \frac{1}{|W|} CT\left(\sum_{x, w \in W} \det(wx) e^{w(\lambda+\rho) - \rho} \overline{e^{x(\mu+\rho) - \rho}}\right) = \frac{1}{|W|} CT\left(\sum_{x, w \in W} \det(wx) e^{w(\lambda+\rho) - x(\mu+\rho)}\right)$$

However, $CT(e^{w(\lambda+\rho) - x(\mu+\rho)}) = \delta_{wx} \delta_{\mu\lambda}$, since for $\lambda, \mu \in R^+$ we have $x^{-1}w(\lambda + \rho) = \mu + \rho \implies x = w, \mu = \lambda$ □

11 Principal $\mathfrak{sl}_2$

Define $\rho^L = \frac{1}{2} \sum_{\alpha \in R^+} \alpha^\vee \in \mathfrak{t}^*$. Like ρ , $(\rho^L, \alpha_i) = 1$ for all simple roots α_i . In particular, if R is simply laced, $\rho = \rho^L$. Then $(\rho^L, \alpha) = \text{height of } \alpha = \sum k_i$ if $\alpha = \sum k_i \alpha_i$

By applying F_{ρ^L} to the Weyl denominator identity for the Langlands dual Lie algebra whose root system is $R^\vee$, we get

$$F_{\rho^L}(chL(\Lambda)) = q^{-(\Lambda, \rho^L)} \prod_{\alpha \in R^+} \frac{(1 - q^{(\Lambda + \rho, \alpha^\vee)})}{(1 - q^{(\rho, \alpha^\vee)})}$$

Note that $(\lambda + \rho, \alpha^\vee)/(\rho, \alpha^\vee) = (\lambda + \rho, \alpha)/(\rho, \alpha)$, so we recovered the same formula for Weyl dimension.

Definition 11.1. We call $F_{\rho^L}(chL(\Lambda))$ the **q -dimension** of $L(\Lambda)$, denoted by $\dim_q L(\Lambda)$

Proposition 11.1. $\dim_q L(\Lambda)$ is a **unimodal polynomial**. More specifically, it lives in $\mathbb{N}[q^2, q^{-2}]^{\mathbb{Z}/2}$ or $q\mathbb{N}[q^2, q^{-2}]^{\mathbb{Z}/2}$ depending on its degree, and the coefficients decrease as the absolute value of the degree gets bigger.

Proof. We will show that $\dim_q L(\Lambda)$ is the character of an $\mathfrak{sl}_2$ -module in which all “strings” have the same parity.

Define $H = 2v^{-1}(\rho^L) \in \mathfrak{t}$. Then $[H, E_i] = 2\alpha_i(v^{-1}(\rho^L))E_i = 2(\rho^L, \alpha_i)E_i = 2E_i$. Let $E = \sum E_i$. Then $[H, E] = 2E$

Now write $H = \sum c_i H_i$ and set $F = \sum c_i F_i$. Then $[H, F_i] = 2F_i$ and so $[H, F] = 2 \sum c_i F_i = 2F$. Also $[E, F] = \sum [E_i, c_i F_i] = \sum c_i H_i = H$. So E, F, H generate a copy of $\mathfrak{sl}_2$. This is called the **principal $\mathfrak{sl}_2$** .

If $\Lambda - \gamma$ is a weight of $L(\Lambda)$, then $L(\Lambda)$ as a $\mathfrak{sl}_2$ -representation by E, F, H has weight $\Lambda(H) = (\Lambda, 2\rho^L)$, $(\Lambda - \gamma)(H) = (\Lambda - \gamma, 2\rho^L)$. By the theory of $\mathfrak{sl}_2$ representations, the difference of these two is even. Therefore, $F_{\rho^L}(chL(\Lambda)) = \sum \dim L(\Lambda)_\lambda q^{-(\lambda, \rho^L)}$ has the required form $\square$

Example 50. Let $\mathfrak{g} = \mathfrak{sl}_n$. In example 44 we have seen that $\wedge^s \mathbb{C}^n = L(w_s)$ and $S^m \mathbb{C}^n = L(mw_1)$ are both irreducible $\mathfrak{sl}_n$ -representations. For $\mathfrak{sl}_n$, $\rho = \rho^L = \frac{1}{2} \sum (n+1-2i)e_i$. Let $\alpha = e_i - e_j \in R^+$ with $i < j$. Then $(\rho, \alpha) = j - i$. We now consider $F_{\rho^L}(chS^k \mathbb{C}^n)$. Then $\Lambda = ke_1$. Therefore, for $\alpha \neq e_1 - e_i \in R^+$, $(\Lambda, \alpha) = 0$. Therefore

$$F_{\rho^L}(chS^k \mathbb{C}^n) = q^{-k(n-1)} \prod_i \frac{(1 - q^{k+i-1})}{(1 - q^{i-1})}$$

If we define $[n] = (q^n - 1)/(q - 1)$ and $[n]! = [n][n-1] \dots [1]$. Then $\prod_i \frac{(1 - q^{k+i-1})}{(1 - q^{i-1})}$ is precisely $\binom{n+k-1}{k}$ in this notation. Proposition 11.1 then shows that polynomials in q of this form is unimodal.

Remark 11.1. When is $L(\lambda) = L(\lambda)^*$? Precisely when the lowest weight of $L(\lambda)$ is $-\lambda$. The lowest weight of $L(\lambda)$ is $w_0\lambda$, where w_0 is the (unique) maximal-length element of W which sends all the positive roots to negative ones.

Now, for a representation V which is isomorphic to its dual V^* , we get a bilinear form on V . This bilinear form is either symmetric or antisymmetric. By above, we see that we can determine which it is by laboriously computing characters; but can we tell simply by looking at λ ?

The answer turns out to be yes. Indeed, the form will pair the highest-weight and the lowest-weight vector, and if we look at the principal $\mathfrak{sl}_2$, it is a string from the highest-weight to the lowest-weight vector. Now, on the principal $\mathfrak{sl}_2$ it is quite easy to see that the form is antisymmetric if the representation is $L(n)$ with n odd, and symmetric if it is $L(n)$ with n even. Thus, all we need to know is (the parity of) $(\lambda, 2\rho^L)$. This is called the **Schur indicator**.

12 Crystals

Let $\mathfrak{g}$ a semisimple Lie algebra, $\Pi = \{\alpha_1, \dots, \alpha_r\}$ simple roots and P the weight lattice

Definition 12.1. A crystal is a set B , $0 \notin B$, together with functions $wt : B \rightarrow P$, $\tilde{e}_i : B \rightarrow B \sqcup \{0\}$, $\tilde{f}_i : B \rightarrow B \sqcup \{0\}$ s.t.

- (1) If $\tilde{e}_i(b) \neq 0$, then $wt(\tilde{e}_i(b)) = wt(b) + \alpha_i$; If $\tilde{f}_i(b) \neq 0$, then $wt(\tilde{f}_i(b)) = wt(b) - \alpha_i$
- (2) For $b, b' \in B$, $\tilde{e}_i b = b' \iff \tilde{f}_i b' = b$

(3) Before we state the third defining property, we need to introduce slightly more notation.

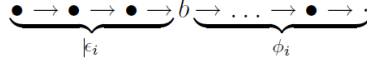
We can draw B as a graph. The vertices are $b \in B$, and the edges are $b \rightarrow b_0$ if $\tilde{e}_i b_0 = b$. We say that this edge is “coloured by i ”. This graph is known as the **crystal graph**.

Example 51. In $\mathfrak{sl}_2$ the string

$$n \rightarrow n-2 \rightarrow n-4 \rightarrow \dots \rightarrow -n$$

is a crystal, where the weight of vertex i is $i\alpha/2$.

Define $\epsilon_i(b) = \max\{n \geq 0 | \tilde{e}_i^n b \neq 0\}$ and $\phi_i(b) = \max\{n \geq 0 | \tilde{f}_i^n b \neq 0\}$



In $\mathfrak{sl}_2$, the sum $\epsilon_i(b) + \phi_i(b)$ is the length of the string. On the other hand, if we are in the highest-weight representation $L(n) = L(nw)$ and $wt(b) = n - 2k$, then $\epsilon(b) = k$ and $\phi(b) = n - k$, i.e. $\phi(b) - \epsilon(b) = (wt(b), \alpha^\vee)$. The third property of the Lie algebras is that this happens in general:

(3) $\phi_i(b) - \epsilon_i(b) = (wt(b), \alpha_i^\vee)$ for all $\alpha_i \in \Pi$

Define $B_\mu = \{b \in B | wt(b) = \mu\}$

Definition 12.2. For B_1 and B_2 crystals, we define the tensor product $B_1 \otimes B_2$ as follows: as a set, $B_1 \otimes B_2 = B_1 \times B_2$, $wt(b_1 \otimes b_2) = wt(b_1) + wt(b_2)$, and

$$\tilde{e}_i(b_1 \otimes b_2) = \begin{cases} \tilde{e}_i b_1 \otimes b_2 & \text{if } \phi_i(b_1) \geq \epsilon_i(b_2) \\ b_1 \otimes \tilde{e}_i b_2 & \text{if } \phi_i(b_1) < \epsilon_i(b_2) \end{cases}, \tilde{f}_i(b_1 \otimes b_2) = \begin{cases} \tilde{f}_i b_1 \otimes b_2 & \text{if } \phi_i(b_1) > \epsilon_i(b_2) \\ b_1 \otimes \tilde{f}_i b_2 & \text{if } \phi_i(b_1) \leq \epsilon_i(b_2) \end{cases}$$

That is, we are trying to recreate the picture of the decomposition of $L(n) \otimes L(m)$ in Figure 10 in each colour

We should check that $B_1 \otimes B_2$ as defined above is a crystal, and $(B_1 \otimes B_2) \otimes B_3 \cong B_1 \otimes (B_2 \otimes B_3)$.

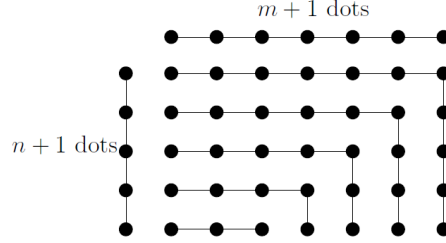


Figure 10: Crystal for $\mathfrak{sl}_2$

Remark 12.1. As we see above, the tensor product is associative. In general, however, it is not commutative.

Definition 12.3. $B^\vee$ is the crystal obtained from B by reversing the arrows. That is $B^\vee = \{b^\vee | b \in B\}$ with $wt(b^\vee) = -wt(b)$ and $\tilde{e}_i(b^\vee) = \tilde{f}_i(b)$ (and vice versa), and $\epsilon_i(b^\vee) = \phi_i(b)$ (and vice versa). Note that $(B_1 \otimes B_2)^\vee = B_2^\vee \otimes B_1^\vee$

Remark 12.2. We will see in a moment precisely how we can attach a crystal to the basis of a representation. If B parametrises a basis for V , then $B^\vee$ parametrises a basis for V^* . The condition on the weights can be seen by noting that if $L(\Lambda)$ has highest weight Λ , then $L(\Lambda)^*$ has lowest weight $-\Lambda$.

Theorem 12.1. (Kashiwara). Let $L(\Lambda)$ be an irreducible highest-weight representation with highest weight $\Lambda \in P^+$. Then

1. There exists a crystal $B(\Lambda)$ whose elements are in one-to-one correspondence with a basis of $L(\Lambda)$, and elements of $B(\Lambda)_\mu$ parametrise $L(\Lambda)_\mu$, so that

$$chL(\Lambda) = \sum_{b \in B(\Lambda)} e^{wt(b)}$$

2. For any simple root α_i (i.e. a simple $\mathfrak{sl}_2 \subset \mathfrak{g}$), the decomposition of $L(\Lambda)$ as an $(\mathfrak{sl}_2)_i$ -module is given by the i -coloured strings in $B(\Lambda)$. (In particular, as an uncoloured graph, $B(\Lambda)$ is connected, since it is spanned by $(\prod \tilde{f}_i)v_\lambda$).
3. $B(\lambda) \otimes B(\mu)$ is the crystal of $L(\lambda) \otimes L(\mu)$ decomposes into connected components in the same way as $L(\lambda) \otimes L(\mu)$ decomposes into irreducible representations

Remark 12.3. We will mention three proof approaches below. The first of them is due to Kashiwara, and is in the lecturer's opinion the most instructive.

Example 52. Let $\mathfrak{g} = \mathfrak{sl}_3$ and $V = \mathbb{C}^3 = L(w_1)$. We compute the crystals for $V \otimes V$ and $V \otimes V^*$

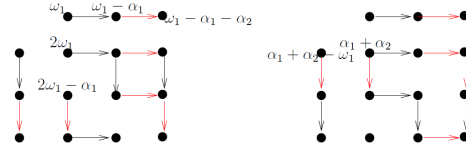


Figure 11: Crystals for V and $V \otimes V^*$. Recall that $V \otimes V = S^2V + \wedge^2V = S^2V + V^*$ and $V \otimes V = \mathfrak{sl}_3 + \mathbb{C}$

Remark 12.4. Note that while the crystals give the decomposition of the representation into irreducibles, they do not correspond directly to a basis. That is, there is no $\mathfrak{sl}_2$ -invariant basis that we could use here. Kashiwara's proof of the theorem uses the **quantum group** $U_q\mathfrak{g}$, which is an algebra over $\mathbb{C}[q, q^{-1}]$ and a deformation of the universal enveloping algebra $U\mathfrak{g}$. The two have the same representations, but over $\mathbb{C}[q, q^{-1}]$ there is a very nice basis which satisfies $e_i b = \tilde{e}_i b + q \cdot (\text{some mess})$. Therefore, setting $q = 0$ ("freezing") will give the crystal.

Lusztig's proof uses some of the same ideas, where I (the lecturer) didn't catch how it goes.

Soon, we will look at Littelmann paths, which give a purely combinatorial way of proving this theorem (which, on the face of it, is a purely combinatorial statement).

Definition 12.4. A crystal is called **integrable** if it is a crystal of a highest-weight module with highest weight $\Lambda \in P^+$.

For two integrable crystals B_1, B_2 , we do in fact have $B_1 \otimes B_2 = B_2 \otimes B_1$. This is false in general.

Remark 12.5. There is a combinatorial condition due to Stembridge which determines whether a crystal is integrable; it is a degeneration of the Serre relations, but we will not state it precisely.

Consider the crystal for the standard representation of $\mathfrak{sl}_n$, $L(w_1) = \mathbb{C}^n$:

$$e_1 = \omega_1 \xrightarrow{1} \omega_1 - \alpha_1 \xrightarrow{2} \omega_1 - \alpha_1 - \alpha_2 \xrightarrow{3} \dots \xrightarrow{n-1} \omega_1 - \alpha_1 - \dots - \alpha_{n-1}.$$

We can use this to construct the crystals for all representations of $\mathfrak{sl}_n$ as follows:

Let $\lambda \in P^+$ and write $\lambda = \sum_{i=1}^{n-1} k_i w_i$. Then $L(\lambda)$ is a summand of $L(w_1)^{\otimes k_1} \otimes \dots \otimes L(w_{n-1})^{\otimes k_{n-1}}$, since the highest weight of this representation is λ (with highest weight vector $\otimes v_{w_i}^{\otimes k_i}$, where v_{w_i} was the hw-vector of $L(w_i)$). Moreover, recall that $\wedge^i \mathbb{C}^n = L(w_i)$ is a summand of $(\mathbb{C}^n)^{\otimes i}$. We conclude that $L(\Lambda)$ occurs in some $(\mathbb{C}^n)^{\otimes N}$ for N sufficient large. Therefore, the crystal for $\mathbb{C}^n$ together with the rule for taking tensor products of crystals determines the crystal of every representation of $\mathfrak{sl}_n$.

We introduce the **semi-standard Young tableau** of a representation. (This is due to Hodge, Schur, and Young.) Write

$$B(w_1) = \boxed{1} \xrightarrow{1} \boxed{2} \xrightarrow{2} \dots \xrightarrow{n-1} \boxed{n}$$

for the crystal of the standard representation of $\mathbb{C}^n = L(w_1)$ with weights $e_1, \dots, e_n$. For $i < n$ let

$$b_i = \boxed{1} \otimes \boxed{2} \otimes \dots \otimes \boxed{i} \in B(w_1)^{\otimes i}$$

(This corresponds to the vector $v_1 \wedge v_2 \wedge \dots \wedge v_i \in \wedge^i \mathbb{C}^n$, where v_k are the basis vectors of $\mathbb{C}^n$). Since w_i is the highest weight of $\wedge^i \mathbb{C}^n \cong L(w_i)$, b_i of weight $w_i = e_1 + \dots + e_i$ is a highest-weight vector of $B(w_1)^{\otimes i}$. Hence, the connected component of $B(w_1)^{\otimes i}$ containing b_i is $B(w_i)$.

Note the weights of $L(w_i) = \wedge^i \mathbb{C}^n$ are precisely $e_{i_1} + \dots + e_{i_i}$ where $1 \leq i_1 < \dots < i_i \leq n$. Hence the connected component of $B(w_i)$ consist of $\{\boxed{a_1} \otimes \dots \otimes \boxed{a_i} \mid 1 \leq a_1 < \dots < a_i \leq n\} \subset B(w_1)^{\otimes i}$

We write elements of form $\boxed{a_1} \otimes \dots \otimes \boxed{a_i}$ as column vectors

$$\begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_i \end{bmatrix}$$

is $\begin{bmatrix} 1 \\ 2 \\ \vdots \\ i \end{bmatrix}$

Write $\lambda = \sum k_i w_i$ and embed $B(\lambda) \hookrightarrow B(w_1)^{\otimes k_1} \otimes \dots \otimes B(w_{n-1})^{\otimes k_{n-1}}$ by mapping the highest weight vector $b_\lambda \mapsto b_1^{\otimes k_1} \otimes \dots \otimes b_{n-1}^{\otimes k_{n-1}}$ (which is a highest-weight vector in the image). Now, we can represent any element in $B(w_1)^{\otimes k_1} \otimes \dots \otimes B(w_{n-1})^{\otimes k_{n-1}}$ by a sequence of column vectors as in Figure 12, where the entries in the boxes are strictly increasing down columns and (non-strictly) decreasing along rows.

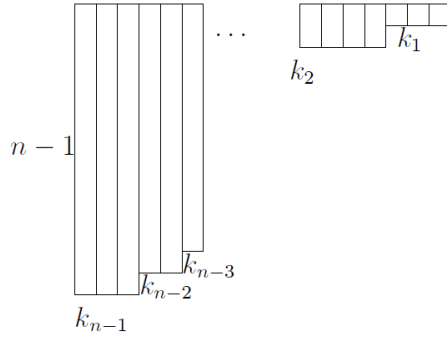


Figure 12: Generic element of $B(w_1)^{\otimes k_1} \otimes \dots \otimes B(w_{n-1})^{\otimes k_{n-1}}$, aka a semistandard Young tableau. Here $k_{n-1} = 2, k_{n-2} = 2, k_{n-3} = 1$ and so on

Definition 12.5. A *semi-standard Young tableau* of shape λ is an array of numbers with dimensions as above, such that the numbers strictly increase down columns, and (non-strictly) decrease along rows.

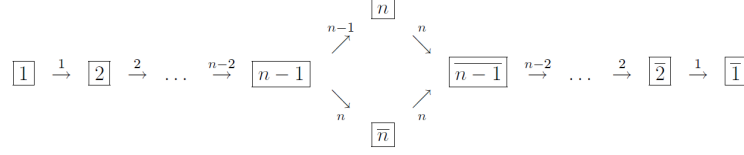
Therefore, the connected component of $B(\lambda)$ in $B(w_1)^{\otimes k_1} \otimes \dots \otimes B(w_{n-1})^{\otimes k_{n-1}}$ are precisely the semi-standard Young tableaux of shape λ .

We will now construct the Young tableaux for all the classical Lie algebras.

Example 53. $\bullet : \mathfrak{so}_{2n+1}$ of type B_n : for the standard representation $\mathbb{C}^{2n+1}$ we have the crystal

$$\boxed{1} \xrightarrow{1} \boxed{2} \xrightarrow{2} \dots \boxed{n} \xrightarrow{n} \boxed{0} \xrightarrow{n} \boxed{\bar{n}} \dots \xrightarrow{1} \boxed{\bar{1}}$$

• : $\mathfrak{so}_2$ of type D_n : the crystal of the standard representation $\mathbb{C}^{2n}$ is



• : $\mathfrak{sp}_{2n}$ of type C_n : the crystal of the standard representation $\mathbb{C}^{2n}$ is

$$[1] \xrightarrow{1} [2] \xrightarrow{2} \dots [n] \xrightarrow{n} [\bar{n}] \dots \xrightarrow{1} [\bar{1}]$$

For B_n we have one more representation, the **spin representation**. Recall the Dynkin diagram of B_n is



Definition 12.6. Let $w_1, \dots, w_n$ be a dual basis of $\alpha_1^\vee, \dots, \alpha_n^\vee$. We call $L(w_n)$ the **spin representation**.

We compute that $w_i = e_1 + \dots + e_i$ for $1 \leq i \leq n-1$ and $w_n = \frac{1}{2}(e_1 + \dots + e_n)$. And $\rho = \sum (n-i+\frac{1}{2})e_i$. Hence $w_n + \rho = \sum (n-i+1)e_i$. Then $(e_i - e_j, w_n) = 0$ for $i < j$, while $(e_i + e_j, w_n + \rho)/(e_i + e_j, \rho) = \frac{2n+2-i-j}{2n+1-i-j}$ and $(e_i, w_n + \rho)/(e_i, \rho) = \frac{2n+2-2i}{2n+1-2i}$. We can plug them in the Weyl dimension formula and deduce that $\dim L(w_n) = 2^{n-1}$

Define $B = \{(i_1, \dots, i_n) | i_j \in \{\pm 1\}\}$ with $wt((i_1, \dots, i_n)) = \sum \frac{1}{2}i_j e_j \in P$, and

$$\tilde{e}_i(i_1, \dots, i_n) = \begin{cases} (i_1, \dots, +1_j, -1_{j+1}, \dots, i_n) & \text{if } (i_j, i_{j+1}) = (-1, 1) \\ 0 & \text{else} \end{cases}$$

for $1 \leq i \leq n-1$ since $\alpha_i = e_i - e_{i+1}$, and

$$\tilde{e}_n(i_1, \dots, i_n) = \begin{cases} (i_1, \dots, +1_n) & \text{if } i_n = -1 \\ 0 & \text{else} \end{cases}$$

since $\alpha_n = e_n$. This is the crystal of the spin representation $L(w_n)$.

Remark 12.6. Note that $\dim L(w_n) = \dim \wedge \mathbb{C}^n$. This is for a very good reason. Indeed $\mathfrak{gl}_n \subset \mathfrak{so}_{2n+1}$ via $A \mapsto \begin{pmatrix} A & & \\ & 0 & \\ & -JA^T J^{-1} \end{pmatrix}$, and $L(w_n)|_{\mathfrak{gl}_n} = \wedge \mathbb{C}^n$

For type D_n , the situation is more complicated. We can define representations

$$V^+ = L(w_n), V^- = L(w_{n-1})$$

these are called the **half-spin representations**. The corresponding crystals are $B^\pm = \{(i_1, \dots, i_n) | i_j \in \{\pm 1\}\}$ with $\prod i_j = 1$ and $\prod i_j = -1$ for B^- . We define the weight and most $\tilde{e}_i, \tilde{f}_i$ as before, with the change that

$$\tilde{e}_n(i_1, \dots, i_n) = \begin{cases} (i_1, \dots, +1_{n-1}, +1_n) & \text{if } (i_{n-1}, i_n) = (-1, -1) \\ 0 & \text{else} \end{cases}$$

since $\alpha_n = e_{n-1} + e_n$ now.

13 Littelmann Paths

Set $P_{\mathbb{R}} = P \otimes_{\mathbb{Z}} \mathbb{R}$. By a path we mean a piecewise linear continuous map $[0, 1] \rightarrow P_{\mathbb{R}}$. We consider paths up to reparametrisation, i.e. $\pi = \pi \circ \phi$, where $\phi : [0, 1] \rightarrow [0, 1]$ is a piecewise-linear isomorphism. (That is, we care about the trajectory, not about the speed at which we traverse the path.)

Let $\mathcal{P} = \{\text{path } \pi \text{ such that } \pi(0) = 0, \pi(1) \in P\}$. Define a crystal structure on $\mathcal{P}$ as follows: we set

$$wt(\pi) = \pi(1)$$

To define $\tilde{e}_i(\pi)$, let

$$h_i = \min \mathbb{Z} \cap \{(\pi(t), \alpha_i^\vee) | 0 \leq t \leq 1\} \leq 0$$

That is, h_i is the smallest integer in $(\pi(t), \alpha_i^\vee)$ with $0 \leq t \leq 1$, and since $\pi(0) = 0$, $h_i \leq 0$

If $h_i = 0$, we set $\tilde{e}_i(\pi) = 0$ (this is not the path that stays at 0, but rather the extra element in the crystal).

If $h_i < 0$, take the smallest $t_1 > 0$ such that $(\pi(t_1), \alpha_i^\vee) = h_i$. Then take the largest $t_0 < t_1$ s.t. $(\pi(t_0), \alpha_i^\vee) = h_i + 1$

The idea is to reflect the segment of the path π from t_0 to t_1 in the $(\lambda, \alpha_i^\vee) = h_i + 1$ plane. Therefore, we define the path $\tilde{e}_i(\pi)$ as follows:

$$\tilde{e}_i(\pi) = \begin{cases} \pi(t) & 0 \leq t \leq t_0 \\ \pi(t_0) + s_i(\pi(t) - \pi(t_0)) & t_0 < t \leq t_1 \\ \pi(t) + \alpha_i & t > t_1 \end{cases}$$

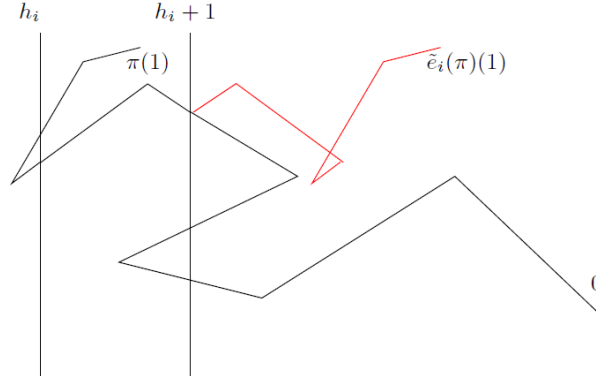


Figure 13: π is marked in black; $\tilde{e}_i(\pi)$, from the point when it deviates from π , is marked in red.

So applying $\tilde{e}_i$ essentially increases h_i by 1, and we have $\epsilon_i(\pi) = -h_i$

Example 54. Let's compute the crystals of some representations of $\mathfrak{sl}_2$:

$$\tilde{e}_i(-\bullet_{-\alpha_i/2} \longleftarrow \bullet_0) = (\bullet_0 \longrightarrow \bullet_{\alpha_i/2}), \quad \tilde{e}_i(\bullet_0 \longrightarrow \bullet_{\alpha_i/2}) = 0$$

and

$$\tilde{e}_i(-\bullet_{-\alpha_i} \longleftarrow \bullet \longleftarrow \bullet_0) = (-\bullet_{-\alpha_i/2} \longrightarrow \bullet_0) \quad \tilde{e}_i(-\bullet_{-\alpha_i/2} \longrightarrow \bullet_0) = (\bullet_0 \longrightarrow \bullet \longrightarrow \bullet_{\alpha_i})$$

(the path in the graph is naturally parametrised)

If π is a path, let $\pi^\vee$ be the reversed path, i.e. $t \mapsto \pi(1-t) - \pi(t)$. Define $\tilde{f}_i(\pi) = (\tilde{e}_i(\pi^\vee))^\vee$. These make $\mathcal{P}$ a crystal.

Define $\mathcal{P}^+ = \{\text{paths } \pi \text{ such that } \pi[0, 1] \subset P_{\mathbb{R}}^+ = \{x \in P_{\mathbb{R}} | (x, \alpha_i^\vee) \geq 0 \forall i\}\}$. If $\pi \in \mathcal{P}_{\mathbb{R}}^+$, then $\tilde{e}_i(\pi) = 0$ for all i by definition.

For $\pi \in \mathcal{P}^+$, let B_π be the subcrystal of $\mathcal{P}$ generated by π (i.e. $B_\pi = \{\tilde{f}_{i_1} \dots \tilde{f}_{i_j} \pi\}$)

Theorem 13.1. (*Littleman*)

1. For $\pi, \pi' \in \mathcal{P}^+$, $B_\pi \cong B_{\pi'}$ as crystals $\iff \pi(1) = \pi'(1)$
2. There is an isomorphism between B_π and $B(\pi(1))$, where $B(\pi(1))$ is the crystal of $L(\pi(1))$. (Of course, the isomorphism sends $\pi \mapsto \pi(1)$)
3. Moreover, for the path $\pi(t) = \lambda t, \lambda \in P^+$, Littleman gives an explicit combinatorial description of the paths in B_π .

Example 55. Let's compute the crystal of the adjoint representation of $\mathfrak{sl}_3$ of type A_2 : we consider the most basic paths starting from 0 to each root and the effect of $\tilde{e}_\alpha, \tilde{e}_\beta$. For example

$$\tilde{e}_\beta \left(\begin{array}{ccc} & & \bullet \\ & \swarrow & \\ -(\alpha + \beta) & & \end{array} \right) = (-\alpha \longleftarrow \bullet)$$

as what we expect by applying e_β to $\mathfrak{g}_{-\alpha-\beta}$

The tensor product of crystals has a very nice (and natural!) realisation in terms of concatenating paths:

Definition 13.1. For $\pi, \pi' \in \mathcal{P}$, define $\pi * \pi'$ to be the concatenation (traverse π , then translate π' to start at $\pi(1)$ and traverse it).

One checks that $*$: $\mathcal{P} \otimes \mathcal{P} \rightarrow \mathcal{P}$ is a morphism of crystals. That is, the tensor product of two crystals simply consists of the concatenated paths in them.

Remark 13.1. We have now defined $B(\Lambda)$ explicitly without using $L(\Lambda)$. One can prove the Weyl character formula

$$chL(\Lambda) = \frac{\sum_{w \in W} \det w \cdot e^{w(\Lambda + \rho)}}{e^\rho \prod_{\alpha \in R^+} (1 - e^{-\alpha})}$$

from this, without referring to $L(\Lambda)$ or quantum groups. To do this, one builds $chL(\lambda)$ and $L(\Lambda)$ itself one root at a time, and uses the **Demazure character formula**.

More specifically: $\forall w \in W$ consider $L_w(\Lambda)$, the $\mathfrak{n}^+$ -submodule of $L(\Lambda)$ generated by $v_{w\Lambda}$, where v_Λ is the highest weight vector. (Equivalently, it is the submodule generated by the 1-dimensional space $wL(\Lambda)_\Lambda$)

Theorem 13.2. (*Demazure character formula*)

$$chL_w(\Lambda) = D_w(e^\Lambda)$$

where we write $w = s_{i_1} \dots s_{i_r}$ a reduced (i.e. shortest) decomposition, and set

$$D_w = D_{s_{i_1}} \dots D_{s_{i_r}}, D_{s_i}(f) = (1 + e^{s_i}) \frac{f}{1 - e^{-\alpha_i}}$$

That is

$$D_{s_i}(e^\lambda) = \begin{cases} e^\lambda + e^{\lambda - \alpha_i} + \dots + e^{s_i(\lambda)} & \text{if } (\lambda, \alpha_i^\vee) \geq 0 \\ 0 & \text{if } (\lambda, \alpha_i^\vee) = -1 \\ -(e^{\lambda + \alpha_i} + \dots + e^{s_i(\lambda) - \alpha_i}) & \text{if } (\lambda, \alpha_i^\vee) \leq -1 \end{cases}$$

“I’ve got about six minutes. Is that enough time to tell you about the Langlands program?”

Suppose that G is an algebraic group with Lie algebra $\mathfrak{g}$ which is semisimple. For a given $\mathfrak{g}$, there is the smallest group G above it, $G^{ad} = G/Z_G$ (whose centre is $\{1\}$). There is also the largest group, G^{sc} the simply connected cover of G , which is still an algebraic group. We have $\pi_1(G^{ad}) = (P/Q)^*$ and $Z_{G^{sc}} = P/Q$.

(For example, SL_n is the simply connected group, and PSL_n is the adjoint group.)

In this course we have studied the category $Rep\mathfrak{g} = RepG^{sc}$.

Recall that for G a finite group, the number of irreducible representations is the number of conjugacy classes, but conjugacy classes do not parametrise the representations. For us, however, the representations are parametrised by $P^+ = P/W$.

It is possible, for G an algebraic group, to define ${}^L G$, the Langlands dual group, where the Lie algebra $\mathfrak{g}$ and the root system R correspond to $\mathfrak{g}^\vee$ and $R^\vee$ (and in fact the torus and the dual torus will swap in the dual group).

If we consider the representations of $G(F)$ where $F = \mathbb{F}_q$ or $F = \mathbb{C}((t))$, these correspond roughly to conjugacy classes of homomorphisms $(W(F) \rightarrow {}^L G(\mathbb{C}))/{}^L G(\mathbb{C})$, where $W(F)$ is a “thickening” of the Galois group of F .

References

[Poo17] Bjorn Poonen. *Rational Points on Varieties*. American Mathematical Society, 2017.